

Seventy-six conjectures in mathematics

A local–global random model, five structural programmes, and six conjectures across fields

The Conjecture Engine*

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Summary sheet: the seventy-six conjectures

Part I: conjectures from the local–global random model, in importance–novelty order.

1. **Prime-power contamination calculus (Conjecture 1).** For a balanced pair race $(n, n + d) \bmod m$, the surviving prime-square orientations $(q^2 - d, q^2)$, $(q^2, q^2 + d)$ occupy provably computable residue classes, and the operator $\mathcal{C}_{d,m}(a; x) = \sum_o \sum_q \Lambda(q^2) \Lambda(\text{prime member})$ determines the drift vector: for twins, $\mathcal{M}_x(D_1(t) \log^2 t / \sqrt{t}) \rightarrow c_T = \frac{1}{2} C(x, x^2 - 2) \pmod{5}$ and $\rightarrow 2c_T$ on class 7 $\pmod{8}$, symmetric differences $\rightarrow 0$; in general each class deficit satisfies $\mathcal{M}_x(\text{deficit} \cdot \log^2 t / \sqrt{t}) \rightarrow \lim \mathcal{C}_{d,m}(a; x) / \sqrt{x}$. Convergence of these logarithmic means at drift scale is part of the conjecture (Rubinstein–Sarnak-type structure for pattern races).
2. **Pinned singular-series variance (Conjecture 2).** With $R(h) = C_4(0, 2, h, h + 2) / \mathfrak{S}(2)^2$ and $G(H) = \sum_{h \leq H} (1 - h/H)(R(h) - 1)$: (i) [conditional proposition] the moving-window twin-pair count obeys $\text{Var} / \mathbb{E} = 1 + \mathfrak{S}(2)(2G(H) - 1) / \log^2 x + o(\log^{-2} x)$; (ii) $-G(H) / \log H \rightarrow \infty$: pair counts are more sub-Poisson than prime counts.
3. **Least-prime ordering deficit (Conjecture 3).** With $U(a, q) = \text{Li}(p(a, q)) / \varphi(q)$: (i) $U \Rightarrow \text{Exp}(1)$ marginally (attributed), Gumbel maximum under joint independence; (ii) the dyadic averages of $\Theta(q) = (1 - \mathbb{E}_a[U]) \log q$ over prime $q \in [Q, 2Q]$ converge to a limit $\Theta > 0$: the ordered primes fill residue classes faster than any exchangeable model, in excess of the derived discreteness and injectivity baselines.
4. **Goldbach lane race (Conjecture 4).** For $n \equiv 2 \pmod{4}$, prime squares enter ordered Goldbach representations only through the $(1, 1)$ lane $\bmod 4$, so with $D = R_3 - R_1$ and the explicit weighted census D_{sys} : $\mathbb{E}_x[D - D_{\text{sys}}] = o(\mathbb{E}_x[D_{\text{sys}}])$ under logarithmic sampling; the drift vanishes on the internal null subprogression $n \equiv 10 \pmod{12}$; the sign density of D tends to $\frac{1}{2}$.

*This paper was produced by an automated conjecture engine (Claude, Anthropic), operated as an exercise in disciplined conjecture-making. All constants, counts, and verification results are reproducible from the accompanying repository (`conjecture-engine`): `run_all.py` regenerates every number in this paper. Primality of integers beyond 3.3×10^{24} was tested with Baillie–PSW and is therefore a probable-prime classification; all smaller primality claims are deterministic (fixed-base Miller–Rabin).

5. **Least Goldbach summand (Conjecture 5).** With $s(n)$ the least prime $p \nmid n$ with $n - p$ prime, $U(n) = \mathfrak{S}(n) \sum_{p \leq s(n), p \nmid n} 1/\log(n - p)$, and the dyadic-log ensemble $\mathbb{E}_X$: (i) $U \Rightarrow \text{Exp}(1)$; (ii) $\Theta_G(X) = (1 - \mathbb{E}_X U) \log X \rightarrow \Theta_G > 0$, the Goldbach sibling of the ordering deficit.
6. **Polynomial–exponential entanglement (Conjecture 6).** Primes $n^2 + 2^n$ force $n \equiv 3 \pmod{6}$; with $\kappa_S = 3 D_S / \prod_{p \in S} (1 - 1/p)$ built from the CRT-exact survivor density D_S over the joint period of S and the lane bonus 3 at the primes 2 and 3: κ_{S_z} converges along the canonical exhaustion $S_z = \{5 \leq p \leq z\}$ to κ , the counting function is $\sim \sum_{n \equiv 3(6), n \leq N} \kappa / \log(n^2 + 2^n) \asymp \log N$; whether the limit factors exactly (as observed through $p \leq 19$) is an open question.
7. **Multibase Fermat quotients (Conjecture 7).** For $p \nmid a$: the Eisenstein–Lerch homomorphism confines multiplicatively dependent bases to a rational subtorus, and (iv) for multiplicatively independent bases $(q_p(a_1), \dots, q_p(a_r))/p$ equidistributes on the full torus as p varies; (v) at most finitely many simultaneous Wieferich primes; plus single-base clauses: KS-discrepancy LIL at constant $1/\sqrt{2}$, shrinking-target law $\#\{p \leq x : q_p < K\} \sim \sum \min(1, K/p)$, Wieferich-count LIL and an almost-sure CLT at the du/u weighting.
8. **Uniform de Polignac covariance field (Conjecture 8).** $\pi_d(x) = \mathfrak{S}(d) \text{Li}_2(x)(1 + o(1))$ uniformly for even $d \leq (\log x)^A$; the moving-window residual field has covariance $K_3(d, d')H/\log^3 x + (1/\log^4 x) \sum_{|h| \leq H} (H - |h|) K_4(d, d'; h)$, triple term dominant on the polylogarithmic window class, with Poisson/Gaussian regimes as finite- x corrections to independence.
9. **Race sub-diffusivity (Conjecture 9).** Balanced-race steps at event index have negative autocorrelations obeying $\rho_k(x) = -(c_k \log \log x + d_k)/\log x (1 + o(1))$; under uniformity in k , tail summability, and $\sum c_k, \sum |d_k| < \infty$, the diffusivity is $\sigma^2(x) = 1 - 2(\sum c_k \log \log x + \sum d_k)/\log x (1 + o(1))$, those hypotheses being model-layer, not asserted asymptotically; the asymptotic diffusive-vs-rigid dichotomy is registered open, with the rigid branch selected only at Conjecture 1(i-b) for its contaminated races.
10. **Uniform quadratic de Polignac and its constants (Conjecture 10).** $\#\{n \leq N : n^2 + 1, n^2 + 1 + d \text{ prime}\} = C(d) I_d(N)(1 + o(1))$ uniformly for even $d \leq (\log N)^B$; a separate power-saving rate clause; and the family law: moments of $C(d)$ converge to derived Euler products (mean 2.7456..., sd 1.6840...), the empirical law converging to the random product with CRT-independent local factors.
11. **First occurrences of prime gaps (Conjecture 11).** On realized gaps: $\log p(g)/\sqrt{g}$ bounded between positive constants; $\liminf = \sqrt{e^\gamma/2}$ conditional on the realization hypothesis; and $\log p(g) = \sqrt{g} + \frac{1}{2} \log g - \frac{1}{2} \log \mathfrak{S}^*(g) + E_g$ with $\Pr[E_g \leq t] \rightarrow 1 - \exp(-e^{2(t-\mu)})$, the min-type Gumbel at scale $\frac{1}{2}$ forced by the slope-2 hazard expansion.
12. **Stern lane race (Conjecture 12).** In $n = p + 2k^2$, square contamination $n = q^2 + 2k^2$ sits in the k -even lane iff $n \equiv 1 \pmod{8}$, k -odd iff $n \equiv 3$, and is impossible for $n \equiv 5, 7$ (provable null classes); on contaminated classes the clean-minus-contaminated difference obeys $\mathbb{E}_X[D - D_{\text{sys}}] = o(\mathbb{E}_X[D_{\text{sys}}])$ with the $\Lambda(q^2)$ -weighted census D_{sys} .
13. **Microscopic variance law (Conjecture 13).** [Conditional on quantitative Hardy–Littlewood] For windows of length $H = \lambda(\log x)^\alpha$, $\text{Var } X/\mathbb{E}X = 1 - (\log H + \gamma + \log 2\pi -$

- 1)/ $\log x + o(1/\log x)$ uniformly in $1 \leq \alpha \leq A$, with Gallagher's Poisson law at the boundary $\alpha = 1$.
14. **Null-mechanism race (Conjecture 14).** For $(n^2 + 1, n^2 + 3)$, square contamination is algebraically impossible (single bounded exception $(n, q) = (1, 2)$); the race mod 5 is driftless at the contamination scale, $\mathcal{M}_x(D(t) \log^2 t / \sqrt{t}) \rightarrow 0$ —the negative control for item 1—and the logarithmic occupation of leadership tends to $\frac{1}{2}$ with Gaussian fluctuations $\sqrt{\log 2 / \log x}$, conjectured directly (the event-index invariance principle is a separately falsifiable local hypothesis, which does not imply it).
 15. **Cubic-shift constants (Conjecture 15).** For non-cube a , the three-state local law gives $\mathbb{E}_a[\omega(p)] = 1$ exactly, so the limit law of $C(a)$ has mean exactly 1 (an L^2 -martingale argument), derived sd $0.2762\dots$; counts are uniform over non-cube $a \leq (\log N)^B$.
 16. **Conjecture F as a family (Conjecture 16).** $Q_A(N) = C(A)I_A(N)(1 + o(1))$ uniformly over odd $A \leq (\log N)^B$; same-index window covariance $[C(A, A') - C(A)C(A')]H/4\log^2 N$ (negative for locally exclusive pairs), with the off-index pinned sum at the same $1/4\log^2 N$ order left open.
 17. **Twin-member Goldbach, orientation-resolved (Conjecture 17).** Under n -uniform Hardy–Littlewood for the four role systems, $R_T(n) = [\sum_o \mathfrak{S}_4^{(o)}(n)] \int_5^{n-5} dt / (\log^2 t \log^2(n-t)) (1 + o(1))$, with the exact identities $\mathfrak{S}_4^{(\text{lu})} = \mathfrak{S}_4^{(\text{ul})}$ and $\mathfrak{S}_4^{(\text{uu})}(n) = \mathfrak{S}_4^{(\text{ll})}(n-4)$.
 18. **Triplet contamination (Conjecture 18).** For $(n, n+2, n+6) \bmod 5$, the unique surviving square configuration is the doubly-thinned $(q^2 - 2, q^2, q^2 + 4)$ (the other two configurations dying for every prime $q > 3$), contaminating class 2: $\mathcal{M}_x((\pi_3(t; 5, 1) - \pi_3(t; 5, 2)) \log^3 t / \sqrt{t}) \rightarrow c_3$, the Bateman–Horn constant of $(q, q^2 - 2, q^2 + 4)$.
 19. **Sexy-pair contamination matrix (Conjecture 19).** For $(n, n+6)$ both orientations survive on complementary q -classes, feeding start classes 3 and 1 (mod 5 and mod 8) with independent constants c_A, c_B (the $(q, q^2 \mp 6)$ Bateman–Horn constants) in the drift-scale sense; class 2 (mod 5) and classes 5, 7 (mod 8) are provably clean.
 20. **Fibonacci–Lucas twins (Conjecture 20).** The odd prime-divisor pools of F_p and L_p are disjoint (ranks p vs $2p$; $\gcd = 2$ iff $3 \mid p$); only finitely many p have both prime; the naive joint accounting assigns prior mass about 1.4×10^{-2} to the index range beyond 10^4 , in which the catalogued index 148091 falls, recorded descriptively and conditionally on the probable-prime status of both F_{148091} and L_{148091} , and fatal to any completeness list.
 21. **Factorial twins (Conjecture 21).** Window rigidity: $n! + a$ is composite for $2 \leq |a| \leq n$ (theorem), so ± 1 are the only bounded offsets; $n = 3$ uniqueness is attributed; the joint-fluctuation clause is a declared-model CLT for $F_+ - F_-$ under explicit covariance and higher-cumulant hypotheses.
 22. **Boundary trichotomy (Conjecture 22).** For $\deg F \geq 2$ with positive leading coefficient, $F(m) - F(j)$ prime forces $m - j = 1$ outside a bounded region; for $x^3 + cx$ the boundary lane $3m^2 - 3m + 1 + c$ is dead-parity for odd c , dead-3-adic for even $c \equiv 2 \pmod{3}$, admissible exactly for $c \equiv 0, 4 \pmod{6}$, with uniform Bateman–Horn counts over admissible $c \leq (\log M)^B$.

23. **Power-obstruction ladder (Conjecture 23).** $m^k = p + j^k$ is impossible for composite k (theorem); for prime k it forces $j = m - 1$ and reduces to primality of the irreducible $D_k(m) = m^k - (m - 1)^k$, which follows Bateman–Horn uniformly over the prime- k lanes.
24. **Alternating cyclotomic chain (Conjecture 24).** Infinitely many primes p with $\Phi_3(p)$ and $\Phi_6(\Phi_3(p))$ prime, at the derived Bateman–Horn rate; alternation is the unique admissible continuation in the anchored construction space.
25. **Twin cyclotomic bases (Conjecture 25).** Of the $\varphi(k) = 2$ family, only $k = 3$ survives ($k = 4$ parity-dead, $k = 6$ a translate), and $\Phi_3(n)$, $\Phi_3(n + 1)$ are simultaneously prime infinitely often with the computed Bateman–Horn constant.

Part II: structural conjectures, in programme order.

26. **Connected motif generating functional (Conjecture 26).** A full connected diagram calculus for all joint cumulants of prime motifs.
27. **Complete overlap-renormalization filtration (Conjecture 27).** A classification of every possible covariance scale and rigid eigenspace in a finite motif family.
28. **Regularized mesoscopic local–spectral trace formula (Conjecture 28).** Independently mollified Euler-product and multiple-zero functionals agree after named counterterms, and the limit is regularization independent.
29. **Anchored arithmetic polymer expansion (Conjecture 29).** Palm cluster activities are graded by the number of new prime constraints beyond one anchored motif.
30. **Topological expansion of non-Gaussianity (Conjecture 30).** The first non-Gaussian terms are graded by the homology of overlap-incidence complexes.
31. **Connected first-arrival functional (Conjecture 31).** The entire least-prime point process has connected kernels obtained from same-class prime correlations.
32. **Tested Gauss-polyspectral reciprocity (Conjecture 32).** Scalar tensor observables have exact Gauss-transport identities under the random-modulus law and dual arithmetic limits.
33. **Rubinstein–Sarnak terminal chaos dichotomy (Conjecture 33).** Terminal extremes are directed by the limiting low-zero chaos measure, with an explicit self-averaging versus Cox criterion.
34. **Nonlinear spectral response calculus (Conjecture 34).** Low-zero and exceptional-zero perturbations pass through first-arrival statistics by universal Volterra response operators.
35. **Local-information capacity of odd class groups (Conjecture 35).** Cohen–Lenstra independence has a sharp two-resource boundary governed by cell entropy and conductor complexity.
36. **Kummer–Haar law on relation tori (Conjecture 36).** Saturated multiplicative relations give the exact horizontal support and Haar law of Fermat-quotient vectors.

37. **Rank-dimensional finite-logarithm large sieve (Conjecture 37).** A positive power range of frequencies obeys a large sieve whose dual dimension is saturated rank.
38. **Mesoscopic-to-lattice shrinking-target transition (Conjecture 38).** Haar universality persists for growing targets, while bounded lattice targets acquire a separate arithmetic regulator intensity.
39. **Global-lattice p -adic regulator-matrix law (Conjecture 39).** A fixed integral Galois-relation lattice controls horizontal regulator matrices and their determinantal rare events.
40. **Functorial horizontal logarithms on tori (Conjecture 40).** A finitely generated global subgroup of a torus has a functorial matrix-valued finite-logarithm Haar law.
41. **Entropy-conductor profile of arboreal resolution (Conjecture 41).** Exact class-measure entropy and Artin-conductor profiles determine observable depth and finite-index shifts.
42. **Arboreal family large sieve and cumulant independence (Conjecture 42).** Growing preimage quotients satisfy square-root trace bounds and higher connected Frobenius factorization over parameters.
43. **Coloured dynatomic Galois-factor process (Conjecture 43).** Exact-period components generate independent coloured, Frobenius-marked factor processes, with growing-degree support restrictions.
44. **Complexity-uniform primitive valuation process (Conjecture 44).** The complete primitive valuation vector has an adelic product law under an explicit height-discriminant complexity regime and uniform squarefull tails.
45. **Divisor-sensitive dynamical gcd classification (Conjecture 45).** Positive normalized gcd height is exactly eventual entry into a preperiodic subvariety carrying the relevant common divisor.
46. **Frobenius-marked Poisson-Dirichlet process (Conjecture 46).** Macroscopic polynomial-value factors carry fixed-point-size-biased Galois marks.
47. **Three-scale polynomial factorization (Conjecture 47).** Small valuations, mesoscopic scale-invariant factors, and macroscopic marked factors form one conservation-corrected product structure.
48. **Adelic Gibbs gluing for reducible values (Conjecture 48).** Reducible polynomial factor processes are an archimedean- p -adic Gibbs mixture of independent component processes.
49. **Full multivariate adelic saddle law (Conjecture 49).** Joint smoothness is governed by a complete joint Perron action, saddle displacement, and Hessian, not a scalar Euler correction.
50. **Buchstab-Bateman-Horn component flow (Conjecture 50).** Rough prime, semiprime, and higher-almost-prime polynomial values follow universal Buchstab components with Galois marks.

Part III: six conjectures across fields, in statement order.

51. **Relative polynomial pretentiousness inverse principle (Conjecture 51).** A persistently nonzero logarithmic correlation of completely multiplicative unimodular functions along multiplicatively independent irreducible polynomials forces one existential block of primitive characters and archimedean parameters with finite relative pretentious distances, the balance identity $\sum_j t_j \deg P_j = 0$, and a nonvanishing joint character average at a common modulus.
52. **Monotone threshold purification of vector paths (Conjecture 52).** Every monotone fractional activation schedule of m vectors of norm at most one in $\mathbb{R}^d$ admits a rounding to single irreversible activation times whose path error is $O(\sqrt{d})$ uniformly in m , containing the open Euclidean Steinitz prefix-sum problem and asserting removal of the logarithm from Banaszczyk's $O(\sqrt{d + \log m})$.
53. **Entropy dimension of random-free graphons (Conjecture 53).** If the row space of a $\{0, 1\}$ -valued graphon is Ahlfors s -regular then $H(G(n, W))/(n \log n) \rightarrow s$, the hypothesis class being nonvacuous at every $s \in (0, 2]$ by Cantor-type threshold constructions.
54. **A Hessian principle for isolated microcanonical graph phases (Conjecture 54).** Under isolated-phase, transversally Lipschitz, and nonemptiness hypotheses, the tangent block averages of a microcanonical coloured graph ensemble fluctuate at scale n^{-1} around their mean with centred Gaussian limit of covariance A^{-1} , the inverse of the contracted entropy Hessian.
55. **Sharp stability of entropy idempotence on finite abelian groups (Conjecture 55).** The squared total variation distance from a probability measure to the nearest uniform coset measure is bounded by a universal constant, conjecturally exactly $2/\log 2$ as pinned by discretized Gaussians on $\mathbb{Z}/p\mathbb{Z}$, times the convolution entropy defect $H(\mu * \mu) - H(\mu)$.
56. **Stationary convolution-entropy rigidity (Conjecture 56).** An ergodic stationary process over a finite abelian group preserves entropy under independent self-convolution exactly when its quotient by the translational stabilizer has zero entropy, so in particular every ergodic binary process of intermediate entropy gains entropy under independent XOR.

Part IV: twenty conjectures on the class-counting polynomial, in statement order.

57. **Irreducibility and connectivity (Conjecture 57).** $H_G(U, Y) = C_G(1 + U, Y)$ is irreducible in $\mathbb{Z}[U, Y]$ exactly when G is connected, the disconnected direction being immediate from Rossmann's multiplicativity and the connected direction verified by exact factorization for all 996 connected graphs on at most seven vertices.
58. **Recovery of the subgraph-component polynomial (Conjecture 58).** Equality of class-counting polynomials forces equality of Tittmann–Averbouch–Makowsky subgraph-component polynomials, though the two arrays, $(|S|, |N[S]| - c(G[S]))$ against $(|S|, c(G[S]))$, admit no apparent functional relation.
59. **Two-field rigidity (Conjecture 59).** The adjoint-rank distributions of the graphical Lie algebras over $\mathbb{F}_2$ and $\mathbb{F}_3$ together determine them over every finite field.

60. **Hybrid rigidity (Conjecture 60).** The $\mathbb{F}_2$ rank distribution together with the subgraph-component polynomial determines the class-counting polynomial, while the degree sequence in place of Q_G does not suffice.
61. **Binary edge-deck reconstruction (Conjecture 61).** The multiset of binary specializations $C_{G-e}(2, Y)$ over the edges determines a graph with at least four edges up to isomorphism, a statement in the orbit of the edge reconstruction conjecture and the high-risk member of the part.
62. **Bipartite binary rigidity (Conjecture 62).** Within bipartite graphs the single specialization $C_G(2, Y)$ determines C_G .
63. **Connected-pseudoforest binary rigidity (Conjecture 63, resolved false).** The proposed one-field rigidity for trees and connected unicyclic graphs is refuted by an explicit seven-vertex unicyclic pair sharing the binary specialization with distinct degree multisets, the unique violating class through nine vertices, and the tree case survives exhaustively through fourteen vertices.
64. **Chordal binary rigidity (Conjecture 64).** The same one-field rigidity holds within chordal graphs.
65. **Recovery of the domination number (Conjecture 65).** Equality of class-counting polynomials forces equality of domination numbers, a statement strictly between Rossmann's connected-domination theorem and the false total-domination analogue.
66. **Recovery of the independent domination number (Conjecture 66).** Equality of class-counting polynomials forces equality of independent domination numbers.
67. **Global double log-concavity (Conjecture 67).** For every graph the coefficient sequence a of $f_G(1 + Z)$ is log-concave and remains log-concave after one application of $L(a)_i = a_i^2 - a_{i-1}a_{i+1}$, the third iterate of L failing for a thirteen-vertex graph.
68. **Strict log-concavity for connected graphs (Conjecture 68).** For connected graphs with at least one edge the inequality $a_i^2 > a_{i-1}a_{i+1}$ is strict at every interior position of the support.
69. **Strict second-order log-concavity (Conjecture 69).** Under the same hypotheses $L(a)$ is strictly log-concave at every interior position of its support.
70. **Rank-slice strict log-concavity (Conjecture 70).** For connected graphs every non-monomial rank slice $[Y^r]F_G(1 + Z, Y)$ has strictly log-concave coefficients at interior positions, the second-order strengthening of the slice statement failing for an eight-vertex graph.
71. **Interval support at fixed rank (Conjecture 71).** The support sizes attaining a fixed rank form an interval of integers, while the transposed statement, attainable ranks at fixed support size, is false.
72. **Coefficientwise tree extremality (Conjecture 72).** Among n -vertex trees the shifted polynomial $f_T(1 + Z)$ is sandwiched coefficientwise between path and star, with equality only at the extremes.

- 73. **Star rank-majorization (Conjecture 73).** For $n \geq 5$ and every prime power q the decreasingly ordered adjoint-rank distribution of the star majorizes that of every n -vertex tree, the hypothesis being sharp since the star fails to majorize the path at $(n, q) = (4, 2)$.
- 74. **Tree rank-variance extremality (Conjecture 74).** For $n \geq 4$ and every prime power the variance of $\text{rank}(\text{ad}_x)$ over a uniformly random element is uniquely minimized by the path and uniquely maximized by the star among n -vertex trees.
- 75. **Leaf-count rigidity (Conjecture 75).** Equality in the lemma's bound $s(T) \geq n - \ell(T) + 2$ holds exactly for paths and stars.
- 76. **Diameter rigidity (Conjecture 76).** Equality in the lemma's bound $s(T) \geq \text{diam}(T) + 1$ holds exactly for paths and stars.

Abstract

We state seventy-six conjectures in four parts. The twenty-five conjectures of Part I are each derived from the local-global random model of the primes: Cramér's model corrected by Hardy–Littlewood singular series, with Bateman–Horn as the organizing framework and Borel–Cantelli accounting for sparse events. Each statement is calibrated to the strongest form the heuristic supports. Every computable constant is computed from its definition, with local admissibility asserted programmatically; every statement was tested against exact counts (probable-prime counts, so labelled, where the integers involved exceed the deterministic certification range), with success measured by the *shape* of agreement (predicted constant, square-root residuals, no drift) rather than by the size of the range searched. The claimed content includes: a prime-power contamination calculus for pattern races—twin races modulo 5 and 8 driven by the pairs $(q^2 - 2, q^2)$, cousin, sexy-pair, and triplet predictions derived from the calculus before testing, a Goldbach lane race with a verified internal null lane, a Stern-representation lane race whose null classes are provable by direct congruence, and an algebraic null control; the measured sub-diffusivity of balanced races (universal negative step correlations tied to the Lemke Oliver–Soundararajan repulsion); canonical ordering deficits for least primes in progressions and for least Goldbach summands; derived covariance kernels for moving-window residual fields, with the diagonal deficit reduced to a pinned singular-series average whose growth already exceeds the single-prime Montgomery–Soundararajan term; distribution laws for the constants of the quadratic-pair and cubic-shift families, with exact mean-value lemmas; an entanglement-aware local-global law for $n^2 + 2^n$ with an exact-rational CRT factorization; a boundary trichotomy for polynomial ladders; a multibase subtorus law for Fermat quotients; and singular-series waiting-time refinements for first prime gaps. Every statement was checked against the literature by an independent search, with novelty labelled conservatively: statements already known in essence are attributed and kept *inside* the conjecture whose claimed content they calibrate, since those benchmarks are what make the derived constants credible. Every falsifiable-by-instance statement was then stress-tested computationally well beyond its original range, and every constant was recomputed independently from its definition. Part II adds twenty-five structural conjectures in five programmes—connected prime-pattern fields, arithmetic first-arrival fields and class groups, finite logarithms and algebraic tori, arboreal arithmetic dynamics, and adelic factorization processes—each introducing a canonical operator, process, invariant, classification, or phase boundary, with its mechanism, nearest literature boundary, first decisive theorem, and failure mode. Part III adds six conjectures across fields: a relative pretentious inverse principle for polynomial arguments, a threshold purification law for monotone vector paths, an entropy-dimension law for random-free graphons, a Hessian principle for microcanonical graph phases, a sharp stability law for entropy idempotence on finite abelian groups, and a convolution-entropy rigidity for stationary processes. Part IV (Conjectures 57–76) adds twenty conjectures

on a single further object, Rossmann’s class-counting polynomial of graphical Lie algebras: rigidity and reconstruction laws for the polynomial and its binary specialization, recovery of the subgraph-component polynomial and of domination invariants, a strict log-concavity ladder with proven failure boundaries, and extremal laws for trees, each tested exhaustively at small order, on adversarial random holdouts, and at every known collision class of the polynomial. One of the twenty, the pseudoforest one-field rigidity claim, has since been refuted by an explicit seven-vertex pair and is recorded as resolved false with its counterexample.

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1 Introduction

This collection states seventy-six conjectures in four parts. Part I (Conjectures 1–25) is a single sustained experiment: the primes, beyond their local structure, are modelled as a random set of density $1/\log n$, and the model is pushed to its sharpest consequences. Part II (Conjectures 26–50) leaves the single model for five structural programmes—connected prime-pattern fields, arithmetic first-arrival fields and class groups, finite logarithms and algebraic tori, arboreal dynamics, and adelic factorization—each contributing one canonical object and its laws. Part III (Conjectures 51–56) leaves number theory itself, carrying the same discipline into multiplicative number theory’s pretentious frontier, discrepancy theory, graph limits, additive information theory, and ergodic theory. Part IV (Conjectures 57–76) turns to a single object on the algebraic side, the class-counting polynomial of graphical Lie algebras, and states for it a rigidity and reconstruction programme, a log-concavity programme, and a tree extremal programme, with the proved layer, an exact rank formula and two lower bounds for trees, separated from the twenty conjectural laws.

One standard governs all four parts. A good conjecture is not merely a statement that has resisted counterexample: it is a determinate proposition about a canonical object, with a stated mechanism, and with enough exposed structure to be refuted in pieces. Concretely, every statement here is asked to (i) survive every congruence and size obstruction; (ii) come with a quantitative form whose constants are derived, not fitted; (iii) demand exactly the fluctuations its mechanism earns and no fewer (the Mertens conjecture died of demanding fewer); (iv) sit inside a known hierarchy where one exists (for Part I, Hardy–Littlewood k -tuples $\subset$ Schinzel’s Hypothesis $H \subset$ Bateman–Horn), so that failure would propagate; and (v) expose instance-falsifiable content—congruences, null lanes, exact identities, derived constants—so that computation can refute the derivation even where the asymptotic itself lies beyond finite refutation. Numerical range is the least important column in the ledger: most deep phenomena drift at the rate $\log \log x$, and $\log \log 10^{18} \approx 3.7$, so verification “to 10^{18} ” is weak evidence by itself. What persuades is the *shape* of the agreement—a count that tracks a derived constant through several decades with residuals that look like noise.

Part I is presented in three sections: the principal conjectures (numbers 1–9), family laws and second-order refinements (numbers 10–17), and instances and structural companions (numbers 18–25). The material they draw on—Bateman–Horn systems, barely-divergent sparse

sequences, representation problems with Borel–Cantelli accounting, and statistical laws of the prime sequence—runs through all three. Section 20 reports the stress tests and the independent recomputation of every constant.

One structural lesson organizes several of the statements below. A probability model integrates over density, and is therefore blind to algebraic families of density zero; the admissibility clause, checked at every prime for every system, is what repairs that blindness. The cubes inside the representation problem $n = p + k^3$ (Theorem 1) are the sharpest instance: they obstruct the problem outright while contributing nothing to any Borel–Cantelli sum taken over all n . This is the classical lesson of admissibility, and it is why the statements below separate what is provable by congruence or factorization from what a probabilistic accounting supplies. Parts II to IV inherit the same separation in a different key: each statement names its canonical object and averaging law, identifies its source, states a first decisive theorem where one exists, and records an honest failure mode.

Three checks stand behind every statement, and §20 reports their outcomes for Part I. First, an independent literature search at neighbourhood depth—defining objects, derived sequences, OEIS comments where relevant, and the abstracts and bodies of near-neighbour papers, read rather than skimmed from search summaries—establishes what is already known. Where a statement exists in essence it is attributed and labelled: some verbatim (Dubner’s twin-sum conjecture, the Stern list for $p + 2k^2$, Caldwell–Gallot’s $e^\gamma \log N$ laws), one with an exact constant catalogued as an OEIS entry (A188596), and the rest attributed accordingly. No such statement stands as a conjecture of its own; each is retained *inside* the conjecture whose claimed content it calibrates, restated in our uniform framework and re-verified at our bounds, and where a statement strengthens a named open problem the containment is stated as its headline. Second, every falsifiable-by-instance statement was verified against exact counts and then re-tested well past its original bound, the computation being designed as an attempt at refutation rather than at confirmation. Third, constants and counts were recomputed from the definitions alone by independent implementations sharing no code with the primary computation. Our novelty policy throughout is conservative: where priority is uncertain we attribute rather than claim.

Taxonomy: claimed content and its calibration layer

Within Part I, one criterion governs the taxonomy. An unrecorded specialization of Bateman–Horn is analogous to evaluating a classical special function at a new argument—worthwhile as data, but not a new conjecture in the conceptual sense; no conjecture’s claimed content is a bare specialization, and the taxonomy is two-layered *within* each conjecture. The *calibration layer*—the previously-known statements this paper begins from (Dubner’s conjectures, the Stern list, the Caldwell–Gallot laws, Martin’s refined Goldbach, Kourbatov’s record framework, the classical single instances of each family)—appears as attributed remarks inside the conjectures, each with its derived constant re-verified at our bounds; membership in the standard hierarchy of conjectures is itself a virtue, and the calibration layer is what makes the novel layer’s constants credible. The *claimed layer* is graded as follows: mechanism-level content (the contamination calculus 1(v) with its instances 18, 19, 14, 12, 1, 4; the ordering deficits 5, 3; race sub-diffusivity 9; the entanglement law 6); derived second-order and distributional laws (10, 15, 2, 8(ii), 11(iii), 13, 7(iv), 16(ii)); and structural classifications (23, 25, 24, 20(i), 21(i), 22)—with framework attributions stated where an instance lives inside someone else’s theory (Kowalski’s for 15(ii), Montgomery–Soundararajan’s for 2 and 13). Every statement was searched against the literature at neighbourhood depth and verified at the scales reported in §20. We regard the prime-power

contamination calculus (Conjecture 1(v)) and its mechanism family—1 and 4, the fresh cousin-race predictions it generated, and the negative control 14—together with Conjecture 3(ii) and the derived covariance kernels of 8(ii) and 16(ii) as the paper’s core, joined by 13 and 23; Conjectures 20, 6 and 11 carry open modelling questions flagged in place.

One tagging convention runs through every statement below: clauses marked [theorem] or “elementary” are proved; clauses marked “conditional proposition” follow from the hypotheses stated with them; the remaining clauses are conjectures; and open questions are labelled as such and are not counted among the seventy-six conjectures.

Computational-check standards

All counts below are exact (sieves; deterministic Miller–Rabin for $n < 3.3 \times 10^{24}$; Baillie–PSW beyond, flagged as such). Every comparison reports the Poisson-normalized residual $z = (\text{obs} - \text{pred})/\sqrt{\text{pred}}$. Every quoted z -score is such a residual and is used *descriptively*: where the counts compared are correlated, overlapping, or examined after selection, these residuals are not calibrated tail probabilities, and no significance claim in this paper rests on them. For sparse counts the cumulative comparison is contaminated at small x by the phantom mass of the main-term integral near its lower limit; in those cases we also report per-decade increments, which are clean. Constants are computed by truncated Euler products with the truncation wobble (difference between cutoffs) reported as an error bar; for conditionally convergent quadratic and cubic products this wobble, not the last digit, is the effective precision.

Primality certification is stratified, and every count inherits the stratum of the integers it involves. Primality of every integer below 3.3×10^{24} is decided *deterministically*, by fixed-base Miller–Rabin over a witness set verified for that range. For larger integers the classification is made by the Baillie–PSW test, and is a *probable-prime* classification, not a proof: no Baillie–PSW pseudoprime is known—and none exists below 2^{64} , where the test is therefore deterministic—but their nonexistence is unproved. Consequently every count involving such integers is a probable-prime count, and is labelled as one wherever it appears. The computations affected are the factorial scans of Conjectures 21 and 2 ($n! \pm 1$ for n beyond about 26), the $n^2 + 2^n$ scan of Conjecture 6 beyond $n \approx 80$, the quartic values of the cyclotomic chain of Conjecture 24 beyond $p \approx 2 \times 10^6$, and the Fibonacci–Lucas values of Conjecture 20 at all but the smallest indices. Every other count below—sieve censuses, race counts, window fields, and the inputs to every constant—lies entirely inside the deterministic range.

Data availability. The accompanying repository (`conjecture-engine`) contains the primary verification programs and the independent reimplementations used for recomputation, the machine-readable results files and exception lists, the Euler-product truncation points used for every constant, and the window-selection rules and random seeds of every sampled computation; `run_all.py` regenerates every number in this paper. Probable-prime classifications are recorded as such in the results files, separately from the deterministically certified ones, so that each count can be read at its own certification stratum.

2 Notation

p, q denote primes; γ is Euler’s constant, $e^\gamma = 1.781072\dots$; φ is Euler’s totient; $\phi = (1 + \sqrt{5})/2$; F_n the Fibonacci numbers; $p\#$ the primorial. For irreducible $f_1, \dots, f_k \in \mathbb{Z}[x]$ with positive

leading coefficients, the Bateman–Horn singular series is

$$C(f_1, \dots, f_k) = \prod_p \frac{1 - \omega(p)/p}{(1 - 1/p)^k}, \quad \omega(p) = \#\{n \bmod p : p \mid f_1(n) \cdots f_k(n)\},$$

and the corresponding main term is

$$I(N) = \int_2^N \frac{dt}{\prod_i \log f_i(t)},$$

which absorbs the usual $1/\prod_i \deg f_i$. The Bateman–Horn conjecture [2] asserts $\#\{n \leq N : \text{all } f_i(n) \text{ prime}\} \sim C \cdot I(N)$ whenever $C \neq 0$; each Bateman–Horn conjecture below is an instance with its constant evaluated. For nonlinear systems the product converges only conditionally; throughout, it is taken in the canonical ordering of increasing primes, in which convergence follows from the equidistribution of $\omega(p)$ about its Chebotarev mean. Our numerical truncation wobble is an error bar for this canonically ordered product. We write $\text{Li}_k(x) = \int_2^x (\log t)^{-k} dt$. The twin singular series is $\mathfrak{S}(d) = 2C_2 \prod_{p|d, p>2} \frac{p-1}{p-2}$ for even d , with $2C_2 = 1.3203236\dots$; the Goldbach series $\mathfrak{S}(n)$ is the same product over odd $p \mid n$.

Eight of the conjectures below are built over the Bateman–Horn move—choose an admissible system, compute its singular series, state the count—but none is a bare instance of it: 10 and 15 are family laws with derived constant distributions, 23 and 25 carry structural classifications, 24 is a searched chain, and 18, 19, 14 are contamination-calculus predictions (two fresh races and the negative control) whose mechanism is the contamination calculus of Conjecture 1(v). Each conjecture carries its classical calibration instance as an attributed remark. The heuristic is uniform, so we give it once. Model the primality of the values $f_i(n)$ as independent events of probability $1/\log f_i(n)$, corrected by the factor $(1 - \omega(p)/p)/(1 - 1/p)^k$ at each prime p to account for the joint local behaviour; multiplying the corrections and integrating gives $C \cdot I(N)$.

For locally integrable F we write $\mathcal{M}_x(F) = \frac{1}{\log x} \int_2^x F(t) \frac{dt}{t}$ for the logarithmic mean; all race conjectures below state their drift and null clauses through $\mathcal{M}_x$, so that “no persistent leader” and “systematic surplus” are claims about a defined functional rather than about pointwise behaviour. Drift clauses are stated at the *drift scale*—for a pair race, $\mathcal{M}_x(D(t) \log^2 t / \sqrt{t})$ converging to an explicit constant—never as $\mathcal{M}_x((D - T)/\sqrt{\pi}) \rightarrow 0$, which is vacuous for the drift coefficient (both $T/\sqrt{\pi}$ and its multiples have logarithmic mean $\rightarrow 0$). Convergence of $\mathcal{M}_x$ at drift scale is itself part of each such conjecture: a pure random-walk fluctuation model would make these means divergent in variance, so their convergence encodes the Rubinstein–Sarnak-type almost-periodicity that the zero-oscillation hypothesis names.

Admissibility ($\omega(p) < p$ for all p) is asserted programmatically during every constant computation. The natural-looking pair $\{n^2 + n + 1, n^2 + n + 3\}$ has $\omega(3) = 3$ and is rejected; the $k = 4$ branch of Conjecture 25 and the naive iterated chain behind Conjecture 24 were both rejected by the same machinery, and both rejections became part of the statements.

Novelty labels, from the independent literature search: (a) previously stated in essence; (b) classical family, quantified instance apparently unstated.

How to read the seventy-six. The count is a numbering convention, not a claim of seventy-six independent mechanisms. The contamination calculus is one operator, stated with the twin races and projected onto the Goldbach, Stern, triplet, and sexy-pair patterns, together with one provably null pattern as its control, and those six statements stand or fall largely together. Several conjectures pair a proved classification with a Bateman–Horn lane that is standard once

stated, and several keep attributed benchmarks inside them as calibration; the tagging convention above separates these layers, and the novelty labels mark what is claimed as new. Part II is five programmes, each contributing one central object and its laws. Part IV is one polynomial invariant carrying three programmes, and its twenty statements stand or fall in blocks rather than singly, with one, Conjecture 63, already fallen and recorded as resolved false. Depth and independence therefore vary by design across the seventy-six, and the summary sheet's ordering, not the raw count, carries the priority claim.

Part I

Conjectures from a local–global random model

3 Principal conjectures

The twin-race statement is split into its provable algebraic core and its conjectural clauses, stated through the logarithmic-mean functional $\mathcal{M}_x$ of the notation section.

Lemma 1 (Orientation elimination and class assignment). *In $\psi_2(x) = \sum_{n \leq x} \Lambda(n)\Lambda(n+2)$, the total contribution of terms in which n or $n+2$ is a proper prime power is, apart from $O(x^{1/3} \log^2 x)$ from cubes and higher powers, carried entirely by the patterns $(q^2 - 2, q^2)$ with q prime and $q^2 - 2$ prime: the mirror pattern $(q^2, q^2 + 2)$ is annihilated by $3 \mid q^2 + 2$ for every prime $q > 3$. Moreover, for every prime $q \neq 5$, $q^2 - 2 \equiv 2$ or $4 \pmod{5}$ (never 1), and for every odd q , $q^2 - 2 \equiv 7 \pmod{8}$. (Two bounded exceptions: at $q = 5$ the term $(23, 25)$ lands outside the twin-start classes altogether, and at $q = 3$ the mirror term $(9, 11)$ survives, the annihilation beginning only at $q > 3$; each is a single term, absorbed in the bounded error.)*

The proof is elementary (squares mod 3, 5, 8; the prime-power count). The lemma orients the race: prime-square contamination can enter only specific residue classes, and the model converts that into a drift prediction for the *unweighted* twin counts.

Conjecture 1 (Twin races modulo 5 and 8, and the prime-power contamination calculus; (b), apparently new). *Twin starts $p > 5$ lie in classes $p \equiv 1, 2, 4 \pmod{5}$. Let $D_1(x) = \pi_t(x; 5, 1) - \frac{1}{2}(\pi_t(x; 5, 2) + \pi_t(x; 5, 4))$ and*

$$T(x) = \frac{1}{2 \log^2 x} \sum_{\substack{q \leq \sqrt{x} \\ q^2 - 2 \text{ prime}}} \log q \log(q^2 - 2).$$

Then: (i) [drift law, at drift-scale normalization] the clause asserts two sub-claims, labelled separately because they can fail independently. (i-a) [mechanism] class 1 carries the systematic surplus T : one has the decomposition $D_1(t) = T(t) + R(t)$ in which the remainder R carries no deterministic component at the drift scale, so that the entire drift-scale deterministic content of D_1 is the contamination term and its coefficient is thereby identified. (i-b) [averaging] at the drift-scale normalization the remainder averages away:

$$\mathcal{M}_x \left(\frac{R(t) \log^2 t}{\sqrt{t}} \right) \rightarrow 0, \quad \text{equivalently} \quad \mathcal{M}_x \left(\frac{D_1(t) \log^2 t}{\sqrt{t}} \right) \rightarrow c_T,$$

where

$$c_T := \lim_{x \rightarrow \infty} \frac{T(x) \log^2 x}{\sqrt{x}} = \frac{1}{2} C(x, x^2 - 2)$$

is the half of the Bateman–Horn constant of the pair $(q, q^2 - 2)$ (the weighted census sum is $\sim C\sqrt{x}$, so the limit exists and is explicit). A normalization at the noise scale, $\mathcal{M}_x((D_1 - T)/\sqrt{\pi_t}) \rightarrow 0$, would be vacuous for the coefficient of (i-a): since $T/\sqrt{\pi_t} \asymp 1/\log t$, one has $\mathcal{M}_x(T/\sqrt{\pi_t}) \asymp \log \log x / \log x \rightarrow 0$, so the condition holds with T replaced by 0, $2T$, or any fixed multiple, and does not express “exactly the removed mass.” The statement therefore normalizes at the drift scale, where the claimed limit is a nonzero constant. Convergence at this normalization is a Rubinstein–Sarnak-type regularity assertion: under a pure random-walk model the logarithmic mean would not converge (its variance grows like $\log x$), so (i-b) asserts almost-periodic structure for the twin race and is the precise content of the zero-oscillation hypothesis, while (i-a) is what fixes the limit at c_T rather than 0 or $2c_T$, and is the calculus. The convergence asserted in (i-b) selects the rigid branch of the dichotomy registered at Conjecture 9(iii); the two statements stand or fall together at drift scale; (ii) [zero deterministic drift in the symmetric differences] the symmetric classes carry no deterministic term at the same normalization: $\mathcal{M}_x((\pi_t(\cdot; 5, 2) - \pi_t(\cdot; 5, 4)) \log^2 t / \sqrt{t}) \rightarrow 0$, and likewise for each of the three pairwise differences among classes $\{1, 3, 5\} \bmod 8$ in clause (iv)—now a statement distinct from (i), since at drift scale “zero” and “ c_T ” are different limits; (iii) [sign density, conditional] conditional on the Gaussian fluctuation model for the normalized remainder in (i)–(ii), the logarithmic density of $\{D_1 > 0\}$ exists and equals $\frac{1}{2}$, approached from above with a finite- x excess of order $\log \log x / \log x$: the pointwise drift-to-noise ratio is $T/\sqrt{\pi_t} \asymp 1/\log t$, and its logarithmic average $\mathcal{M}_x(1/\log t) \sim \log \log x / \log x$ carries an extra $\log \log$ (the scale is not $1/\log x$). In contrast to Chebyshev’s races, the twin race has no persistent leader; (iv) [mod-8 companion] since $q^2 - 2 \equiv 7 \pmod{8}$ always (Lemma 1), the entire deterministic term falls on the single class $7 \pmod{8}$: with $D_7(x) = \frac{1}{3} \sum_{a \in \{1, 3, 5\}} \pi_t(x; 8, a) - \pi_t(x; 8, 7)$,

$$\mathcal{M}_x\left(\frac{D_7(t) \log^2 t}{\sqrt{t}}\right) \rightarrow 2c_T$$

(twice the mod-5 constant, undiluted—at this normalization the factor 2 is a falsifiable prediction, the mechanism and averaging claims separating here exactly as in (i-a)–(i-b)), while classes 1, 3, 5 are mutually symmetric. The mod-5 and mod-8 races are two projections of one mechanism, so their joint behaviour is a family test in the sense of the shape-of-agreement criterion; (v) [contamination calculus for balanced races] for a pattern $(n, n + d)$ and modulus m , call the race balanced if the prime-prime singular series is identical across the competing residue classes (as CRT gives for the classes compared in (i)–(iv)) and no mechanism other than prime powers contributes at the $\sqrt{x}$ scale under the zero-oscillation hypothesis. For balanced races the deterministic drift vector is computed by one operator. Define the contamination operator

$$\mathcal{C}_{d,m}(a; x) = \sum_{\text{orientations } o \in \{(q^2 - d, q^2), (q^2, q^2 + d)\}} \sum_{\substack{q \leq \sqrt{x}, \text{ } o \text{ prime-compatible} \\ \text{start}(o) \equiv a \pmod{m}}} \Lambda(q^2) \Lambda(\text{prime member of } o),$$

where prime-compatible means the non-square member is prime and no fixed prime annihilates the orientation. The provable layer is the orientation census and class assignment (which o survive, and where their starts land—Lemma 1 and its analogues); the conjectural layer is the transfer: the unweighted class counts are depressed by $\mathcal{C}_{d,m}(a; x) / \log^2 x$ relative to the clean classes, in the drift-scale limiting-log-mean sense of (i), with the mechanism and averaging claims separated as in (i-a)–(i-b)—for each class a , the deficit obeys $\mathcal{M}_x(\text{deficit}_a(t) \log^2 t / \sqrt{t}) \rightarrow c(a) := \lim \mathcal{C}_{d,m}(a; x) / \sqrt{x}$,

an explicit Bateman–Horn constant per class, so every coefficient in the vector is identified—i.e. “exactly the removed mass,” which requires the weighted identity, partial summation uniform in classes, higher-power bounds (exponents $r \geq 3$ contribute $O(x^{1/3+\epsilon})$ against the operator’s $\asymp \sqrt{x}$ and are negligible—the one easy layer of the transfer), and the zero-oscillation hypothesis, and is the calculus’s single analytic conjecture rather than a rule. The drift is $\asymp \sqrt{x}/\log^2 x$ against a count fluctuation $\asymp \sqrt{x}/\log x$, i.e. smaller by $1/\log x$ in standard-deviation units at every height—so the calculus predicts occupation biases under long logarithmic aggregation, never a fixed nonzero standardized mean, and every verification in this family is designed around that fact. (Races with unequal primary constants, or with interfering orientations beyond those enumerated, are outside the calculus until those terms are separated.) In particular, for the cousin pattern $d = 4$ the calculus makes two predictions with no free choices: the orientation $(q^2 - 4, q^2)$ is dead apart from a single term ($q^2 - 4 = (q - 2)(q + 2)$ is composite for $q > 3$, while at $q = 3$ the value 5 is prime: one bounded term, whose start 5 lies outside the mod-5 cousin classes and contributes one bounded term to class $5 \bmod 8$), the orientation $(q^2, q^2 + 4)$ survives (no prime annihilates it), $q^2 + 4$ prime forces $q \equiv \pm 2 \pmod{5}$ for $q \neq 5$ (at $q = 5$ the value 29 is prime, but the start $25 \equiv 0 \pmod{5}$ lies outside the cousin start classes $\{2, 3, 4\}$), so the contaminated cousin-start class is $4 \pmod{5}$ (of $\{2, 3, 4\}$) and—since $q^2 \equiv 1 \pmod{8}$ —the class $1 \pmod{8}$ (of $\{1, 3, 5, 7\}$): both deficits converge, in the drift-scale limiting-log-mean sense of (i)–(ii), to $c^c = \lim_{x \rightarrow \infty} T^c(x) \log^2 x / \sqrt{x}$, the Bateman–Horn constant of the pair $(q, q^2 + 4)$, where $T^c(x) = \frac{1}{\log^2 x} \sum_{q \leq \sqrt{x}, q \neq 5, q^2 + 4 \text{ prime}} \log q \log(q^2 + 4)$.

Clause (v) answers the structural question of what general principle the mod-5 and mod-8 races instantiate, and the cousin predictions it yields were derived first and tested second. At $x = 10^9$ the predicted drift is far inside the noise ($T^c/\sqrt{\pi_c} \approx 0.09$, as the $1/\log x$ law requires), so the sharp statistic is leadership under logarithmic averaging, calibrated against the null of Conjecture 14(ii) (log-occupation $\frac{1}{2}$ with Gaussian spread $\sqrt{\log 2 / \log x} \approx 0.18$ at 10^9 , at the occupation-fraction constant): the class-4 deficit race led on the predicted side with log-density 0.99 (+2.7 null standard deviations), the class-1 mod-8 race with 0.92 (+2.3), both signs agreeing with the prediction, while the control differences among uncontaminated classes stayed within one noise unit of zero in log-mean. This is moderately strong directional evidence—two independent projections each between two and three null standard deviations on the predicted side—though the races share underlying primes and the two statistics are positively correlated, so we do not call it sharp confirmation; the family now has five projections of one mechanism (twin mod 5, twin mod 8, cousin mod 5, cousin mod 8, and the Goldbach lanes of Conjecture 4) plus two algebraic nulls (Conjecture 14 and the dead cousin orientation), which is the shape-of-agreement standard applied to the calculus itself.

A limiting sign density strictly between $\frac{1}{2}$ and 1 would be inconsistent with the scale analysis: when drift/noise $\rightarrow 0$, a Gaussian model forces the sign density to $\frac{1}{2}$, and a nontrivial limiting density would require a persistent normalized bias that no mechanism here produces. The fluctuation in (iv) is not written as $O^*(\sqrt{\pi_t})$, since O -notation asserts a provable bound where only a conjectural fluctuation statement is meant. Two model assumptions remain open and are stated as such: the passage from the Λ -weighted count to unweighted twin counts is a partial-summation argument whose error terms need writing out (higher prime powers are disposed of by Lemma 1); and the *zero-oscillation caveat*—the drift law treats the oscillatory explicit-formula contributions to class differences as having vanishing logarithmic mean, the exact analogue of the Rubinstein–Sarnak GRH-plus-linear-independence framework [6], and a conspiracy among hypothetical zeros could in principle mimic or cancel part of the drift; clause (i) is conjectured

on the standard side of that hypothesis.

The mechanism is the twin analogue of the classical explanation of Chebyshev's bias: prime-square terms contaminate the classes that can contain them. (Biases of twin primes in residue classes have been studied before—Sahoo [27] for the mod-4 race, and [28] for prime pairs versus isolated primes—but we could find no prior statement of the mod-5 race, the $(q^2 - 2, q^2)$ mechanism, or the quantified surplus below.) Mod 3 kills the pattern $(q^2, q^2 + 2)$ outright, an amusing local accident that halves the mechanism and (as a mod-3 computation shows, overturning the natural first guess of a quadratic-residue bias against class 4) concentrates the surplus entirely on class 1. Data to 4×10^9 : class counts (3980017, 3982505, 3981922), D_1 within one noise unit of T throughout, no persistent leader, and the control race $\pi_{t,2} - \pi_{t,4}$ wandering like a fair coin. Mod-8 data to 10^9 : class counts (856684, 855807, 856046, 855967) for $a = 1, 3, 5, 7$; $D_7 = +212$ at 10^9 against predicted $T = +254$ with noise ~ 1068 , positive through most of the range (log-density 0.775), and the three pairwise $\{1, 3, 5\}$ controls all within one noise unit of zero—the two projections of the mechanism behave coherently, none of it provable at present.

Conjecture 2 (The pair-level Montgomery–Soundararajan reduction; apparently new). *For the window field of Conjecture 8(ii), the diagonal variance of the twin-pair count reduces to a pinned singular-series average. With $R(h) = C_4(0, 2, h, h + 2)/\mathfrak{S}(2)^2$ for $h \geq 1$ (zero when the offset set is degenerate or inadmissible—all odd h , and $h = 2$ since $\{0, 2, 4\}$ dies at 3; consecutive twin pairs cannot overlap beyond $(3, 5, 7)$, so no overlap term exists) and*

$$G(H) = \sum_{1 \leq h \leq H} \left(1 - \frac{h}{H}\right) (R(h) - 1),$$

the derived identity (clause (i)) is

$$\frac{\text{Var}_t \pi_2(t; H)}{\mathbb{E}_t \pi_2(t; H)} = 1 + \frac{\mathfrak{S}(2) (2G(H) - 1)}{\log^2 x} + o\left(\frac{1}{\log^2 x}\right) :$$

with $p = \mathfrak{S}(2)/\log^2 x$ the per-site intensity, the variance is $Hp(1 - p) + 2p^2HG(H)$, so $\text{Var}/\mathbb{E} = 1 - p + 2pG(H)$ —the Bernoulli diagonal term $-p = -\mathfrak{S}(2)/\log^2 x$ is of exactly the order of the claimed error and cannot be dropped (it is the same diagonal that, in the single-prime case, merges into Montgomery–Soundararajan's constant $\gamma + \log 2\pi - 1$). With that term retained, the entire deficit left open at Conjecture 8(ii) is the single computable function G plus the explicit Bernoulli correction—the pair analogue of Montgomery–Soundararajan's reduction of the prime-count variance to $\sum_{h \leq H} \mathfrak{S}(h)(H - h)$ [19]. Clause (i) is a conditional proposition, not an independent conjecture: given quantitative Hardy–Littlewood for 4-tuples, the reduction follows. The conjectural clause is (ii): $-G(H)/\log H$ diverges—the pinned average grows strictly faster than the single-prime secondary term $\frac{1}{2}(\log H + \gamma + \log 2\pi - 1)$, so pair counts in windows are more sub-Poisson than prime counts—with the caveat that data to $H = 3000$ cannot distinguish divergence from a pure log with a large constant (nor $c \log H \log \log H$ from $(\log H)^\alpha$): what the computation establishes is only that the local slope exceeds the single-prime value sevenfold across the computable range; the divergence and its exact form are registered, not evidenced beyond that.

The pinning leaves the local densities in exact balance. At every odd prime p the relative local factor of $R(h)$ takes the value $p/(p - 2)$ for $h \equiv 0$, the value $p(p - 3)/(p - 2)^2$ for $h \equiv \pm 2$, and the value $p(p - 4)/(p - 2)^2$ otherwise, and their exact average over $h \bmod p$ equals 1, since $(p - 2) + 2(p - 3) + (p - 3)(p - 4) = (p - 2)^2$. Any persistent negativity of $G(H)$ is therefore

not a first-order local-density effect but a global correlation effect across primes, which is what makes its growth rate delicate.

G was computed exactly to $H = 3000$ (4-tuple constants by Euler products to 2×10^4): $G(3000) = -20.9$, with local logarithmic slope increasing from 3.4 to 3.7 across the range—seven times the single-prime slope $\frac{1}{2}$, consistent with (ii) over the computable range while leaving the limiting form undetermined (a quadratic-in-log fit is unstable on subranges). The window measurement brackets the extrapolation: at $H = 10^5$, $x \in [10^8, 2 \times 10^8]$ the pure-log form predicts $\text{Var}/\mathbb{E} = 0.97$, the $\log^2$ form 0.72, and the observed value is 0.815—intermediate growth, exactly where the computable range points. (Caldwell–Gallot’s factorial-prime law $\#\{n \leq N : n! + 1 \text{ prime}\} \sim e^\gamma \log N$ [4], the divergent single-sided companion of Conjecture 21, is included as an attributed calibration benchmark: 15 primes to $n = 700$ against 11.7, $z = +0.98$ —a probable-prime count, $n! + 1$ leaving the deterministic range at $n \approx 26$.)

Computational checks. The pinned singular-series computation and window comparison are reported at the statement; its factorial benchmark stands at 15 primes with $n \leq 700$ against 11.7 ($z = +0.98$; a probable-prime count for n beyond about 26), the divergent single-sided companion of Conjecture 21’s convergent side.

Conjecture 3 (Least primes in progressions; (a) for part (i) [20, 21, 30, 29], (b) for part (ii)). *Let $p(a, q)$ be the least prime $\equiv a \pmod{q}$ and $U(a, q) = \text{Li}(p(a, q))/\varphi(q)$. Then: (i) $U \Rightarrow \text{Exp}(1)$ marginally as $q \rightarrow \infty$; and, under the additional hypothesis that the class processes are asymptotically jointly independent in the extreme-value sense (which marginal convergence alone does not supply), $\max_a U - H_{\varphi(q)}$ converges to a Gumbel law ($H_n = \sum_{k \leq n} 1/k$), the order-statistics form of Wagstaff’s $\max_a p(a, q) \sim \varphi(q) \log^2 q$; (ii) [canonical deficit, dyadic-average form] define, with no reference to any control model, $\Theta(q) = (1 - \mathbb{E}_a[U(a, q)]) \log q$. The conjecture is stated for dyadic averages over prime moduli—pointwise convergence along every prime q may be too strong, since arithmetic fluctuations in q could persist:*

$$\frac{1}{\#\{q \in [Q, 2Q] \text{ prime}\}} \sum_{\substack{Q \leq q \leq 2Q \\ q \text{ prime}}} \Theta(q) \rightarrow \Theta > 0 \quad (Q \rightarrow \infty).$$

(The formal claim is model-free: existence and strict positivity of the dyadic-average limit. The stronger reading—that Θ exceeds the discreteness level any parity-aware control produces—is the diagnostic layer below, not part of the display, since a control constant has no place in a model-free statement.) The limiting distribution of $\Theta(q)$ over the dyadic block is left open, and in particular we do not assert $\text{sd}(\Theta(q)) \rightarrow 0$: fluctuations driven by the arithmetic of q and of $q - 1$ (through the singular-series pair terms $\mathfrak{S}(kq)$ and the class structure mod q) may well persist at bounded size, so $\Theta(q)$ need not concentrate around its dyadic mean.

The decomposition $\Theta(q) = \theta_{\text{disc}}(q) + \theta_{\text{corr}}(q)$ is a *diagnostic*, not a canonical invariant: its two parts move if the benchmark moves, and clause (ii) is therefore stated through the model-free $\Theta(q)$ alone. The named benchmark (the parity-aware Cramér–Bernoulli control: candidates $a, a+q, a+2q, \dots$ independently “prime” with hazard $h_i = (q/\varphi(q)) \cdot s_i / \log(a+iq)$, $s_i = 2$ for odd candidates and 0 for even, matching density and parity) defines $\theta_{\text{disc}}(q) = (1 - \mathbb{E}_a[U^{\text{model}}]) \log q$ exactly for each q , and $\theta_{\text{corr}} := \Theta - \theta_{\text{disc}}$ is the part of the measured deficit that the control cannot produce; it is this diagnostic that localizes the anomaly in the *ordering* of the primes (Remark 1). Measured on prime moduli $q \in [1500, 6000]$: $\Theta = 1.671 \pm 0.009$, of which the control accounts for 0.847.

One further deterministic baseline must be subtracted before any of the residue is called an anomaly: for prime q , the primes $p < q$ occupy *distinct* classes mod q —an injective initial phase that no exchangeable allocation reproduces. Its size has a closed form in index time (where $\text{Li}(p_i) \approx i$, so U is the first-hit index over φ): forcing the first $\pi(q)$ arrivals to be collision-free and leaving the rest exchangeable gives $\mathbb{E}[U] = 1 - \pi(q)^2/2\varphi^2 + O(\pi(q)/\varphi^2)$, hence

$$\Theta_{\text{inj}}(q) = \frac{\pi(q)^2}{2\varphi(q)^2} \log q (1 + o(1)) = \frac{1 + o(1)}{2 \log q},$$

closed form 0.092, 0.082, 0.074 at $q = 1499, 3001, 5987$, with Monte Carlo agreeing within sampling error and the exchangeable control at 1.000. The injective phase is therefore real but an order of magnitude too small to be the explanation: it accounts for ≈ 0.08 of the measured $\theta_{\text{corr}} = 0.824 \pm 0.009$, and carries a $1/\log q$ decay that the flat measured values do not show. (One might expect the effect to sit at the full deficit scale, since the *affected-class fraction* is $\pi(q)/\varphi \asymp 1/\log q$; but the deficit is the affected fraction times the collision probability an exchangeable model would have suffered in that phase, $\asymp \pi(q)/2\varphi$, giving $1/2 \log^2 q$ in $\mathbb{E}[U]$, not $1/\log q$.) The injective phase is in any case the $y < q$ head of the pair-correlation expansion already registered below—collisions require $q \mid p_2 - p_1$, impossible before $y = q$ —so subtracting it is the first term of the stated programme, not a rival explanation. The residue $\theta_{\text{corr}} - \Theta_{\text{inj}} \approx 0.74$ remains the unexplained ordering anomaly.

A derivation route for the deficit, which we adopt as the conjecture’s stated programme, runs as follows. Let $V_q(y)$ be the number of reduced classes mod q not yet visited by a prime $\leq y$; then $\mathbb{E}_a[U] = \int_0^\infty \overline{V_q(y_\tau)}/\varphi(q) d\tau$ with the Li-time change y_τ defined by $\text{Li}(y_\tau) = \tau\varphi(q)$, and inclusion–exclusion expands $V_q(y)$ in prime-tuple correlations *within* a class:

$$\mathbb{E} \binom{N_a}{2} \text{ summed over } a = \#\{p_1 < p_2 \leq y : q \mid p_2 - p_1\} \approx \sum_{1 \leq k \leq y/q} \mathfrak{S}(kq) \int_2^{y-kq} \frac{dt}{\log t \log(t + kq)},$$

so the first correction to the exchangeable occupancy model is carried by the Hardy–Littlewood singular series of gaps divisible by q . The truncation is structural, not a convenience: a gap kq requires $p_1 \leq y - kq$, so the k -sum stops at y/q and the endpoint-shortened integrals supply the triangular weights, while the untruncated sum $\sum_{k \geq 1} \mathfrak{S}(kq)$ diverges (the singular series has mean one on average over its argument) and the formally factorized expression $\text{Li}_2(y) \sum_k \mathfrak{S}(kq)$ is meaningless. Evaluating this expansion to second order, and comparing it with the measured Θ , is the concrete open computation that would either derive the ordering anomaly from pair correlations or prove it lies deeper; either outcome retires Question 1’s mystery one level.

The stratified experiment behind these numbers separates moduli by factorization type in the matched range $q \in [1500, 6000]$: over 40 *prime* moduli, $\theta_{\text{disc}} = 0.847$ (essentially constant across q) and $\theta_{\text{corr}} = 0.824 \pm 0.009$; over 40 *7-smooth* moduli in the same range, $\theta_{\text{disc}} = 4.32 \pm 0.22$ (the large $q/\varphi(q)$ hazards make discreteness dominant) while $\theta_{\text{corr}} = -2.32 \pm 0.22$ reverses sign. The comparison with Leung [29] is thereby given its answer-shaped experiment: Leung derives the exponential law under a uniform Hardy–Littlewood hypothesis and reports discrepancies for *smooth* moduli, and our smooth stratum indeed behaves anomalously—but the positive ordering term survives, with small error bars, exactly where smooth-modulus effects cannot reach. Whether θ_{corr} tends to a constant, and the exact functional form in $1/\log q$, remain open: data at $q \leq 6000$ cannot distinguish $1/\log q$ from $(\log \log q)/\log q$ -type refinements—the aggregate $\Theta \approx 1.5\text{--}1.9$ over $q \in [10^2, 6 \times 10^3]$, with $\mathbb{E}_a[U]$ bottoming near 0.737 around $q \approx 200$

and recovering through 0.776 by $q \approx 6000$, is calibration, not a determination of the form. The averaging family is fixed as: all moduli of the stated type in dyadic ranges of q , per stratum.

Remark 1. *The deficit is reproducible and is not an artefact of the time-change: a Cramér pseudo-prime control (independent Bernoulli “primes” with the correct densities and parity) shows roughly half of it—the discreteness of large per-candidate hazards—while the other half vanishes under any reshuffling of the actual prime residues (iid labels, or a random permutation of the true residue sequence, both restore $\mathbb{E}[U] = 1.00$). The ordered sequence of primes fills residue classes measurably faster than any exchangeable model, and the stratified experiment above shows the effect is not a smooth-modulus artefact: it is largest and cleanest on prime moduli. We measure θ_{corr} ; we cannot yet derive it.*

Conjecture 4 (The Goldbach lane race; apparently new). *For $n \equiv 2 \pmod{4}$ let $R_1(n)$, $R_3(n)$ be the ordered Goldbach representation counts in the lanes $p \equiv q \equiv 1$ and $p \equiv q \equiv 3 \pmod{4}$ (the lane convention of the Martin benchmark recorded at Conjecture 5), and $D(n) = R_3(n) - R_1(n)$. Prime-square terms $n = q^2 + p$ (q odd prime) land exclusively in the (1, 1) lane, since $q^2 \equiv 1 \pmod{4}$ and then $n - q^2 \equiv 1 \pmod{4}$ automatically. Consequently the (3, 3) lane leads on average. Define*

$$D_{\text{sys}}(n) = \frac{2}{\bar{\ell}(n)} \sum_{\substack{q \text{ odd prime, } q \leq \sqrt{n} \\ n - q^2 \text{ prime}}} \log q \log(n - q^2), \quad \bar{\ell}(n) = \frac{n - 6}{\int_3^{n-3} \frac{dt}{\log t \log(n - t)}},$$

(the analytic mean of $\log p \log(n - p)$ under the lane profile; the empirical lane mean is its estimator). Write $\mathbb{E}_x$ for the average over n drawn from $\{n \equiv 2 \pmod{4}, n \leq x\}$ with weight proportional to $1/n$ (the logarithmic sampling measure, now explicit). Then: (i) [drift law] $\mathbb{E}_x[D - D_{\text{sys}}] = o(\mathbb{E}_x[D_{\text{sys}}])$ as $x \rightarrow \infty$: the (3, 3) lane leads on average by exactly the square contamination; (ii) [weighted mean and internal null lane] $\mathbb{E}_x[D_{\text{sys}}]$ is given by the Hardy–Littlewood (Conjecture H) local model for $n - q^2$, whose local factors vary strongly with n —most drastically at $p = 3$: for $n \equiv 1 \pmod{3}$, $3 \mid n - q^2$ for every prime $q > 3$, so the contamination collapses to a single boundary family, the $q = 3$ term (when $n - 9$ is prime): the complementary possibility $n - q^2 = 3$ is empty on this ensemble, since $n \equiv 2 \pmod{4}$ forces $n - 3 \equiv 3 \pmod{4}$, never a square. The family is negligible in logarithmic mean, so the race carries no drift at the systematic scale on the subprogression $n \equiv 10 \pmod{12}$: with the positive comparison scale $A_G(x) := \mathbb{E}_x^{(\neq 1 \pmod{3})}[D_{\text{sys}}]$, the logarithmic-ensemble mean of the systematic term over the contaminated lane $n \not\equiv 1 \pmod{3}$ (a comparison against this subprogression’s own systematic term, which vanishes, would be empty), the claim is $\mathbb{E}_x[D] = o(A_G(x))$ on $n \equiv 10 \pmod{12}$, while the drift concentrates on $n \not\equiv 1 \pmod{3}$: an internal null lane, predicted by the same mechanism that predicts the lead; (iii) [sign density] the per- n drift-to-noise ratio $\kappa(n) = D_{\text{sys}}(n)/\sqrt{\mathfrak{S}(n)J(n)}$ (where $J(n) = \int_3^{n-3} dt/(\log t \log(n - t))$, so that $\mathfrak{S}(n)J(n)$ is the total ordered representation count) satisfies $\kappa(n) \asymp c(n)/\log n \rightarrow 0$; under the Gaussian fluctuation model the logarithmic density of $\{D > 0\}$ is therefore $\frac{1}{2}$ in the limit, with the finite- x sign fraction predicted to be $\mathbb{E}_x[\Phi(\kappa)]$ (Φ the standard normal distribution function)—decaying, but far from $\frac{1}{2}$ at any accessible height.

Derivation status: the identity behind D_{sys} is the Λ -weighted lane symmetry minus its prime-power part; the transfer to unweighted counts is a partial-summation argument whose error terms (higher prime powers, which enter only at the $n^{1/3}$ scale, and the variation of the weight across

the lane) remain to be written out—the same open transfer flagged at Conjecture 1—and clause (iii) inherits the zero-oscillation caveat stated there. One further disclosure: the census D_{sys} is computed from the primes in the same sample it is compared against, so the level agreement is in-sample; the out-of-sample content of the verification is the null lane and the class assignment, which the mechanism fixes in advance.

This is the Goldbach-partition analogue of Chebyshev’s bias, produced by the same explicit-formula mechanism as Conjecture 1 but through $q^2 + p$ patterns rather than twin patterns; the single-sign prediction (every square contamination falls into one lane) makes it cleaner than the classical race. We could find no prior statement. The neighbouring phenomenon for consecutive primes is the Lemke Oliver–Soundararajan mod-3 law, whose verification here— $\hat{c}$ drifting $0.400 \rightarrow 0.370$ over $10^7 \rightarrow 4 \times 10^9$ with $(1, 1)/(2, 2)$ symmetry 0.99989—serves as calibration for the present race.

Computational checks. Over 500 log-spaced samples $n \leq 10^8$: mean $D = 76.5$ against mean empirical $D_{\text{sys}} = 64.5$, the $(3, 3)$ -lane lead significant at 5.2 standard errors of the mean. Clause (ii)’s weighted model, computed in presieve-exact form (trial division of each $n - q^2$ to 10^3 with the exact Mertens normalization), reproduces the empirical contamination with no fitted parameter: model mean 66.3 against empirical 67.8 (ratio 0.98) on a 400-sample run. The internal null lane behaves as predicted: on $n \equiv 1 \pmod{3}$ (123 samples) the model and empirical D_{sys} both vanish and the measured lead drops to 31 ± 23 (consistent with zero), while on $n \not\equiv 1 \pmod{3}$ (277 samples) the lead is 88 ± 21 against predicted contamination 96—the drift lives exactly where the mechanism puts it. Clause (iii): computed $\mathbb{E}_x[\Phi(\kappa)] = 0.572$ (κ ranging 0–0.64 with mean 0.18) against observed sign fraction 0.588; weak, and quantitatively the weakness the model requires.

Conjecture 5 (The least Goldbach summand: exponential law and ordering deficit; the time-changed law and deficit constant apparently unstated). *Let $s(n)$ be the least prime $p \nmid n$ with $n - p$ prime (the exclusion $p \nmid n$ removes a trivial obstruction: $p \mid n$ forces $p \mid n - p$, so $n - p$ is composite except in the diagonal case $n = 2p$, which would otherwise distort the hazard at one atypical point), and time-change by the expected-arrivals clock $U(n) = \mathfrak{S}(n) \sum_{p \leq s(n), p \nmid n} 1/\log(n - p)$. The ensemble is fixed first, since $U(n)$ is deterministic at each n and an expectation therefore needs a declared measure: for a scale X ,*

$$\mathbb{E}_X U = \frac{\sum_{X < n \leq 2X, 2|n} U(n)/n}{\sum_{X < n \leq 2X, 2|n} 1/n}, \quad \Theta_G(X) = (1 - \mathbb{E}_X U) \log X.$$

Then: (i) [limit law] under $\mathbb{E}_X$ -sampling, $U \Rightarrow \text{Exp}(1)$ as $X \rightarrow \infty$; (ii) [canonical ordering deficit] $\Theta_G(X) \rightarrow \Theta_G > 0$ —the Goldbach sibling of the least-prime deficit of Conjecture 3(ii), produced by the same occupancy mechanism: the ordered primes fill the available representations faster than an exchangeable model, so the first arrival comes early and $\mathbb{E}_X[U] < 1$.

The extremal theory of $s(n)$ is well developed—Granville, van de Lune and te Riele conjectured $s(n) = O(\log^2 n \log \log n)$, and Oliveira e Silva–Herzog–Pardi [12] tabulate first occurrences of minimal Goldbach summands to 4×10^{18} against prime k -tuple predictions—but the time-changed distributional law (i) and the deficit constant (ii) appear unstated (we could not rule out that the untransformed version of (i) is implicit in the comparisons of [12]). Measured on 2,000 log-sampled $n \leq 10^8$: $\mathbb{E}[U] = 0.796$, with $\Theta_G = 3.12 \pm 0.35$ on $[10^6, 10^7)$ and 3.42 ± 0.45 on $[10^7, 10^8)$ —positive, stable, and roughly *twice* the least-prime-in-progressions deficit $\Theta = 1.67$ of Conjecture 3, a comparison the occupancy expansion there should eventually explain; the Kolmogorov–Smirnov distance to $\text{Exp}(1)$ (0.088 at this size) is dominated by the deficit displacement itself, as (ii)

requires. The derivation route for Θ_G , parallel to the occupancy expansion at Conjecture 3: expand the survival probability $\Pr(s(n) > y) = 1 - \sum_{p \leq y} h_p + \sum_{p_1 < p_2 \leq y} h_{p_1, p_2} - \dots$, where the pair term requires $n - p_1$ and $n - p_2$ simultaneously prime and so carries the Hardy–Littlewood singular series of the shift $p_2 - p_1$; inserting the k -tuple predictions and integrating against the time change isolates the $1/\log X$ coefficient—deriving Θ_G rather than measuring it is this conjecture’s stated programme. $\mathfrak{S}(n)$ alone does not encode the between-candidate dependence that the pair term carries. (The (3, 3)-lane refined Goldbach statement—every $n \equiv 2 \pmod{4}$, $n \geq 6$, is $p + q$ with $p \equiv q \equiv 3 \pmod{4}$, with the halved singular-series count law, formulated by K. Martin [13]—is included as an attributed calibration benchmark: exhaustively verified to 10^9 , count formula matching to 0.2–1.3%; its comparative lane statement is Conjecture 4.)

For sequences whose n th term has size e^{cn} the expected number of primes is $\sum_n \kappa_n/(cn)$: convergence or divergence of this sum is the entire question, and when it diverges it diverges logarithmically, so the right conjecture has counting function $\alpha \log x$ with a derived α .

Conjecture 6 ($n^2 + 2^n$; sequence is OEIS A064539). *Every prime of the form $n^2 + 2^n$ with $n > 1$ has $n \equiv 3 \pmod{6}$ (elementary), and: (i) infinitely many such primes exist, with counting function $\asymp \log N$; (ii) [candidate law, entanglement-aware form] for a finite set S of primes $p \geq 5$ (the primes 2 and 3 are exact in the lane reduction, $\delta_2 = \delta_3 = 0$, and $\text{ord}_2 2$ is undefined) let D_S be the exact joint density, within the lane $n \equiv 3 \pmod{6}$, of n with $p \nmid n^2 + 2^n$ for all $p \in S$, computed over the full joint period $\text{lcm}_{p \in S} \text{lcm}(6, p \cdot \text{ord}_p 2)$, and set $\kappa_S = 3 D_S / \prod_{p \in S} (1 - 1/p)$: the factor $3 = 2 \cdot \frac{3}{2}$ is the exact lane bonus at the primes 2 and 3 (within the lane $\delta_2 = \delta_3 = 0$, so each contributes its full $(1 - 1/p)^{-1}$), included in the definition so that κ_S is the complete local constant of the count in (ii-c). The conjecture is stated along the canonical exhaustion $S_z = \{p \text{ prime} : 5 \leq p \leq z\}$, and its three layers are logically separate and are asserted separately: (ii-a) the sequence κ_{S_z} converges as $z \rightarrow \infty$ (a general net over arbitrary finite S carries an order-of-exhaustion ambiguity we do not need), with limit κ ; (ii-b) whether the limit factors—i.e. whether the exact finite-level factorization observed below persists, so that entanglement is asymptotically absent—is an open question, expressly not counted among this paper’s conjectural assertions, on which (ii-a) takes no position; and (ii-c) the count is $\sim \sum_{n \leq N, n \equiv 3 \pmod{6}} \kappa / \log(n^2 + 2^n)$ —a genuine further step, since a convergent local density does not automatically govern the global count. The working value $\kappa^* = 4.2734\dots$ is a hybrid, not a computation of any single κ_{S_z} : the lane factor 3 times the CRT-exact core S_{19} times independent per-prime factors for $19 < p \leq 300$ —so $\kappa^* = \kappa_{S_{300}}$ exactly if and only if the observed exact factorization persists to 300.*

This is not a Bateman–Horn problem: the local conditions at different primes are tied through the orders $\text{ord}_p 2$, so the per-prime product $\prod_p (1 - \delta_p)/(1 - 1/p)$ would tacitly assume their independence. The law is therefore stated through the CRT-exact quantities κ_S , which make no independence assumption. The computation then returned a small surprise in the other direction: through $p \leq 19$ (joint period 116,396,280) the joint survivor fraction equals the product of the per-prime fractions as an *exact rational identity*, verified in exact arithmetic at every level $S = \{5 \leq p \leq P\}$, $P \in \{5, 7, 11, 13, 17, 19\}$. Entanglement is thus absent at all computed levels; whether exact factorization persists for all p —equivalently, whether the survival events are exactly independent over every joint period, which would itself be a striking equidistribution statement about the orbits of 2—is part of what clause (ii) asks. We separate the robust claim (i) from the candidate law (ii), and note that the eight hits below 6000 are far too few to validate a four-digit constant—the verification below checks consistency, not the constant. The OEIS records further candidate indices beyond our completed range: 29355, 34653, 57285, 99069, and the probable-prime candidate 1933695. We could not independently re-verify these, but all five

satisfy the lane constraint $n \equiv 3 \pmod{6}$, and taking them at face value the κ^* -model predicts 2.9 hits in $(6 \times 10^3, 10^5]$ against 4 reported and 5.9 in $(6 \times 10^3, 1.94 \times 10^6]$ against 5 ($z = +0.65$ and -0.38): the records extend the consistency check by two decades.

The congruence claim is elementary: n odd is forced by parity, and for odd n with $3 \nmid n$ we have $n^2 + 2^n \equiv 1 + 2 \equiv 0 \pmod{3}$. What appears to be a harsh obstruction is, inside the surviving lane, a local bonus: $\delta_2 = \delta_3 = 0$ contribute the factors $2 \cdot \frac{3}{2} = 3$ carried explicitly in the definition of κ_S .

Computational checks (probable-prime count beyond $n \approx 80$). Hits $\{3, 9, 15, 21, 33, 2007, 2127, 3759\}$ up to 6000: observed 8, model 8.55, $z = -0.19$ (over the shorter range $n \leq 4200$: 8 vs 8.19, $z = -0.07$); the lane constraint was verified exhaustively (no prime off $n \equiv 3 \pmod{6}$ to $n = 3000$, independently reconfirmed), and the CRT-exact factorization of clause (ii) was verified through $p \leq 19$ as described at the statement.

Conjecture 7 (Fermat quotients: the multibase subtorus law and the Wieferich ledger; single-base heuristic is (a) [22], the vertical multibase law apparently new). *With $q_p(a) = (a^{p-1} - 1)/p \pmod{p}$ (defined for primes $p \nmid a$; all clauses below range over such p only) and $q_p = q_p(2)$, clauses (0)–(v), stated multibase group first: (0) [structure, classical] $q_p(ab) \equiv q_p(a) + q_p(b) \pmod{p}$ (the Eisenstein–Lerch homomorphism; verified exactly on every tested prime), so for multiplicatively dependent bases the vector $(q_p(a_1), \dots, q_p(a_r))/p$ is confined to the rational subtorus cut out by the relations; (iv) [multibase equidistribution, the claimed law] for multiplicatively independent bases the vector equidistributes on the full torus as p varies—the vertical joint law, complementing the fixed- p , varying-base statistics of Ostafe–Shparlinski and Cobeli–Zaharescu—with correlations vanishing and the same LIL calibration as (i) below; (v) [simultaneous Wieferich] the expected count of $p \leq x$ with $q_p(2) = q_p(3) = 0$ has convergent sum $\sum 1/p^2$: at most finitely many simultaneous Wieferich primes exist (the single-base folklore of this accounting is Conrad’s), and the empirical list is empty to 10^7 . Three further clauses on the base-2 quotient alone: (i) [global equidistribution] q_p/p equidistributes on $[0, 1]$ with discrepancy at exactly the calibrated random-model scale: in its sharp form,*

$$\limsup_{x \rightarrow \infty} \frac{\sqrt{\pi(x)} D_{KS}(x)}{\sqrt{\log \log \pi(x)}} = \frac{1}{\sqrt{2}},$$

the Chung–Smirnov law-of-the-iterated-logarithm constant for an i.i.d. uniform sequence—demanding both no more (a bound $O(\pi(x)^{-1/2})$ would be stronger than the random model supports, the same over-demand that killed the Mertens conjecture) and no less than the model earns: the upper bound alone would be consistent with hidden rigidity, and the matching lower bound asserts the quotients fluctuate like genuine noise; (ii) [shrinking targets] for every fixed $K \geq 1$, and uniformly for $K = K(x) \leq \log x$, $\#\{p \leq x : q_p < K\} \sim \sum_{p \leq x} \min(1, K/p)$ —the fixed- K case down to $K = 1$ is what governs the zero event and does NOT follow from (i), since $\{q_p = 0\}$ is a target of shrinking measure $1/p$; (iii) [Wieferich count, fluctuation-calibrated, with the probability space declared] $W(x) = \#\{\text{Wieferich } p \leq x\}$ is a deterministic function, so its fluctuation clauses are stated in the two legitimate forms, an unqualified CLT for a fixed sequence having no meaning: the deterministic envelope conjecture

$$\limsup_{x \rightarrow \infty} \frac{W(x) - \log \log x}{\sqrt{2 \log \log x \log \log \log x}} = 1$$

(the iterated-logarithm scale at variance $V(x) \sim \log \log x$; a bound $O_\varepsilon(V^{1/2+\varepsilon})$ would be weaker than the model’s exact prediction), and the almost-sure central limit law, at the correct logarithmic

weighting: with $u = \log \log x$ the variance-time of the count, sample u with density du/u on $[e, L]$, $L = \log \log X$ (equivalently: $\log u = \log \log \log x$ uniform); then the sampled distribution of $(W(x) - u)/\sqrt{u}$ converges to $N(0, 1)$ as $X \rightarrow \infty$. The weighting is what makes the clause true, and the two natural alternatives both fail. A block-sampled version on $[X, X^A]$ fails because only $\log A = O(1)$ new events enter the block. A uniform-in- u version on $[1, L]$ fails too, by Brownian scaling: with $W(e^u) - u \approx B(u)$ and $u = Ls$, the self-similarity $B(Ls)/\sqrt{Ls} \stackrel{d}{=} \tilde{B}(s)/\sqrt{s}$ means the uniform- u empirical distribution converges not to Φ but to the random occupation functional $\int_0^1 \mathbf{1}\{\tilde{B}(s)/\sqrt{s} \leq z\} ds$ —values at proportional times stay correlated, so uniform sampling never averages over enough independent scales. The du/u weighting is the classical almost-sure-CLT weighting (the continuous analogue of the $1/k$ weights), for which the model does predict almost-sure convergence of the sampled law to Φ . Both clauses are conjectured on the Crandall–Dilcher–Pomerance side of the model choice that clause (iv)’s discussion records.

Two things do not follow. Clause (iii) is not a consequence of (i): a square-root-size error in the aggregate distribution is astronomically larger than $\log \log x$, and only the shrinking-target clause (ii) at bounded K speaks to the zero event. And an $O(1)$ fluctuation claim would violate this paper’s calibration principle of demanding exactly the fluctuations the model earns: with event probabilities $1/p$ the count has standard deviation $\sim \sqrt{\log \log x}$. Nor is any location predicted for a third Wieferich prime: the expected count reaching 3 “at doubly exponential heights” is an expected-count statement whose additive constant is uncalibrated, not a forecast. Consistency data: $\{1093, 3511\}$ below 10^8 against expected ≈ 2.7 ; $\sqrt{n}$ KS = 0.82, 0.82, 0.52, 0.63 at $x = 10^5, \dots, 10^8$ (bounded, no drift); census at $K = 100$: observed 169 against model 161.2.

The multibase clauses were verified at 10^7 (664,577 primes): the homomorphism (0) held *exactly* on every spot-checked prime; for the independent pair $(2, 3)$, correlation $+0.0023 \pm 0.0012$, a 20×20 joint occupancy χ^2 of 441 on 399 degrees of freedom ($+1.5\sigma$), joint small-quotient census 46 against model 41.2 ($z = +0.75$), and no simultaneous Wieferich prime. Two positions are taken here: the vertical multibase law (iv) sides with the Crandall–Dilcher–Pomerance accounting against Gras’s competing probabilistic model (which predicts $q_p(a) = 0$ rarer than $1/p$, hence possibly finitely many Wieferich primes even for a single base)—the disagreement is what gives clauses (iii)–(iv) content; and the horizontal literature (fixed p , base varying: Ostafe–Shparlinski’s joint distributions and the Fermat-quotient-matrix statistics of Cobeli–Zaharescu and coauthors) is the attributed neighbour whose vertical counterpart is what this conjecture claims.

Conjecture 8 (Uniform quantitative de Polignac; (a)-core). (i) $\pi_d(x) = \#\{p \leq x - d : p, p + d \text{ prime}\} = \mathfrak{S}(d) \text{Li}_2(x) (1 + o(1))$ uniformly for even $d \leq (\log x)^A$. (ii) [residual field, with derived covariance kernel] The same primes participate in many pair counts, and the covariance of two of them is carried by triple configurations sharing one prime. For even $d \neq d'$ define the overlap constant

$$K(d, d') = C_3(0, d, d') + C_3(0, d', d + d') + C_3(0, d, d + d') + C_3(0, |d - d'|, \max(d, d')),$$

where C_3 denotes the Hardy–Littlewood triple constant of the offset set (degenerate or inadmissible triples contributing 0). The field is randomized canonically—covariances of the deterministic cumulative counts are otherwise undefined: for t uniform on $[x, 2x]$ and a window length H , let $\pi_d(t; H) = \#\{t < n \leq t + H : n, n + d \text{ prime}\}$ and let $Z_d(t; H)$ be its studentization. The mean count per window is $\asymp H/\log^2 x$, and the regime is set by that ratio (a phase condition: a bare $H \rightarrow \infty$ does not fix the regime):

$$\text{Cov}_t(\pi_d, \pi_{d'}) \sim \frac{K(d, d') H}{\log^3 x}, \quad \rho(d, d') \sim \frac{K(d, d')}{\sqrt{\mathfrak{S}(d)\mathfrak{S}(d')} \log x},$$

valid for $\max(d, d') = o(H)$, which is the standing restriction of this clause and holds throughout the window class fixed below. In general each overlap orientation o —the in-window displacements $h_o \in \{0, d, -d', d - d'\}$ and their reflections—enters with multiplicity $(H - |h_o|)_+$ rather than H , so that the shared-prime term reads

$$\frac{1}{\log^3 x} \sum_o (H - |h_o|)_+ K_3^{(o)}(d, d'), \quad \sum_o K_3^{(o)}(d, d') = K(d, d'),$$

and an orientation with $|h_o| > H$ is absent from the window altogether; under $\max(d, d') = o(H)$ one has $(H - |h_o|)_+ = H(1 + o(1))$ for every orientation and the displayed form is recovered, so the uniform-in- d statement carries that restriction throughout. Here the shared-prime (triple) term is the first covariance contribution but not the whole of it: the full expansion is

$$\text{Cov}_t(\pi_d, \pi_{d'}) = \frac{K(d, d') H}{\log^3 x} + \frac{1}{\log^4 x} \sum_{|h| \leq H} (H - |h|) K_4(d, d'; h) + \text{error},$$

with K_4 the centred singular series of the off-index four-point sets $\{0, d, h, h + d'\}$ —the cross analogue of the pinned diagonal average of Conjecture 2—where degenerate offset relations are handled inside the kernels: whenever the four-point set collapses (repeated entries, as at $h = 0$, $h = d$, $h = -d'$, or $d' - d = \pm h$) the configuration is a triple or pair and its contribution belongs to the K -term (or to the mean), so K_4 is defined as the centred singular series of genuinely four-element sets, with special pairs such as $d' = 2d$ acquiring their extra overlap orientations in $K(d, d')$'s census rather than double-counted across kernels. The triple term dominates only where the pinned sum satisfies $\sum(1 - |h|/H)K_4 = o(\log x)$; the dominance range is stated as an explicit function class: for $H = \lambda(\log x)^\alpha$, any fixed $\lambda > 0$ and any $\alpha \geq 1$ with $\alpha > A$ —the inequality on the exponents that makes $\max(d, d') = o(H)$ hold uniformly for all $d, d' \leq (\log x)^A$ —the per-shift centred average grows only polylogarithmically in H (the Conjecture-2 scale), so $\sum(1 - |h|/H)K_4/H = O((\log \log x)^{2+\varepsilon}) = o(\log x)$ and the triple term dominates throughout the polylogarithmic window class; the conjecture asserts the two-term form on exactly that class and takes no position outside it. When $H/\log^2 x \rightarrow \lambda$ the window counts converge jointly to Poisson laws with these vanishing covariances, and when $H/\log^2 x \rightarrow \infty$ every fixed finite collection $\{Z_{d_1}, \dots, Z_{d_r}\}$ is asymptotically jointly normal with independent limiting coordinates: both kernels are finite- x corrections to independence (test below), not nondegenerate limiting kernels. Both distributional conclusions are asserted under an explicit hypothesis beyond the displayed covariances, namely uniform Hardy–Littlewood tuple estimates on the window class fixed above, strong enough to control all higher joint cumulants of the window counts: every joint cumulant of order at least three vanishes in the respective normalizations. The diagonal correction to $\text{Var}_t(\pi_d)$ is Conjecture 2's pinned average, stated there.

The kernel is verified twice over. Direct window-field measurement ($x = 10^8$, $H = 10^5$, 2000 windows, all even $d \leq 40$, 190 pairs): the empirical correlation matrix matches the predicted kernel entrywise with correlation 0.86, mean off-diagonal 0.33 observed against 0.37 predicted, the small deficit having the sign of the omitted diagonal corrections, which are also directly visible as $\text{Var}/\text{mean} = 0.94$ on average across d . Second, the deterministic cumulative profile $z_d(x) = (\pi_d(x) - \mathfrak{S}(d) \text{Li}_2(x)) / \sqrt{\mathfrak{S}(d) \text{Li}_2(x)}$ —the estimator the uniformity verification in (i) uses—shows the same structure at cumulative scale (there $\rho \sim K \text{Li}_3 / \sqrt{\mathfrak{S} \mathfrak{S}' \text{Li}_2}$): mean off-diagonal $\rho = 0.43$ over all even $d, d' \leq 60$ at $x = 10^8$, hence predicted profile spread $\sigma^2 \leq 1 - 0.41 = 0.59$ before the diagonal correction, against observed ≈ 0.27 , implying a diagonal factor ≈ 0.5 of Montgomery–Soundararajan sign and order.

The force of the statement is its uniformity: the singular series takes a thousand different values over $d \leq 2000$, spanning a factor of about three between $d = 2$ and the primorial-rich d , and the counts must reproduce the entire profile simultaneously with square-root errors. A conspiracy would need to bend a thousand-dimensional vector, not one number. At $x = 10^8$: correlation 0.999999, slope 0.99966, $\max_d |z_d| = 1.80$; stressed to $d \leq 6000$, $\max |z_d| = 2.01$ over three thousand values.

(A further Hardy–Littlewood calibration, on a prime quintuplet constellation, gives observed 15,236 against predicted 15,230.3 at 4×10^9 , ratio 1.0004, $z = +0.05$.)

Conjecture 9 (The sub-diffusivity of balanced prime races; apparently new). *Model a balanced race (Conjecture 1(ii)’s symmetric class differences) as the walk of its class-assignment steps, the process defined at event index: enumerate the pattern occurrences $n_1 < n_2 < \dots$ in the two competing classes, set $\xi_i = \pm 1$ by class membership of n_i , and let $\rho_k(x)$ denote the lag- k sample autocorrelation of (ξ_i) over the events with $n_i \leq x$. Then the walk is measurably sub-diffusive: (i) [decaying-repulsion law] the step autocorrelations are negative at small lags—the race analogue of the Lemke Oliver–Soundararajan consecutive-pattern repulsion [24], which supplies the mechanism—and, since that repulsion is scale-dependent rather than fixed, the conjectured form is a decay law*

$$\rho_k(x) = -\frac{c_k \log \log x + d_k}{\log x} (1 + o(1)),$$

with constants c_k, d_k in principle derivable from the LOS correlation series (underived here; and the measured values $\rho_1 \approx -0.037$, $\rho_2 \approx -0.009$ at 10^9 —universal across twin and cousin races mod 5 and 8 to within 0.004, each ~ 50 standard errors from zero—are observations at one height, consistent with $c_1 \approx 0.25$ there, not asymptotic constants); (ii) [long memory, measured] short lags do not exhaust the deficit: the running maxima $M(x) = \max_{t \leq x} |D(t)|/\sqrt{\text{count}}$ of the four races land at quantiles 0.02–0.17 of the simulated independent-step null (median 2.34 at this span), and still only 0.03–0.23 after correcting for $\rho_{1,2}$: the negative correlation persists across lags, so the cumulative diffusivity $\sigma^2(x) = 1 + 2 \sum_k \rho_k(x)$ sits well below 1 at this height. The model layer is declared: (ξ_i) is a deterministic, nonstationary family and the $\rho_k(x)$ are empirical statistics of it, so the Green–Kubo identity $\sigma^2 = 1 + 2 \sum_k \rho_k$ is invoked under an explicit local-stationarity model hypothesis—the step process restricted to a window $[x, 2x]$ is modelled as a stationary sequence whose autocorrelations are the measured $\rho_k(x)$ —and the variance statement can equivalently be taken as a direct definition of $\sigma^2(x)$ through block averages of $(S_{n+m} - S_n)^2/m$ over the events up to x , a form in which no stationarity is assumed at all. The Green–Kubo sum requires more than the fixed-lag law of (i): the per-lag decay controls each $\rho_k(x)$ but not the sum unless the decay is uniform in k with summable tail, since the number of relevant lags may grow with x . We therefore state the needed hypothesis as part of the clause: $\rho_k(x)$ obeys the law of (i) uniformly for $k \leq K(x)$ with $\sum_{k > K(x)} |\rho_k(x)| = o(\log \log x / \log x)$ for some $K(x) \rightarrow \infty$, and—since the tail bound alone leaves the growing head sum uncontrolled—the repulsion constants are summable, $\sum_k c_k < \infty$ and $\sum_k |d_k| < \infty$, so that $\sigma^2(x) = 1 - 2(\sum_k c_k \log \log x + \sum_k d_k) / \log x (1 + o(1))$ is a genuine consequence rather than an unjustified interchange of limits. The measured correlation length at 10^9 (a few tens of lags, with $|\rho_k|$ decreasing geometrically in the observable range) is consistent with both hypotheses. Under them, $\sigma^2(x)$ returns to 1 asymptotically, making sub-diffusivity a finite-height phenomenon of $\log \log x / \log x$ size, like every other second-order law in this paper; (iii) [registered dichotomy, open] asymptotically the two possibilities are mutually exclusive, and which of them holds is the open question. Either the fixed-lag expansion of (i) extends uniformly in k with summable constants—in which case the hypotheses of (ii) hold, (ii)

applies, and the race is asymptotically diffusive with $\sigma^2(x) \rightarrow 1$ (running maxima $\sim \sqrt{2 \log \log x}$, Darling–Erdős class)—or that expansion fails at long lags, and only in that case can the race inherit the almost-periodic rigidity that Rubinstein–Sarnak-type structure [6] forces on classical races (maxima $\asymp (\log \log \log x)^A$, the Montgomery–Ng class). Rigidity therefore requires a failure of the hypotheses of (ii) at long lags, and cannot coexist with them. No explicit formula is known for twin patterns, the two branches differ only beyond any computable height, and we register the dichotomy without choosing in this conjecture. What is conjectured here is therefore the fixed-lag law (i) outright, and the variance law of (ii) conditionally on its stated uniformity and summability hypotheses, which belong to the finite-height model layer and are not asserted asymptotically: asserting them would silently select the diffusive branch of (iii). The pattern-specific selection is made elsewhere—Conjecture 1 asserts the rigid branch for its races, through (i-b) for the contaminated combinations and through the drift-scale nulls of its clause (ii) for the symmetric differences—and consistency between the two conjectures then requires exactly that the hypotheses of (ii) fail at long lags for those races.

The repulsion constants ρ_1, ρ_2 are pattern-independent across our four races—a universality the LOS correlation series should predict quantitatively, and the conjecture’s stated derivation programme—and the suppressed maxima are the first (to our knowledge) direct measurement of race sub-diffusivity, an effect invisible to endpoint statistics. (The twin-gap record benchmark— $G_t(x) \asymp \log^3 x$ with working constant $1/(2C_2)$, inside Kourbatov’s extreme-value framework [15]—is included as an attributed calibration benchmark: records to 4×10^9 wander in $[0.41, 0.56] \log^3 x$, approaching from below on a log log clock, largest observed twin-to-twin gap 5292 after the twin at 2,466,641,069; the Cramér–Granville fragility of such constants [17] is why the constant was always flagged as first-order.)

4 Family laws and second-order refinements

Conjecture 10 (Uniform quadratic de Polignac; family law apparently unstated, the $d = 2$ instance is (a), cf. OEIS A080149, [32]). *For even $d \geq 2$ let $\pi_d^q(N) = \#\{n \leq N : n^2 + 1, n^2 + 1 + d \text{ both prime}\}$ and $C(d) = C(x^2 + 1, x^2 + 1 + d)$. Then: (i) [uniform family law] for every fixed $B > 0$, $\pi_d^q(N) = C(d) I_d(N)(1 + o(1))$ uniformly over even $d \leq (\log N)^B$; (ii) [registered rate, strictly stronger] there is $\eta_B > 0$ with*

$$\sup_{\substack{2 \leq d \leq (\log N)^B \\ 2 \mid d}} \left| \frac{\pi_d^q(N)}{C(d) I_d(N)} - 1 \right| \ll_B (\log N)^{-\eta_B}$$

—an error clause that no version of Bateman–Horn supplies, registered separately from (i) since its analytic content is independent; (iii) [family statistic: the law of the constants] each local factor of $C(d)$ depends only on $d \bmod p$, and the factors at any finite set of primes are exactly independent as d varies (CRT), with the passage to the infinite product controlled by the convergent tail-variance sum—which supplies the uniform integrability (via L^2 -boundedness of the normalized partial products) that the moment identities require, since almost-sure convergence alone would not carry means through the limit; so the moments of $C(d)$ over even $d \leq D$ converge, as $D \rightarrow \infty$, to derived Euler products; in particular

$$\frac{1}{\#\{2 \mid d \leq D\}} \sum_{2 \mid d \leq D} C(d) \rightarrow \bar{C} = 2.7456 \dots, \quad \text{sd}(C(d)) \rightarrow 1.6840 \dots,$$

and the empirical distribution of $C(d)$ converges to the law of the random product $\prod_p f_p(U_p)$ with independent uniform residues U_p . In particular ($d = 2$, the classical instance): $\#\{n \leq N : n^2 + 1, n^2 + 3 \text{ both prime}\} \sim C(2) I(N)$, $C(2) = 2.954014 \dots$

This is the quadratic analogue of the uniform de Polignac law of Conjecture 8: not one count but the whole profile of constants must be matched at once, and the family, unlike its linear parent, has no known prior statement. Every even shift is admissible (for $p \geq 5$, $\omega(p) \leq 4 < p$; the classes mod 3 and 5 vary with d , which is what makes the $C(d)$ -profile nonconstant, spanning a factor of about twenty over $d \leq 300$, from 0.47 at $d = 298$ to 9.06 at $d = 162$). At $d = 2$ the local analysis reads: n even, $n \not\equiv 0 \pmod{3}$, $\omega(p) = 2 + \left(\frac{-1}{p}\right) + \left(\frac{-3}{p}\right)$ for $p \geq 5$, and the conjecture implies infinitely many twin pairs $(m+1, m+3)$ with m a perfect square. Profile verification at $N = 10^6$ over all 150 even shifts $d \leq 300$: correlation 0.99976 between observed and predicted counts, regression slope 1.0022, residual mean $z = +0.24$ with spread 0.84 and $\max_d |z| = 2.28$ —the shape-of-agreement standard applied across the family. Clause (iii) was verified by computing both sides independently: the derived Euler-product moments give mean 2.7456 and standard deviation 1.6840, against 2.7434 and 1.6726 measured over the 150 constants with $d \leq 300$ (ratios 0.9992 and 0.9932)—a family-level statistic whose constants are derived, not fitted. Clause (iii), unlike (i) and (ii), may prove accessible unconditionally by the exact-CRT and martingale route recorded at the cubic family below; the hard content of the conjecture is the uniform law and the rate clause. One question the family raises is recorded as open rather than conjectured: the maximal range $D(N)$ for which the uniform law can hold (the analogue of the Elliott–Halberstam question for this family—whether there is a sharp transition at some power scale N^δ rather than a logarithmic one).

Computational checks (the classical single instance). At $d = 2$ and bound 10^7 : observed 32,898 against predicted 32,862.7, ratio 1.0011, $z = +0.19$; the family-profile verification across the 150 shifts $d \leq 300$ is reported above.

Conjecture 11 (First occurrences of prime gaps, liminf form; (b)). *Let $\mathcal{G}$ be the set of even g that occur as gaps between consecutive primes (all even g , under Polignac’s conjecture), and for $g \in \mathcal{G}$ let $p(g)$ be the prime starting the first such gap. Then: (i) $\log p(g)/\sqrt{g}$ is bounded between positive constants on $\mathcal{G}$; (ii) [liminf law, stated as the dual of the maximal-gap limsup, conditional on the realization hypothesis below]*

$$\liminf_{\substack{g \rightarrow \infty \\ g \in \mathcal{G}}} \frac{\log p(g)}{\sqrt{g}} = \sqrt{e^\gamma/2} = 0.943682 \dots$$

The inversion of Granville’s $\limsup G(X)/\log^2 X = 2e^{-\gamma}$ into first-occurrence coordinates is not automatic: the limsup controls how large maximal gaps get, not which gap sizes are realized as first occurrences near the envelope. Clause (ii) is therefore conditional on the realization hypothesis: infinitely many $g \in \mathcal{G}$ first occur inside a maximal-gap event at the Cramér–Granville envelope scale, i.e. with $g = (2e^{-\gamma} + o(1)) \log^2 p(g)$. Under that hypothesis the displayed value is forced by algebra; without it only the lower bound $\liminf \log p(g)/\sqrt{g} \geq \sqrt{e^\gamma/2}$ follows from the envelope. The value is registered as that dual (against the slope-1 laws), not claimed as an independent phenomenon; (iii) [second-order, singular-series dependence] sampling g uniformly from the realized gaps $\mathcal{G} \cap [G, 2G]$ and letting $G \rightarrow \infty$,

$$\log p(g) = \sqrt{g} + \frac{1}{2} \log g - \frac{1}{2} \log \mathfrak{S}^*(g) + E_g, \quad \mathfrak{S}^*(g) = \prod_{\substack{p|g \\ p > 2}} \frac{p-1}{p-2},$$

where the error E_g converges in distribution: for the sampled g ,

$$\Pr[E_g \leq t] \longrightarrow 1 - \exp(-e^{2(t-\mu)}) \quad (G \rightarrow \infty)$$

for some centring constant μ —the min-type (reversed) Gumbel law at scale $\frac{1}{2}$, i.e. the law of $\frac{1}{2} \log W$ for W a unit exponential. The scale is forced by the hazard computation: the cumulative intensity of gap- g events is $\Lambda_g(y) \asymp (\mathfrak{S}(g)/y^2) \exp(y - g/y)$ in $y = \log x$ (the factor $e^{-g/y}$ is the no-intervening-prime probability), so $\frac{d}{dy} \log \Lambda_g = 1 + g/y^2 - 2/y = 2 + o(1)$ at the centring point $y_0 \approx \sqrt{g}$, and the first-passage identity $\Lambda_g(y) = W$ gives $E_g = y - y_0 = \frac{1}{2} \log W + o(1)$. The scale is $\frac{1}{2}$ and not 1 precisely because the slope of the log-intensity at y_0 is 2 (both the y -term and the g/y -term contribute), and the same slope-2 expansion is what produces the $\frac{1}{2}$ coefficients of the centring. (Gumbel-type modelling of gap extremes and first occurrences is Kourbatov–Wolf territory [15, 25]; the claim of this clause is only the $\mathfrak{S}^*$ -centring and the scale- $\frac{1}{2}$ first-passage form, not the extreme-value framework. Sampling uniformly over the realized gaps in $[G, 2G]$ conditions on realization, which the independent first-passage model does not represent; the clause conjectures that this selection does not shift the limiting law, and that assumption is part of what a refutation would refute.) Smooth gaps, whose tuple constant is large, appear earlier by exactly half a logarithm of that constant—the waiting-time first-passage computation makes the coefficient $-\frac{1}{2}$, not a free parameter, while μ is left as the one unresolved constant of the clause. (The restriction to $\mathcal{G}$ is needed: $p(g)$ is not known to be defined for every even g , and quantifying over all g would smuggle Polignac’s conjecture into the statement. The $\mathfrak{S}^*$ -clause records that the Hardy–Littlewood factor of g materially shifts waiting times; it does not move the liminf constant, since $\log \mathfrak{S}^*(g) = O(\log \log g)$.)

Clause (iii) was tested on the full first-occurrence table to 10^9 : over the 100 even gaps realized in $[60, 500]$, regressing $\log p(g) - \sqrt{g} - \frac{1}{2} \log g$ on $\log \mathfrak{S}^*(g)$ (with a $\sqrt{g}$ trend term absorbing the liminf-versus-typical drift) gives coefficient -0.466 against the predicted $-\frac{1}{2}$, partial correlation -0.24 : the singular series of the gap is visible in when the gap first appears, at the predicted strength.

The full limit does not follow by inverting Granville’s $\limsup G(x)/\log^2 x = 2e^{-\gamma}$: that inversion is invalid, since a limsup envelope for maximal gaps controls only where gaps *can first appear at the earliest*, i.e. a lower envelope for $p(g)$ —hence a liminf claim, not a limit. Whether $\log p(g)/\sqrt{g}$ converges at all, and if so whether the limit is Shanks’s classical 1 [16] (as Wolf’s refinement $p(g) \sim \sqrt{g} e^{\sqrt{g}}$ and the Kourbatov–Wolf residue-class laws [25] also assert) or the envelope value, involves at least three distinct phenomena—whether a given g occurs near the extreme scale at all, the distribution of exact gap lengths near records, and the difference between record gaps and non-record first occurrences—and is left open here. The two candidate constants differ by 6%, beneath the resolution of any feasible computation: our regression slope over $g \geq 100$ falls from 1.307 at 10^9 to 1.297 at 4×10^9 , far above both and decreasing, as it must be for either.

Conjecture 12 (The Stern lane race; apparently new). *In Stern’s representation problem $n = p + 2k^2$ (n odd), race the k -even and k -odd lanes. Prime-square contamination of the Λ -weighted lane counts comes from $n = q^2 + 2k^2$ —the norm form $x^2 + 2y^2$ of $\mathbb{Q}(\sqrt{-2})$ with prime x —and since $q^2 \equiv 1 \pmod{8}$ while $2k^2 \equiv 0$ or $2 \pmod{8}$ by the parity of k : (i) [class assignment, direct congruence] contamination sits in the k -even lane iff $n \equiv 1 \pmod{8}$, in the k -odd lane iff $n \equiv 3 \pmod{8}$, and for $n \equiv 5, 7 \pmod{8}$ no square contamination exists at all: $q^2 + 2k^2 \equiv 1$ or $3 \pmod{8}$ always, a two-line congruence (genus theory enters only for the*

converse question of which n are represented), so the null classes are provable, not conjectured; (ii) [drift law, in the ensemble form of Conjecture 4] Fix the ensemble: for a scale X , let $\mathbb{E}_X$ average over odd $n \in (X, 2X]$ of the given residue class mod 8 with weight proportional to $1/n$ (the logarithmic sampling measure, as at Conjectures 5 and 4). Define, per class, with $R_o(n)$ and $R_e(n)$ the numbers of representations $n = p + 2k^2$ with k odd and k even respectively, the clean-minus-contaminated difference with explicit sign: $D(n) = R_o - R_e$ for $n \equiv 1 \pmod{8}$ (even lane contaminated), $D(n) = R_e - R_o$ for $n \equiv 3 \pmod{8}$, and $D(n) = R_e - R_o$ (sign immaterial) on the null classes $n \equiv 5, 7 \pmod{8}$. Set $D_{\text{sys}}(n) = [\sum_{(q,k): q^2+2k^2=n, q \text{ prime}, k \geq 1} \log q] / \lambda_{\text{an}}(n)$ —the weight is $\Lambda(q^2) = \log q$, each representation counted once in the $k \geq 1$ convention (the ordered-representation factor 2 of Conjecture 4 has no counterpart here, and doubling would be a normalization error)—where $\lambda_{\text{an}}(n)$ is the analytic lane mean of $\log p$ over the candidate profile, defined with no reference to which candidates are prime:

$$\lambda_{\text{an}}(n) = \frac{\int_0^{\sqrt{(n-2)/2}} du}{\int_0^{\sqrt{(n-2)/2}} \frac{du}{\log(n-2u^2)}},$$

the mean of $\log(n-2u^2)$ under the density proportional to $1/\log(n-2u^2)$ on the candidate interval (the upper limit keeps $n-2u^2 \geq 2$, the convention of $I(N)$, avoiding the nonintegrable point $n-2u^2 = 1$), which is the profile a Bateman–Horn heuristic assigns to the prime representations. The definition is out-of-sample: the verification’s empirical lane mean over the observed prime representations is an estimator of λ_{an} , used as a consistency check, not part of the definition. The law, coefficient-identifying as at Conjecture 4(i): on the contaminated classes

$$\mathbb{E}_X[D - D_{\text{sys}}] = o(\mathbb{E}_X[D_{\text{sys}}]) \quad (X \rightarrow \infty),$$

and on the null classes—where D_{sys} vanishes identically, so that a comparison against it would be empty—the statement is made against the positive comparison scale

$$A(X) := \mathbb{E}_X^{(1)}[D_{\text{sys}}],$$

the logarithmic-ensemble mean of the systematic census taken over the contaminated class $n \equiv 1 \pmod{8}$: writing D_{null} for the difference D on a null class $n \equiv 5$ or $7 \pmod{8}$, the claim is $\mathbb{E}_X[D_{\text{null}}] = o(A(X))$.

This is the calculus applied to a representation problem with norm-form contamination—the drift weight is carried by the arithmetic of $\mathbb{Q}(\sqrt{-2})$ rather than by a polynomial lane—and it comes with two provable null classes. On depth: an *asymptotic* for the contaminating count $\#\{(q, k) : q^2 + 2k^2 = n, q \text{ prime}\}$, averaged over n , is a norm-form-with-prime-argument problem of genuine analytic difficulty (the same species as primes represented with restricted variables); $D_{\text{sys}}(n)$ is *computed exactly per sample*, so the verification tests the transfer conjecture against the true census, not against an unproved averaged asymptotic. Verified on 2,500 log-sampled odd $n \leq 10^8$ (mean ~ 330 representations each): contaminated strata pooled $D = +0.27 \pm 0.33$ against predicted $D_{\text{sys}} = +0.15$ (at the $\Lambda(q^2)$ weight); null strata pooled -0.22 ± 0.33 , consistent with zero (one of the four strata sits at -2σ individually, within a four-test family). (Stern’s exceptional list—the ten known odd integers that are not $p + 2k^2$, the largest 5993, with the seven prime members the Stern primes; completeness of the list is conjectural, OEIS A060003—is included as the attributed calibration benchmark anchoring this conjecture: no further exception occurs below 10^9 in our computation.)

Conjecture 13 (Microscopic variance law; (b); the constant is Montgomery–Soundararajan’s [19]). *Fix $\lambda > 0$ and $c \in (0, 1]$, and let X count primes in $[t, t + \lambda \log x)$ for t uniform on $[x, (1+c)x]$. Then, under the Hardy–Littlewood k -tuple conjectures, $X \Rightarrow \text{Poisson}(\lambda)$ (Gallagher’s argument [18], which is conditional on exactly those conjectures), and—conditional on the quantitative hypothesis spelled out after the statement (uniform Hardy–Littlewood with $o(h)$ control of the averaged singular series), without which the display below is a registered extrapolation rather than a consequence—at finite x*

$$\frac{\text{Var } X}{\mathbb{E} X} = 1 - \frac{\log(\lambda \log x) + \gamma + \log 2\pi - 1}{\log x} + o\left(\frac{1}{\log x}\right),$$

with $\gamma + \log 2\pi - 1 = 1.41509\dots$, the leading correction being independent of the sampling constant c . [Interpolation clause] More generally, for windows of length $H = \lambda(\log x)^\alpha$ the same formula with $\log H$ in place of $\log(\lambda \log x)$,

$$\frac{\text{Var } X}{\mathbb{E} X} = 1 - \frac{\log H + \gamma + \log 2\pi - 1}{\log x} + o\left(\frac{1}{\log x}\right),$$

holds uniformly for $1 \leq \alpha \leq A$ (any fixed A with $H \leq x^{o(1)}$): one expression interpolates from Gallagher’s Poisson boundary ($\alpha = 1$, deficit $\asymp \log \log x / \log x$) through the mesoscopic Montgomery–Soundararajan regime (deficit $\asymp \alpha \log \log x / \log x$), with no transition other than the slow growth of $\log H$.

The deficit arises by inserting Montgomery’s singular-series average $\sum_{d \leq h} \mathfrak{S}(d)(h-d) = \frac{h^2}{2} - \frac{h}{2}(\log h + \gamma + \log 2\pi - 1) + o(h)$ into the second factorial moment at the microscopic window $h = \lambda \log x$. The constant $\gamma + \log 2\pi - 1$ is Montgomery–Soundararajan’s, was identified in their framework long before this paper, and the statement is a *microscopic extrapolation and numerical test* of their second-moment formula at Gallagher’s boundary—not a newly discovered second-order law. The hypothesis must also be quantitative: what is required is not the qualitative k -tuple conjectures but uniform Hardy–Littlewood estimates for all shifts up to $O(\log x)$, control of the average singular series at the $o(h)$ level, and bounds for prime powers, endpoint weights, and the variation of $\log t$ across the sampling range. Three further qualifications are built into the statement: Gallagher’s Poisson law is conditional, not a theorem about primes; the sampling measure for t is specified, since different weightings shift lower-order terms; and the error is claimed only as $o(1/\log x)$. The naive claim $\text{Var} / \mathbb{E} = 1$ fails at every accessible scale by 18–28%; the corrected law matches with no fitted parameter, and the observed residual $\approx +0.003$ is consistent with (but not evidence for) a genuine $1/\log^2 x$ term. (Numerical precedent: Sanchis-Lozano [26] fitted the Montgomery–Soundararajan variance shape at $x \sim 10^8$ in the mesoscopic regime $h \gg \log N$; the microscopic statement and test appear to be new.)

λ	$x = 10^9$		$x = 4 \times 10^9$	
	observed	predicted	observed	predicted
1/2	0.8241	0.8189	—	—
1	0.7910	0.7854	0.7998	0.7960
2	0.7527	0.7520	0.7670	0.7646
4	0.7201	0.7185	—	—

The residual $\approx +0.003$ sits at the scale of a $1/\log^2 x$ term, consistent with the hedged reading at the statement.

Conjecture 14 (The null-mechanism race; apparently new). *For the quadratic twin pairs of Conjecture 10 at $d = 2$, square contamination is algebraically impossible: $n^2 + 1 = q^2$ has no solution with $n \geq 1$, and $n^2 + 3 = q^2$ only at $(n, q) = (1, 2)$. The surviving classes of $n \bmod 5$ are $\{0, 1, 4\}$ (the single escape $n = 2$, the pair $(5, 7)$ in the killed class 2, is one bounded term), and the classes 1 and 4 have identical singular series (elementary, via CRT). Let $\pi_a^q(x)$ count solutions with $n \equiv a \pmod{5}$, $D(x) = \pi_1^q(x) - \pi_4^q(x)$, and define the normalized race process $Y(x) = D(x)/\sqrt{\pi_1^q(x) + \pi_4^q(x)}$. Then, in contrast to the twin race of Conjecture 1, this race is driftless: (i) $\mathcal{M}_x(D(t) \log^2 t / \sqrt{t}) \rightarrow 0$ at the drift-scale normalization of Conjecture 1(i)—the scale at which the contaminated races converge to nonzero constants—so this is the genuine negation of Conjecture 1(i) for this pair (the weaker normalization $\mathcal{M}_x(Y) \rightarrow 0$ with $Y = D/\sqrt{\pi^q}$ would not exclude a drift at the contamination scale, hence would not make this race a control; the form stated excludes exactly that); (ii) [occupation law, conjectured directly, with a separately falsifiable local hypothesis] the process is defined precisely at event index: let $n_1 < n_2 < \dots$ enumerate the n with $n^2 + 1$ and $n^2 + 3$ both prime and $n \equiv 1$ or $4 \pmod{5}$, set $\xi_i = +1$ if $n_i \equiv 1$ and -1 if $n_i \equiv 4$, and let $S_N = \xi_1 + \dots + \xi_N$ (so $D(x) = S_{N(x)}$ with $N(x)$ the number of such events to x). The sequence (ξ_i) is deterministic, so the hypothesis is stated for a randomly sited window rather than for S_N itself. Fix a growth scale N_0 and a window length $N = N(N_0)$ with $N = o(N_0)$, and choose the starting index M of the window according to the logarithmic measure on $[1, N_0]$:*

$$\Pr[M = m] = \left(\sum_{1 \leq m' \leq N_0} \frac{1}{m'} \right)^{-1} \frac{1}{m}, \quad 1 \leq m \leq N_0.$$

The hypothesis is an almost-sure invariance principle (ASIP) for the induced random path $(S_{M+j} - S_M)_{0 \leq j \leq N}$ under that sampling law: on a probability space carrying M and a standard Brownian motion B ,

$$S_{M+j} - S_M = B(j) + O(N^{1/2-\delta}) \quad \text{uniformly for } 0 \leq j \leq N,$$

almost surely in M as $N_0 \rightarrow \infty$, for some $\delta > 0$. The logarithmic sampling law is the same clock in which the occupation statement below is read, and randomizing the window start is what supplies the deterministic step sequence with a probability space. The weak functional limit theorem— $S_{\lfloor N\tau \rfloor} / \sqrt{N}$ converging weakly to standard Brownian motion on $\tau \in [0, 1]$ —is retained as the weaker, separately falsifiable consequence of the windowed ASIP. The occupation law itself, however, is not a consequence of the windowed hypothesis, and is conjectured directly. The windowed path $S_{M+j} - S_M$ is recentred at the window start, and recentring erases exactly the absolute level that occupation measures: a deterministic bias $b(n) = \sqrt{n}/\log n$ added to S_n shifts the level by $\sqrt{n}/\log n$ while contributing window increments of size $\sim j/(2\sqrt{n} \log n) \ll \sqrt{j}$, invisible to the windowed ASIP at every window scale, yet it would drive the occupation fraction to 1. A global coupling $D(t) = B(N(t)) + o(\sqrt{N(t)})$ with $N(t) \asymp t/\log^2 t$ would suffice, but for the deterministic sequence (ξ_i) no randomization is available that makes such a coupling well posed, so we do not hypothesize it. The Brownian computation is instead the motivation that identifies the limit and the constant: for a coupled walk, in logarithmic time $u = \log t$ the Lamperti reduction $Z(s) = e^{-s/2} B(e^s)$ is a stationary, ergodic Ornstein–Uhlenbeck process, and the logarithmic occupation measure of leadership converges almost surely to $\frac{1}{2}$. The conjectured law is that conclusion itself:

$$\frac{1}{\log x} \int_2^x \mathbf{1}_{\{D(t) > 0\}} \frac{dt}{t} \rightarrow \frac{1}{2},$$

with fluctuations of the Gaussian scale $\sqrt{\log 2 / \log x}$ (the occupation-indicator covariance $\frac{1}{2\pi} \arcsin e^{-u/2}$ integrates to $\frac{1}{2} \log 2$; note that the integrated covariance for the sign average $2\mathbf{1} - 1$ is four times this, its standard deviation twice, the occupation fraction itself carrying $\sqrt{\log 2 / \log x}$); the classical arcsine spread survives only in the natural event clock, i.e. for occupation fractions of $\{B(m) > 0, m \leq N(x)\}$. A persistent systematic leader—a log-occupation stuck near 0 or 1 beyond the stated Gaussian scale—is the refutation.

The clock matters here. An arcsine-distributed limit for the logarithmic occupation itself is incompatible with the change of clock: the arcsine law lives in the natural time of the walk, whereas under dt/t averaging the Lamperti transform is ergodic and forces the limit $\frac{1}{2}$. The distinction is what makes the statistic informative: the observed log-occupation 0.21 at 10^7 sits 1.4 null standard deviations below $\frac{1}{2}$ (scale $\sqrt{\log 2 / \log x} \approx 0.21$ there)—still consistent with the driftless null—whereas read as an arcsine draw the same number would carry no information at all.

This is the negative control that completes the mechanism family of Conjectures 1 and 4: those races carry a predicted drift *because* prime squares can infiltrate specific residue classes of specific patterns; here the quadratic form blocks squares identically, so the same model predicts *nothing*—a falsifiable contrast. If a persistent drift were found in this race, the mechanism story behind the family would be wrong no matter how well the positive cases fit. The absence of *this* drift source does not by itself prove the race driftless—oscillatory zero terms, higher prime powers, and endpoint corrections are other conceivable sources—so clause (i) is a conjecture under the same zero-oscillation hypothesis as Conjecture 1, not a consequence of the square-blocking algebra alone. What the algebra supplies is the *contrast*: whatever residual model one adopts, it must produce drift in Conjectures 1 and 4 and none here. Computational checks at 10^7 : 32,898 pairs, class counts 10,917 against 10,981 ($D = -64$, i.e. -0.43 noise units), log-mean normalized drift -0.46 (consistent with 0 at one noise unit), log-occupation of leadership 0.21—1.4 null standard deviations below $\frac{1}{2}$ on the clock and constant of clause (ii), within the fair race’s 95% band. (The quadratic triple $n^2 + \{1, 3, 7\}$, $C = 10.64599 \pm 0.006$, is included as a calibration benchmark—observed 3,963 against predicted 3,998.6 at 10^7 , $z = -0.56$, independently recomputed to 3×10^7 ; the longer pattern $n^2 + \{1, 3, 7, 9, 13\}$ has been recorded in the OEIS since 2000, so the tuple family has prior presence and only its singular-series evaluation is plausibly new.)

Computational checks. The race data are reported at the statement above; independent recomputation of this conjecture’s calibration member, the quadratic triple, gave 10.647706, with primes to 10^9 —inside our stated wobble—and an independent count to 3×10^7 with final ratio 1.0059.

Conjecture 15 (The cubic shift family: the distribution of its constants; instance-level law apparently unstated). *For non-cube $a \geq 1$ let $C(a) = C(x^3 + a)$. Only $p \equiv 1 \pmod{3}$ moves the product, and the local root count there takes three values as a varies mod p : $\omega = 3$ with probability $(p-1)/3p$ ($-a$ a nonzero cube), $\omega = 1$ with probability $1/p$ (the case $p \mid a$, easy to overlook and essential to the mean-value identity below), and $\omega = 0$ with probability $2(p-1)/3p$; for $p \equiv 2 \pmod{3}$ the cube map is a bijection and $\omega = 1$ identically. Then: (i) [mean-value one, elementary lemma] $\mathbb{E}_a[\omega(p)] = 3 \cdot \frac{p-1}{3p} + \frac{1}{p} = 1$ exactly at every p , so the derived mean of $C(a)$ is exactly 1—the cubic analogue of the Korevaar–te Riele mean-value-one theorem for linear prime-pair constants, here a one-line computation at each finite level; passing the mean through the infinite product needs more than almost-sure convergence, and the needed ingredient is on hand: the convergent variance sum makes the partial products an L^2 -bounded martingale (each factor has mean 1 and is independent of the earlier ones), so they are uniformly*

integrable and the limit law has mean exactly 1, not merely limit-of-means 1; (ii) [limit law] the empirical law of $\{C(a) : a \leq A, a \text{ a non-cube}\}$ converges, as $A \rightarrow \infty$, to the law of the random Euler product $\prod_{p \equiv 1(3)} f_p(\xi_p)$ with the three-state local variables ξ_p above—independent at any finite set of primes exactly, by CRT, with the infinite product’s tail controlled by the convergent variance sum (unlike the quadratic family of Conjecture 10(iii), no normalization is needed)—with derived standard deviation 0.2762... computed from the full three-state law; (iii) [uniformity] $\#\{n \leq N : n^3 + a \text{ prime}\} = C(a)I_a(N)(1 + o(1))$ uniformly over non-cube $a \leq (\log N)^B$.

The framework here is Kowalski’s theory of averages of Euler products and singular-series distributions [11], which proves such limit laws for k -tuple singular series; the one-parameter cubic shift family, its exact mean-one lemma, and the short-range uniformity appear unstated, and we present (ii) as an instance-level law inside that framework rather than a new mechanism. Computational checks: derived moments (mean exactly 1, sd 0.2762) against the 294 non-cube shifts $a \leq 300$: empirical mean 1.0215 (+1.5 profile standard errors), sd 0.2481 (finite-family tail correlations account for the sd gap: for $p > a_{\max}$ the residues of a are not yet equidistributed); uniformity profile over 57 shifts at $N = 2 \times 10^5$: mean $z = +0.09$, spread 0.72, max $|z| = 2.31$. (The single instance $a = 2$, $C = 1.298435 \pm 0.0003$, verified at 10^7 with $z = +0.32$, is included as the calibration member; no statement of Bunyakovsky type is proven for any cubic, the nearest theorem being Heath-Brown’s $x^3 + 2y^3$ [3], which is what keeps the whole family conjectural.)

Computational checks (the classical single instance). At $a = 2$ and bound 10^7 : observed 287,956 against predicted 287,784.8, ratio 1.0006, $z = +0.32$; the family-profile verification across the 57 shifts is reported above.

Conjecture 16 (Conjecture F as a family; (a), [1], cf. [23]). *For odd A let $Q_A(N) = \#\{n \leq N : n^2 + n + A \text{ prime}\}$ and $C(A) = C(x^2 + x + A)$. Then: (i) for every fixed $B > 0$, $Q_A(N) = C(A)I_A(N)(1 + o(1))$ uniformly over odd $1 \leq A \leq (\log N)^B$; (ii) [window field] the residual field obeys the analogue of Conjecture 8(ii), kernel derived and probability space specified: for t uniform on $[N, 2N]$ and window length $H \leq N^{o(1)}$, let $Q_A(t; H) = \#\{t < n \leq t + H : n^2 + n + A \text{ prime}\}$ (mean $\asymp H/2 \log N$, which sets the regime). Two members share the index window, so with $C(A, A') = C(x^2 + x + A, x^2 + x + A')$,*

$$\text{Cov}_t(Q_A, Q_{A'}) \sim [C(A, A') - C(A)C(A')] \frac{H}{4 \log^2 N},$$

with $C(A, A') = 0$ (hence negative correlation) for the locally exclusive pairs. This is the same-index contribution only: the full covariance adds the off-index pinned sum

$$\frac{1}{4 \log^2 N} \sum_{0 < |h| \leq H} (H - |h|) [C(f_A(x), f_{A'}(x + h)) - C(A)C(A')],$$

the two-parameter analogue of Conjecture 2’s $G(H)$, whose evaluation is this family’s open component. (The order of this sum matters: the joint event here is two single prime values at distinct indices, of probability $\asymp 1/4 \log^2 N$, unlike Conjecture 8’s pair-of-pairs events at $1/\log^4 x$. At that order the off-index sum, which grows like a pinned G -average over $\asymp H$ shifts against the same-index term’s single H , is generically the larger component, so the numerical cancellation of the same-index kernel establishes nothing about the full cross-member covariance, and the localization of the observed profile deficit to the diagonal is an open question rather than an inference.) For $H/\log N \rightarrow \lambda$ the counts are jointly Poisson with local exclusions, and for $H/\log N \rightarrow \infty$ each fixed finite collection of studentized counts is asymptotically jointly normal

with independent limiting coordinates, all correlations of size $\asymp 1/\log N$ making both kernels finite- N corrections to independence. As at Conjecture 8(ii), both distributional conclusions are asserted under an explicit hypothesis beyond the displayed covariances—uniform Hardy–Littlewood tuple estimates over the family, strong enough that every joint cumulant of order at least three of the studentized counts vanishes in the respective normalizations—since the covariance kernels alone do not determine the limit laws. (Uniformity over a range growing with N is the substantive claim; uniformity over any fixed finite set of A is logically automatic from the individual asymptotics.) Among odd $A \leq 199$, Euler’s $A = 41$ has the largest constant, $C(41) = 6.6395463\dots$ (Cohen’s high-precision value [31]).

Numerical evaluation of the kernel (all odd $A, A' \leq 99$ at $N = 10^6$: 1,225 pairs, of which 289 locally exclusive) returned a structurally clean answer: the positive correlations of admissible pairs and the negative correlations of exclusive pairs *cancel*, mean off-diagonal $\rho = -0.003$, so the derived *same-index* cross-member kernel is null. With the off-index sum at its correct order (larger, not smaller, than the same-index term), this cancellation does not license the conclusion that the members are asymptotically independent or that the observed profile variance ≈ 0.37 must be diagonal; the natural reading remains that each $\text{Var}(Q_A)$ is sub-Poisson through within-sequence pair correlations, the quadratic-family analogue of the Montgomery–Soundararajan deficit (Conjecture 13), but the off-index cross terms are an unevaluated competitor—and part of the spread is truncation noise in the constants themselves (the $\pm 10^{-3}$ wobble moves each z_A by ~ 0.3). Deriving the diagonal law and the off-index pinned sums is the family’s remaining open component, and the more important one; the numerical cancellation of the *same-index* cross-member kernel is a finite computation over $A \leq 99$, which localizes the deficit off the same-index diagonal without proving that the averaged kernel or the off-index contributions vanish—an analytic classification of the sign of $C(A, A') - C(A)C(A')$, and the evaluation of the two-parameter pinned sums, remain open. At $N = 10^6$: mean ratio 1.0006, standard deviation 0.0027, $\max |z| = 1.92$, rank correlation 0.99988; for $A = 41$, observed 261,080 against predicted 261,017.6 ($z = +0.12$, using Cohen’s constant). On precision: a truncated Euler product for $C(41)$ at cutoff 2×10^5 reads 6.64092—wrong in the fourth significant decimal—because these conditionally convergent products drift at the 10^{-3} scale over practical cutoffs. Quoted digits for all quadratic-family constants in this paper are limited by the stated truncation wobble, not by the last printed digit. The entire 10^8 -bit primality dataset compresses to one product formula per A .

Conjecture 17 (Twin-member Goldbach, orientation-resolved; the basis conjecture and the integral kernel are Dubner’s [9]; the orientation decomposition apparently new). *For even n let $R_T(n)$ count ordered (a, b) , $a + b = n$, with a and b both members of twin pairs, each ordered role (lower/upper) counted. Each of the four role assignments is a 4-form linear system in $t = a$ with root set $\{0, \pm 2\} \cup \{n, n \pm 2\}$ -type mod each p , hence its own n -dependent singular series $\mathfrak{S}_4^{(o)}(n)$, and*

$$R_T(n) = \left[\sum_o \mathfrak{S}_4^{(o)}(n) \right] \int_5^{n-5} \frac{dt}{\log^2 t \log^2(n-t)} (1 + o(1)),$$

the four constants computed exactly per n from the coincidence pattern of the root sets (which of $n, n \pm 2, n \pm 4$ each small prime divides). The error term is stated as $(1 + o(1))$, not $(1 + O(1/\log n))$: our own measurements show a second-order deficit whose $1/\log n$ coefficient is not yet pinned down (below). Two precision notes: the asymptotic runs over varying n , so what is assumed is a Hardy–Littlewood estimate for 4-form systems uniform in the n -dependent coefficients—strictly stronger than the fixed-system conjecture—and the number 5, the unique prime that is

simultaneously a lower twin member (of $(5, 7)$) and an upper one (of $(3, 5)$), is double-counted consistently by the ordered-role convention, affecting finitely many representations per n and no asymptotic.

Dubner’s paper already contains the integral kernel (his quotient μ_4) and the qualitative basis conjecture; the content claimed here is the orientation-resolved singular series and its n -profile. Two exact identities compress the four orientations: the substitution $t \mapsto n - t$ exchanges the two summand roles, forcing $\mathfrak{S}_4^{(\text{lu})}(n) = \mathfrak{S}_4^{(\text{ul})}(n)$ identically (each of the (ll) and (uu) systems being self-dual under it), and the affine substitution $t \mapsto t + 2$ gives $\mathfrak{S}_4^{(\text{uu})}(n) = \mathfrak{S}_4^{(\text{ll})}(n - 4)$. So only one function of n per symmetry class is free, and the profile is really two-dimensional; determining the average of $\sum_o \mathfrak{S}_4^{(o)}(n)$ over n in closed form is the conjecture’s registered identity-hunting programme. Verified on 150 log-sampled $n \leq 10^8$: the profile *shape* is confirmed at log-log correlation 0.9993 across two decades, while the *level* runs at 0.87 of prediction on $[10^6, 10^7]$ and 0.81 on $[10^7, 10^8]$ —a systematic deficit of second-order size ($\approx 3.4/\log n$ at the top range, the same order as the 18–28% finite-height Poisson deficit of Conjecture 13), which we flag as its open component rather than absorb into a fitted constant: either the 4-tuple second-order correction accounts for it, or the orientation model needs repair, and the two are distinguishable at 10^{10} . The *direction* of the trend is itself the anomaly: a genuine $1 - c/\log n$ correction shrinks as n grows, so a ratio moving *away* from 1 across the decades indicates either a different error shape or a normalization problem, and settling which is part of what the open component must resolve. (The attributed benchmark stands inside the statement: every even $n \geq 4210$ is a sum of two twin members, with the conjectured complete list of 35 exceptions (OEIS A007534)—verified to 10^8 , re-swept to 10^9 , independently re-verified by a different algorithm, the same 35 exceptions each time.)

5 Instances and structural companions

Conjecture 18 (Contamination in prime triplets; apparently new). *The contamination calculus of Conjecture 1(v) extends to k -tuples, and the triplet pattern $(n, n + 2, n + 6)$ is its first instance beyond pairs. Of the three configurations placing a prime square inside the pattern, two die algebraically for $q > 3 - (q^2, q^2 + 2, q^2 + 6)$ by $3 \mid q^2 + 2$, and $(q^2 - 6, q^2 - 4, q^2)$ because $q^2 - 4 = (q - 2)(q + 2)$ is composite (at $q = 3$ the factorization is trivial, $q^2 - 4 = 5$ is prime, and the single bounded exceptional configuration $(3, 5, 9)$ occurs—one term, harmless to every asymptotic)—leaving the single doubly-thinned survivor $(q^2 - 2, q^2, q^2 + 4)$, which requires $q^2 - 2$ and $q^2 + 4$ simultaneously prime and forces $q \equiv \pm 2 \pmod{5}$ for $q \neq 5$ (else $5 \mid q^2 + 4$; at $q = 5$ all three conditions do hold, the configuration being $(23, 25, 29)$, but its start $23 \equiv 3 \pmod{5}$ lies outside the triplet start classes $\{1, 2\}$, so the single term is harmless). Triplet starts lie in classes $n \equiv 1, 2 \pmod{5}$ (single bounded exception $(5, 7, 11)$, start $\equiv 0$), and the surviving configuration has start $q^2 - 2 \equiv 2$: class 2 is contaminated and class 1 leads, in the drift-scale form of Conjecture 1(i):*

$$\mathcal{M}_x \left(\frac{(\pi_3(t; 5, 1) - \pi_3(t; 5, 2)) \log^3 t}{\sqrt{t}} \right) \rightarrow c_3 := \lim_{x \rightarrow \infty} \frac{T_3(x) \log^3 x}{\sqrt{x}}, \quad T_3(x) = \frac{1}{\log^3 x} \sum_{\substack{q \leq \sqrt{x}, q \neq 5 \\ q^2 - 2, q^2 + 4 \text{ prime}}} \log(q^2 - 2) \log q \log(q^2 + 4)$$

(c_3 exists and is the Bateman–Horn constant of the triple $(q, q^2 - 2, q^2 + 4)$, since the census sum is $\sim c_3 \sqrt{x}$; convergence of the logarithmic mean at this normalization is the same Rubinstein–Sarnak-type structure hypothesis as at Conjecture 1(i); a normalization at the noise scale,

$\mathcal{M}_x((D - T_3)/\sqrt{\pi_3})$ with D the class difference displayed above and π_3 the total triplet count to t , would be vacuous for the coefficient, exactly as at the pair races), with drift $\asymp \sqrt{x}/\log^3 x$, one logarithm weaker than the pair races, and drift-to-noise $\asymp \log^{-3/2} x$ against their $\log^{-1}$, a scale the calculus itself predicts and the verification must respect. The mechanism clause and the averaging clause are asserted separately, as at Conjecture 1(i).

This is the calculus applied at tuple level: the configuration census, the double thinning, and the mod-5 class assignment are all forced, and the prediction was derived before the data were taken. At 10^9 : classes 189,837 against 189,670 ($D = +167$ on the predicted side; $T_3 = +25$ against noise 616, so the endpoint is uninformative exactly as the $\log^{-3/2}$ law requires), leadership log-density 0.65 (+0.8 null standard deviations at the occupation constant of Conjecture 14(ii)—directionally consistent, sharpness unavailable at this height by the calculus’s own accounting). (The Chernick chain $\{p, 2p - 1, 3p - 2\}$ —the population behind the universal-form Carmichael numbers, quantified by Dubner [10]—is included as an attributed calibration benchmark: $C = 2.858249$, observed 125,379 against predicted 125,429.4 at 3×10^8 , $z = -0.14$.)

Computational checks. The triplet-race data and the Chernick-chain calibration count are reported at the statement above.

Conjecture 19 (The contamination matrix: sexy pairs with two surviving orientations; apparently new). *For the pattern $(n, n + 6)$, both prime-square orientations survive, on complementary classes of $q \bmod 5$ —the first matrix instance of the calculus (every pair race so far had exactly one): orientation $A = (q^2 - 6, q^2)$ requires $q^2 - 6$ prime, forcing $q \equiv \pm 2 \pmod{5}$ for $q \neq 5$ (at $q = 5$ the value 19 is prime, but its start $19 \equiv 4 \pmod{5}$ lies outside the sexy start classes $\{1, 2, 3\}$), and lands in start class 3 (mod 5), class 3 (mod 8); orientation $B = (q^2, q^2 + 6)$ requires $q^2 + 6$ prime, forcing $q \equiv \pm 1 \pmod{5}$ (the term $q = 5$ falls outside the start classes), and lands in class 1 (mod 5), class 1 (mod 8). Sexy starts occupy classes $\{1, 2, 3\} \bmod 5$ and $\{1, 3, 5, 7\} \bmod 8$; with $T_A(x) = \frac{1}{\log^2 x} \sum_{q^2-6 \text{ prime}, q \neq 5} \log q \log(q^2 - 6)$, $T_B(x) = \frac{1}{\log^2 x} \sum_{q^2+6 \text{ prime}, q \neq 5} \log q \log(q^2 + 6)$, and c_A, c_B their drift-scale limits ($c_A = \lim T_A \log^2 x / \sqrt{x}$, the Bateman–Horn constant of $(q, q^2 - 6)$, likewise c_B for $(q, q^2 + 6)$), the predicted drift vector, in the drift-scale limiting-log-mean sense of Conjecture 1(i) (an $\mathcal{M}_x$ -over- $\sqrt{\pi}$ normalization would be coefficient-blind), is, writing $\pi(a)$ for the number of sexy starts up to t in class a of the modulus in question: mod 5: $\mathcal{M}_x((\pi(2) - \pi(3)) \log^2 t / \sqrt{t}) \rightarrow c_A$, $\mathcal{M}_x((\pi(2) - \pi(1)) \log^2 t / \sqrt{t}) \rightarrow c_B$ (class 2 clean); mod 8: $\mathcal{M}_x((\frac{1}{2}(\pi(5) + \pi(7)) - \pi(3)) \log^2 t / \sqrt{t}) \rightarrow c_A$, $\mathcal{M}_x((\frac{1}{2}(\pi(5) + \pi(7)) - \pi(1)) \log^2 t / \sqrt{t}) \rightarrow c_B$ (classes 5, 7 clean, mutually symmetric, with $\mathcal{M}_x((\pi(5) - \pi(7)) \log^2 t / \sqrt{t}) \rightarrow 0$). Two independently computable drift constants, one certified null class per modulus. The mechanism clause and the averaging clause are asserted separately, as at Conjecture 1(i).*

The matrix structure—two orientations feeding disjoint classes on complementary q -classes, with an interior null—is what distinguishes this from every single-orientation race, and it tests the calculus beyond its balanced one-mechanism cases; the class assignments were verified independently before the data were taken. At 10^9 the predicted components ($T_A = 191$, $T_B = 110$) sit far inside the noise ($\approx 2,100$), as the $1/\log x$ law forces; all four measured components and the 5–7 control lie within one noise unit of their predictions (trivially, at this height), and the four leadership log-densities (0.48, 0.56, 0.53, 0.73) sit at $(-0.1, +0.3, +0.1, +1.2)$ null standard deviations (occupation constant 0.18): three at the null, the mod-8 B -component mildly positive on the predicted side. The matrix is *registered and consistent*, with sharpness unavailable below $\sim 10^{14}$ by the calculus’s own accounting. (The pair $\{p, p^2 - 2\}$, OEIS A062326, $C = 3.383216$, the input that sets Conjecture 1’s drift scale, is included as an attributed calibration benchmark:

verified at 10^7 , $z = -1.31$, independently recomputed to 10^9 ; a refutation of its infinitude would remove the twin race's predicted mechanism.)

Computational checks. The sexy-pair race data are reported at the statement above; independent recomputation of this conjecture's calibration member, the pair $\{p, p^2 - 2\}$, gave 3.383227, with primes to 10^9 —inside our stated wobble—and an independent count to 3×10^7 with final ratio 0.9965.

Conjecture 20 (Fibonacci–Lucas twins: the convergent side stress-tested; the object is OEIS A080327). (i) *[rank-disjointness, elementary, with its exception stated] every odd prime factor r of L_p has rank of apparition exactly $2p$ (from $z(r) \mid 2p$, $z(r) \nmid p$, and $z(r) = 2$ being impossible), and every prime factor of F_p ($p \neq 5$ prime) has rank exactly p : the odd divisor pools of the two numbers are disjoint. The prime 2 is the unique exception: 2 divides both F_p and L_p precisely when $p = 3$ ($F_3 = 2$, $L_3 = 4$; $\gcd(F_p, L_p) = 2$ iff $3 \mid p$)—the restriction to odd divisors is essential, the clause being false at $p = 3$ without it. Disjoint pools remove shared-prime correlation but not all correlation: dependence can still enter through the order structure of the recurrence, which is why (iii) matters; (ii) *[finiteness] only finitely many primes p have F_p and L_p simultaneously prime; (iii) *[calibration clause, quantitative] the naive joint accounting—probability $(e^\gamma \log p / (p \log \phi))^2$ per index—assigns prior mass about 1.4×10^{-2} to the whole index range beyond 10^4 , in which the catalogued index $p = 148091$ falls (OEIS A080327; both F_p and L_p are probable primes—numbers of roughly 30,950 digits, with Lucas numbers proved prime only up to index 56003—so everything recorded here is conditional on both probable-prime classifications, for which no Baillie–PSW pseudoprime is known: no unconditional refutation is claimed from uncertified numbers). That rare-event datum is recorded descriptively, with no significance claim attached to it: the model was examined after the event was known, and no test protocol was fixed in advance. A single surprising event is not a strict refutation, and it does not by itself estimate the size of the correction: an underestimated constant, genuine cross-dependence, and a mis-calibrated finite-range tail are competing explanations that one event cannot separate. What the datum does bear on (conditionally as above) is not the model but the list: it is fatal to any Goldilocks completeness list; the conjectural content of (ii) is that even the corrected accounting has convergent sum.***

This conjecture is the collection's designated stress test of the convergent Borel–Cantelli template that Conjecture 21 exemplifies. Our scan to $p \leq 10^4$ found exactly the joint indices $\{5, 7, 11, 13, 17, 47\}$ with expected further mass 1.4×10^{-2} , while the catalogued index 148091, at which F_p and L_p are both probable primes, lies beyond that range. We therefore conjecture finiteness *without* a completeness clause: the Goldilocks-maximal list that is defensible for factorial twins is indefensible here. The lesson, in the language of Grantham–Granville [7], is that recurrence-sequence constants need their local corrections *before* tail masses are trusted. The single-sided calibration (probable-prime counts: F_p and L_p leave the deterministic range at very small p): 25 Fibonacci prime indices $\leq 10^4$ against the naive screening prediction 29.2 (the finite sum of the declared per-index hazard over primes to 10^4) ($z = -0.8$, a mild deficit, not significant on its own); 29 Lucas prime indices $\leq 10^4$; the corrected c_F needs a Granville-type patch, and clause (iii) quantifies the same need at the joint level.

Computational checks. Joint scan and single-sided calibration reported at the statement (Fibonacci prime indices $\{3, 5, 7, \dots, 9311, 9677\}$, 25 of predicted 29.2, $z = -0.78$; Lucas indices 29; joint $\{5, 7, 11, 13, 17, 47\}$ to 10^4)—probable-prime counts at all but the smallest indices, the Fibonacci and Lucas values leaving the deterministic range almost at once.

Conjecture 21 (Factorial twins; the uniqueness core is (a), OEIS A088054; clauses (i) and (iii) are ours). (i) *[window rigidity, elementary]* for $n \geq 4$ and $2 \leq |a| \leq n$ the number $n! + a$ is composite (any prime factor p of a satisfies $p \leq n$, so $p \mid n! + a$, and $n! + a > n \geq p$); the two small anomalies $2! - 2 = 0$ and $3! - 3 = 3$ are the only exceptions, which is why the clause is restricted to $n \geq 4$. Hence among all offsets of bounded size the only candidates for primality near $n!$ are $a = \pm 1$: the twin question below is the unique bounded-offset constellation question at $n!$, and every admissible offset set is a subset of $\{-1, +1\}$. (ii) *[uniqueness; (a)]* Only finitely many n have $n! - 1$ and $n! + 1$ simultaneously prime, and the complete list is $n = 3$: i.e. $6 = 3! = 3\#$ is the unique factorial lying between twin primes. (iii) *[joint fluctuation model]* the two single-sided counts $F_{\pm}(N) = \#\{n \leq N : n! \pm 1 \text{ prime}\}$ are each $\sim e^{\gamma} \log N$ and, under the independent-indices model stated below, asymptotically independent, with $F_+ - F_-$ normalized by $\sqrt{2e^{\gamma} \log N}$ asymptotically standard normal. The probability space is declared: the clause is about a random model in which the events $E_n^{\pm} = \{n! \pm 1 \text{ prime}\}$ carry the Caldwell–Gallot marginal hazards e^{γ}/n with an unspecified dependence structure, and the Gaussian limit is asserted under three explicit hypotheses on that structure: the same-index bound $\sum_{n \leq N} |\text{Cov}(\mathbf{1}_{E_n^+}, \mathbf{1}_{E_n^-})| = o(\log N)$, which pins the variance at $2e^{\gamma} \log N (1 + o(1))$ and which the joint law $\sim (e^{\gamma}/n)^2$ derived below supplies with a convergent sum—without it the remaining hypotheses leave the second cumulant free, and a model with $\Pr(E_n^+ \cap E_n^-) = e^{\gamma}/2n$ halves the limiting variance; the cross-index covariance bound $\sum_{m < n \leq N} |\text{Cov}(\mathbf{1}_{E_m}, \mathbf{1}_{E_n})| = o(\log N)$ over the four sign pairs (the variance itself is $\asymp \log N$), and—since pairwise control alone cannot preclude higher-order dependence—the higher-cumulant condition $\kappa_r(F_+ - F_-) = o((\log N)^{r/2})$ for each fixed $r \geq 3$, the method-of-moments sufficient condition. None of the three is imported from an independence model; whether they hold for the arithmetic dependence across factorial indices is the open component of this clause.

Each event separately has Caldwell–Gallot probability $\sim e^{\gamma}/n$ [4] and divergent sum (the $n! + 1$ side is the classical law retained as calibration at Conjecture 2); the joint event has probability $\sim (e^{\gamma}/n)^2$, whose sum converges. This is the suite’s exemplar of the convergent side of the Borel–Cantelli dichotomy—the accounting that predicts finitude rather than infinitude. The at-a-common-index independence in (iii) has a derivation: a generic twin pair near $n!$ would carry the Hardy–Littlewood twin factor $2C_2 \prod_{2 < p \leq n} \frac{p-2}{p-1} / \prod_{2 < p \leq n} (\frac{p-1}{p})^2$ -type couplings, but here the screening (coprimality to all $p \leq n$) is *deterministic*—both neighbours always pass it—and the twin coupling factor and the reciprocal sieve factors cancel to first order, leaving the joint probability $\sim (e^{\gamma}/n)^2$ with no singular-series correlation; what remains open, and is flagged in (iii), is dependence *across* distinct indices n , whose divisibility structures are nested. That open component has a concrete first computation, which we register as the conjecture’s programme: for $m < n$ and a prime $p > n$, the residues $m! \bmod p$ and $n! \bmod p$ are deterministically linked by $n! = m! \cdot (m+1) \cdots n$, so the covariance of the events $p \mid m! \pm 1$ and $p \mid n! \pm 1$ is an explicit character-sum average over the multiplier $(m+1) \cdots n$; summing it over p either bounds the cross-index dependence at $o(1/n)$ —vindicating the independent-indices model—or exhibits genuine coupling, and the computation is finite at each height. Verified for $n \leq 700$ ($F_+ = 15$, $F_- = 16$; difference -1 against noise scale ≈ 4.8); the model-expected number of further twin examples beyond 700 is $e^{2\gamma}/700 \approx 4.5 \times 10^{-3}$ (computed tail). The defensible core is finiteness; the exact one-element list is the Goldilocks-maximal form—strictly stronger than the tail estimate can guarantee. On priority: the exact uniqueness conjecture is on the public record, OEIS A088054 stating that 3 is conjecturally the intersection of A002981 and A002982, so clause (ii) is attributed there and the conjecture’s claimed content is clauses (i) and (iii).

The primorial analogue— $p\# \pm 1$ both prime only for $p \in \{3, 5, 11\}$ —is due to Lillie [5], whose

abstract gives both the $O(n^{-2})$ joint probability and the prediction that there are three instances in total; it is included here as an attributed calibration benchmark, independently verified to $p \leq 4000$.

Computational checks (probable-prime count for n beyond about 26). Exhaustive to $n \leq 700$: the only factorial twin is $n = 3$; individually, $n! + 1$ is prime for $n \in \{2, 3, 11, 27, 37, 41, 73, 77, 116, 154, 320, 340, 399, 400\}$ and $n! - 1$ for $n \in \{3, 4, 6, 7, 12, 14, 30, 32, 33, 38, 94, 166, 324, 379, 469, 546\}$ —two divergent single-sided laws straddling their shared constant while the joint count stops at one, the convergent accounting in action. (The primorial companion, attributed to Lillie [5]: exhaustive to $p \leq 4000$, twins exactly $\{3, 5, 11\}$, with $p\# + 1$ prime for eleven p against the Caldwell–Gallot prediction 12.4, $z = -0.40$; again a probable-prime count above the deterministic range.)

Theorem 1 (Cube obstruction). *For $k \geq 2$, the cube k^3 is representable as $p + j^3$ with p prime, $j \geq 1$, if and only if $3k^2 - 3k + 1$ is prime. Consequently the set of integers not representable as $p + k^3$ is infinite: the non-representable cubes alone have counting function $\sim x^{1/3}$. (That the full exceptional set is $\sim x^{1/3}$ follows only when combined with Conjecture 22(iii), and is conjectural.)*

Proof. $k^3 - j^3 = (k - j)(k^2 + kj + j^2)$; for $1 \leq j \leq k - 2$ both factors exceed 1, so the difference is composite. The only candidate is $j = k - 1$, giving $3k^2 - 3k + 1$. Since $3k^2 - 3k + 1$ is composite for a density-one set of k (all but $O(K/\log K)$ of $k \leq K$, by an upper-bound sieve of Brun or Selberg type applied to the quadratic’s prime values), all but a vanishing proportion of cubes are unrepresentable. $\square$

The restriction to non-cubes in the next conjecture is therefore essential: the exception set of $n = p + k^3$ over *all* n is infinite, and the cubes are exactly what makes it so. The effect is visible immediately in the data—all 412 exceptions found in $(10^8, 10^9]$ are perfect cubes—and it is invisible to a Borel–Cantelli sum taken over all n , since the cubes have density zero. The cube lane is not lost, only relocated: by Theorem 1 it is governed by the primality of $3k^2 - 3k + 1$, which is a Bateman–Horn family in its own right and appears as such in Conjecture 22.

Conjecture 22 (The boundary trichotomy for polynomial ladders; the divisibility principle is classical and its cubic case is Cunningham’s cuban observation (OEIS A002407); the family classification is apparently new). *For $F \in \mathbb{Z}[x]$ with $\deg F \geq 2$ and positive leading coefficient, and $m > j \geq 1$, $(m - j) \mid F(m) - F(j)$ (textbook), and the divided difference $(F(m) - F(j))/(m - j)$ —whose leading form is then positive on the cone $m > j \geq 1$ —exceeds 1 outside an effectively bounded region; so $F(m) - F(j)$ prime forces $m - j = 1$ apart from finitely many exceptional pairs (for the family below the exceptional set is provably empty). Both hypotheses are necessary: for $\deg F = 1$ the cofactor can be identically 1 and the collapse fails, and for negative leading coefficient the divided difference tends to $-\infty$ on the cone, so the “exceeds 1” clause would be false as stated (one may equivalently keep general sign and read the claim through $|F(m) - F(j)|$). Every representation problem $F(m) = p + F(j)$ in this range thus collapses to the boundary polynomial $D_F(m) = F(m) - F(m - 1)$. The content is the classification of the boundary lanes. For $F = x^3 + cx$, $c \geq 0$: (i) [trichotomy, elementary] $D_F = 3m^2 - 3m + 1 + c$ and the lane is dead-parity for c odd (D_F always even), dead-3-adic for even $c \equiv 2 \pmod{3}$ ($3 \mid D_F$ always), and admissible exactly for $c \equiv 0, 4 \pmod{6}$; (ii) [uniform boundary law] over the admissible lanes, $\#\{m \leq M : D_F(m) \text{ prime}\} \sim C(D_F) I(M)$ uniformly for $c \leq (\log M)^B$ —a Bateman–Horn family whose members are indexed by the ladder classification; (iii) [attributed core] the case $c = 0$ is the cuban ladder: Cunningham’s classical observation that a prime difference of cubes forces consecutive arguments, with the non-cube exceptional set of $n = p + k^3$ finite (Hardy–Littlewood’s $E_3(X) = O(1)$ [1]).*

The reducible-boundary phenomenon is part of the same classification: $F = x^4$ has $D_F = (2m-1)(2m^2-2m+1)$ —the boundary lane itself factors, which is precisely the composite- k obstruction of Conjecture 23 seen from the ladder side, and $F = x^3 + x^2$ gives $D_F = m(3m-1)$, dead by reducibility. Verified: the collapse is algebraically empty for $x^3 + cx$ ($c \geq 0$), the trichotomy checked to $m = 5000$ on all $c \leq 12$, and the two smallest admissible new lanes counted at $M = 10^6$: $c = 4$ ($C = 2.12956$, $z = +0.27$) and $c = 6$ ($C = 2.68954$, $z = -0.92$). (Empirically the non-cube exception census of (iii) by decade is 4, 27, 168, 763, 2011, 2808, 1181, 88 up to 10^8 , peaking at the sixth decade and collapsing thereafter, largest found 78,526,384.)

Conjecture 23 (The power-obstruction ladder: an elementary structural proposition with a Bateman–Horn corollary; apparently new as a family). *For $k \geq 2$ and $m \geq 2$, write $D_k(m) = m^k - (m-1)^k$, and call m^k representable if $m^k = p + j^k$ for some prime p and $j \geq 1$. Then:* (i) [theorem] *for composite k , no k -th power is representable: if r is a proper divisor of k with $r > 1$ (such r exists precisely because k is composite; the excluded value $r = 1$ would allow only the useless factor $m - j = 1$ at $j = m - 1$), then for every $1 \leq j < m$ the difference $m^k - j^k$ has the proper factor $m^r - j^r > 1$ with cofactor > 1 ;* (ii) [elementary] *for prime k (including $k = 2$), m^k is representable if and only if $D_k(m)$ is prime; and D_k is irreducible for prime k (a root α has $(\alpha/(\alpha-1))^k = 1$ with $\alpha/(\alpha-1) \neq 1$, so $\zeta = \alpha/(\alpha-1)$ is a primitive k -th root of unity and $\mathbb{Q}(\alpha) = \mathbb{Q}(\zeta)$ has degree $k-1 = \deg D_k$, forcing irreducibility; explicitly $D_k(x) = (x-1)^{k-1} \Phi_k(x/(x-1))$. The forcing $j = m-1$ needs only $k \geq 2$);* (iii) [conjecture] *for each prime $k \geq 3$ the representable lane follows Bateman–Horn ($k = 2$ is a theorem, the prime number theorem on the lane $2m-1$): $\#\{m \leq M : D_k(m) \text{ prime}\} \sim C_k \int_2^M dt / \log D_k(t)$.*

The dividing line is prime versus composite k , not odd versus even: for *odd* composite $k = rs$ the factorization of clause (i) applies verbatim, so that $D_9(m)$, for instance, is *never* prime. This was checked directly for $k = 9, 15, 21, 25, 27, 33$ —no prime value of D_k occurs up to $m = 2000$, and the predicted divisor $D_r(m) \mid D_k(m)$ is present identically—and it is the same algebraic mechanism, seen from the ladder side, that Theorem 1 exhibits for cubes. The family generalizes the cubic case $k = 3$ of Conjecture 22.

Computational checks. The even-rung theorem was checked numerically to 10^8 (no fourth power is $p + j^4$); the three verifiable Bateman–Horn lanes $k = 2, 3, 5$ (constants 2 exactly, 3.36181, and 3.67770 ± 0.007 , the quartic product computed by brute-force root counts to 10^5) all sit within $|z| < 0.6$: at bound 10^7 the lane $k = 2$ gives observed 1,270,606 against predicted 1,270,902.8 (ratio 0.9998, $z = -0.26$); at 10^6 the lane $k = 3$ gives 126,826 against 126,641.6 (ratio 1.0015, $z = +0.52$); and at 8×10^5 the lane $k = 5$ gives 56,925 against 57,021.3 (ratio 0.9983, $z = -0.40$).

Conjecture 24 (The alternating cyclotomic chain; apparently new). *There are infinitely many primes p such that, writing $u = \Phi_3(p) = p^2 + p + 1$, the three integers*

$$p, \quad u = p^2 + p + 1, \quad \Phi_6(u) = u^2 - u + 1 = p^4 + 2p^3 + 2p^2 + p + 1$$

are all prime, with $\#\{p \leq x\} \sim C_6^{\text{ch}} I(x)$ for the Bateman–Horn system $\{x, x^2 + x + 1, x^4 + 2x^3 + 2x^2 + x + 1\}$, $C_6^{\text{ch}} = 3.6143 \pm 0.011$. The first chains are (2, 7, 43) and (3, 13, 157).

This is the repunit analogue of a Cunningham chain: a prime base p , its length-3 repunit $u = (p^3 - 1)/(p - 1)$, then $(u^3 + 1)/(u + 1)$. The chain must *alternate* between Φ_3 and Φ_6 : the naive iterate $\{p, \Phi_3(p), \Phi_3(\Phi_3(p))\}$ is *inadmissible*—if $p \equiv 2 \pmod{3}$ then $u \equiv 1 \pmod{3}$, so $3 \mid \Phi_3(u)$ always, while $p \equiv 1 \pmod{3}$ forces $3 \mid u$, so the admissibility check rejects the naive chain outright. The uniqueness claim is made over a specified construction space: among length-3

chains $\{x, \Phi_j(x), \Phi_k(\Phi_j(x))\}$ with $j, k \in \{3, 6\}$ extending the repunit sub-chain (i.e. $j = 3$), the mod-3 arithmetic above forces $k = 6$: alternation is the unique admissible continuation. (Dropping the repunit anchor admits one sibling, the pure- Φ_6 chain $\{x, \Phi_6(x), \Phi_6(\Phi_6(x))\}$ on the complementary lane $x \equiv 1 \pmod{3}$; since $\Phi_6(x) = \Phi_3(x - 1)$, its second element is again a repunit value, and we regard the classification of all admissible words in the two-letter alphabet $\{\Phi_3, \Phi_6\}$ —which words are obstruction-free at every prime, and whether they form a regular language—as a natural open programme.) The length-2 sub-chain $\{p, p^2 + p + 1\}$ is the classical cyclotomic Sophie Germain statement, included here as an attributed calibration benchmark ($C = 1.521661 \pm 0.0005$, catalogue value 1.5217315..., OEIS A188596; verified at $z = -1.06$ at 10^7): infinitely many prime bases have prime repunits of length 3, the base-independent home of the phenomenon whose base-2 shadow is the Mersenne sequence. The length-3 chain stated here is, as far as we can find, unstated, and its quartic layer makes it the highest-degree system in the suite.

Computational checks. At bound 10^7 : observed 1,362 against predicted 1,357.2, ratio 1.0035, $z = +0.13$. The chain census by decade is 4, 11, 41, 224, 1362 at $N = 10^3, \dots, 10^7$ (a probable-prime count for the quartic layer beyond $p \approx 2 \times 10^6$; z from +0.40 to +0.13, no drift), the quartic layer’s singular series computed by brute-force root counts to 10^5 ; the retained sub-chain benchmark $\{p, p^2 + p + 1\}$ stands at 33,661 against 33,855.5 at 10^7 ($z = -1.06$). Independent recomputation of that calibration member, the Sophie Germain sub-chain, gave 1.521730, with primes to 10^9 —inside our stated wobble—and an independent count to 3×10^7 with final ratio 0.9962.

Conjecture 25 (Twin cyclotomic bases: the complete quadratic family; no prior trace found). *For the three cyclotomic polynomials of degree 2 (indices k with $\varphi(k) = 2$, i.e. $k \in \{3, 4, 6\}$), consider $\Phi_k(n)$ and $\Phi_k(n + 1)$ simultaneously prime. Then: (i) [parity obstruction, elementary] for $k = 4$ one member of the pair $(n^2 + 1, n^2 + 2n + 2)$ is even for every n , so the only prime pair is $n = 1$, giving $(2, 5)$ —the same obstruction that forbids consecutive $n^2 + 1$ primes; (ii) [collapse, elementary] $\Phi_6(x) = \Phi_3(x - 1)$ identically, so the $k = 6$ pair at index n is the $k = 3$ pair at index $n - 1$: the family contains exactly one nontrivial statement; (iii) [conjecture, the live instance] there are infinitely many n for which the length-3 repunits in bases n and $n + 1$ are simultaneously prime: $\#\{n \leq N : \Phi_3(n) = n^2 + n + 1 \text{ and } \Phi_3(n + 1) = n^2 + 3n + 3 \text{ both prime}\} \sim C_5 I(N)$, with $C_5 = C(x^2 + x + 1, x^2 + 3x + 3)$.*

Both forms in (iii) are odd for every n (each is Φ_3 of an integer), and mod 3 each vanishes on exactly one class ($n \equiv 1$ and $n \equiv 0$ respectively), so the pair is admissible with $\omega(3) = 2$. Unlike an arbitrary shifted pair, the companion here is structurally forced: it is the same cyclotomic value at the consecutive base, making this the natural “twin” statement inside the cyclotomic-chain family of Conjecture 24. The family analysis (i)–(ii) is what a family *lift* of the statement uncovers, and both clauses were found by the admissibility computation itself: the $k = 4$ branch was rejected as inadmissible at $p = 2$ at singular-series time (both-odd count to 10^6 : zero, prime pairs: $n = 1$ only), and the $k = 6$ run returned counts *identical* to $k = 3$, which is the numerical shadow of the polynomial identity in (ii).

Computational checks. $C_5 = 2.964239 \pm 0.002$; at bound 10^7 , observed 33,274 against predicted 32,976.7, ratio 1.0090, $z = +1.64$.

Part II

Structural conjectures in five programmes

The twenty-five conjectures of this part are organized by mechanism rather than by the contamination calculus of Part I. Each names a canonical object and averaging law, identifies an arithmetic, geometric, spectral, or adelic source, and states a first decisive theorem and a failure mode.

6 Programme I: connected prime-pattern fields

Let $H \subset \mathbb{Z}$ be finite and admissible, and write

$$\Lambda_H(n) = \prod_{h \in H} \Lambda(n+h), \quad Y_H(x, L) = \sum_n w\left(\frac{n-x}{L}\right) (\Lambda_H(n) - \mathfrak{S}(H)),$$

where w is a fixed smooth compactly supported function and $\mathfrak{S}(H)$ is the Hardy–Littlewood singular series $\prod_p (1 - \nu_H(p)/p)(1 - 1/p)^{-|H|}$, with $\nu_H(p)$ the number of residue classes occupied by H modulo p . The window position x is sampled from $[X, 2X]$ with density $dx/(x \log 2)$ and $X^\varepsilon \leq L \leq X^{1-\varepsilon}$. For translated motifs $H_i + t_i$, their *exact overlap type* records every equality among shifted prime constraints. The connected singular series $\mathfrak{S}^c(K_1, \dots, K_r)$ is obtained by Möbius inversion over set partitions of the indexed tuple $(K_1, \dots, K_r)$.

Conjecture 26 (Connected motif generating functional). *Fix admissible motifs $H_1, \dots, H_m$ and put*

$$\mathcal{Z}_{X,L}(\mathbf{z}) = \mathbb{E}_x \exp\left(\sum_{i=1}^m z_i Y_{H_i}(x, L)\right).$$

For every fixed total degree R , the degree- $\leq R$ Taylor polynomial of $\log \mathcal{Z}_{X,L}$ has a uniform asymptotic expansion indexed by connected exact-overlap diagrams of at most R translated motifs. A diagram whose coincidence hypergraph connects the r translated sets $H_{i_1} + t_1, \dots, H_{i_r} + t_r$ contributes, at the leading order of its stratum,

$$W_\tau(L; \mathbf{t}) \mathfrak{S}\left(\bigcup_{j=1}^r (H_{i_j} + t_j)\right) \prod_v (\log X)^{m_v-1},$$

where $W_\tau(L; \mathbf{t}) = L \int_{\mathbb{R}} \prod_{j=1}^r w(u + t_j/L) du$ is the archimedean window-overlap weight of the diagram and m_v is the multiplicity of the prime constraint at v : any set partition that separates a coincidence loses at least one factor of $\log X$, so the Möbius alternation leaves the single-block moment as the leading term of a connected diagram. The same alternation cancels the leading terms of disconnected diagrams, whose contribution is carried by the connected singular series of their disjoint blocks at strictly smaller logarithmic order, reducing for the fully disjoint type to $\mathfrak{S}^c(H_{i_1} + t_1, \dots, H_{i_r} + t_r)$ with no logarithmic enhancement. The log powers are therefore partition dependent and are never factored across the alternation, and every prime-power counterterm is included in the local weight attached to its multiplicity. Ordering terms lexicographically by powers of L and $\log X$, the remainder after any fixed truncation is smaller than the last retained scale, uniformly in the mesoscopic range.

Significance. This is a multitype connected field theory for prime constellations. It simultaneously organizes covariance, odd moments, mixed motifs, overlap singularities, and lower-order prime-power effects. Montgomery–Soundararajan and Kuperberg provide the closest one-field moment frameworks, while constrained singular-series sums provide the nearest local input [19, 34, 35]. The new content is the complete connected multitype functional. The first decisive theorem is the full third-cumulant formula for two distinct pair motifs. A failure would reveal a term of $\log Z_{X,L}$ at a scale indexed by no connected exact-overlap diagram, a correlation source among the motif fields $Y_{H_i}(x, L)$ invisible to the Möbius alternation over set partitions.

Conjecture 27 (Complete overlap-renormalization filtration). *Fix a finite motif family $H_1, \dots, H_m$. Apply the degree-two specialization of Conjecture 26 and order all of its connected covariance strata by their asymptotic scales $s_1 \succ s_2 \succ \dots$ (lexicographically in powers of L , $\log X$, and the renormalized disjoint scales). In the mesoscopic range scales sharing the same total power of L and of logarithms differ only by powers of $\theta = \log L / \log X$ and merge into a single stratum whose limiting form is polynomial in θ . The filtration, the graded convergence, and the eigenvalue asymptotics below are asserted for every fixed $\theta \in (\varepsilon, 1 - \varepsilon)$. Recursively subtract the earlier strata and let $A_\nu = A_\nu(\theta)$ be the limiting covariance form at scale s_ν , asserted nondegenerate for every such θ , and put*

$$\mathcal{W}_\nu = \bigcap_{\mu < \nu} \ker A_\mu.$$

For every fixed truncation of this ordered diagram expansion, the spectral projections of the covariance matrix converge to the associated graded spaces $\mathcal{W}_\nu / \mathcal{W}_{\nu+1}$, and every eigenvalue whose first nonzero term occurs within the truncation is asymptotic to s_ν times a positive eigenvalue of the induced form A_ν . Vectors annihilating all retained forms pass canonically to the next connected diagram scales of Conjecture 26.

Consequently every asymptotic variance scale of a finite motif statistic is generated by a connected overlap or renormalized disjoint diagram from Conjecture 26, and no scale external to that diagrammatic expansion occurs. The filtration is invariant under replacing the motif family by one with isomorphic exact-overlap incidence algebra and identical local Euler weights.

Significance. The claim classifies all mechanisms by which a finite linear combination of prime motifs can become rigid. It predicts a canonical renormalization filtration, not merely one variance formula, and gives an algorithm for constructing observables that expose deeper arithmetic strata. A counterexample would identify a genuinely new source of rigidity outside overlap and disjoint correlation. The first theorem is a three-motif example in which one overlap form has a rational kernel and the next form is nondegenerate on it.

Conjecture 28 (Regularized mesoscopic local-spectral trace formula). *Fix motifs H_i, H_j , a Schwartz function ϕ with $\phi(0) = 0$, an even cutoff $\eta \in C_c^\infty(\mathbb{R})$ with $\eta = 1$ near the origin, and an even mollifier $\kappa \in C_c^\infty(\mathbb{R})$ of integral one. Let $Q_{ij}^{\text{loc}}(\phi; L)$ be the exact-overlap-renormalized singular-series form*

$$\sum_{t \in \mathbb{Z}} \left[L \int_{\mathbb{R}} w(u) w(u + t/L) du \right] \phi(t/L) \mathfrak{S}^c(H_i, H_j + t).$$

Let $\Lambda_{T,\eta}^\#$ be the Weil explicit-formula distribution in the logarithmic variable, with each nontrivial zero of ordinate γ multiplied by $\eta(\gamma/T)$ and with the pole, trivial-zero, archimedean,

and prime-power terms kept as separately named summands. For $\delta > 0$, put

$$\Lambda_{T,\eta,\delta}^\# = \Lambda_{T,\eta}^\# *_{\log} \kappa_\delta, \quad \kappa_\delta(u) = \delta^{-1} \kappa(u/\delta).$$

This is a smooth function, so products of its shifted copies are unambiguous. Define

$$Q_{ij}^{\text{spec}}[\eta, \kappa] = Q_{ij}^{\text{spec}}(\phi; X, L; T, \delta, \eta, \kappa)$$

by inserting these mollified copies into the smoothed motif covariance, applying the same exact-overlap Möbius projection as on the local side, and subtracting the explicitly specified pole, diagonal, trivial-zero, archimedean, and prime-power counterterms before the mollifier is removed.

For every sufficiently large fixed A, B , take

$$T = (X/L)(\log X)^A, \quad \delta = (L/X)(\log X)^{-B}.$$

Then

$$Q_{ij}^{\text{spec}}(\phi; X, L; T, \delta, \eta, \kappa) - Q_{ij}^{\text{loc}}(\phi; L) = o(V_{ij}(X, L))$$

uniformly for $X^\varepsilon \leq L \leq X^{1-\varepsilon}$ and for ϕ in bounded Schwartz sets, where V_{ij} is the first nonzero disjoint covariance scale. The same limit holds jointly for every fixed matrix of motifs and tests. Changing A, B, η , or κ within the stated admissible class changes the renormalized spectral form by $o(V_{ij})$, so the limiting trace functional is canonical.

Significance. The spectral side is a finite smooth expression before any products are taken, and regularization independence is part of the conjecture rather than an implicit convention. The statement extends the Goldston–Montgomery and Chan prime/zero equivalence from one prime-counting field to products of shifted von Mangoldt functions [54, 33]. Failure can occur in two mathematically different ways: a missing mesoscopic covariance functional, or a genuine regularization anomaly. The first decisive theorem is the pair-motif covariance with one factor unshifted and two independent admissible mollifiers.

Conjecture 29 (Anchored arithmetic polymer expansion for prime motifs). *Fix an admissible motif H of size k . A finite set $A \subset \mathbb{Z}$ of occurrence starts is a polymer when $0 \in A$, the overlap graph on A is connected, and $U_A = \bigcup_{t \in A} (H + t)$ is admissible. Define its anchored codimension*

$$d_H(A) = |U_A| - |H|;$$

this is the number of new prime constraints beyond the occurrence anchored at 0. In the regularized spectral model of Conjecture 28, let Z be the limiting low-mode vector (the joint limit, asserted as part of this conjecture, of the mode integrals of the Conjecture 28 functional against frequencies below a fixed cutoff K on the L -scale, the Conjecture 28 limits taken first and $K \rightarrow \infty$ last), let $\lambda(t \mid Z)$ be the conditional one-point occurrence intensity, and set

$$\lambda_H^0 = \frac{\mathfrak{S}(H)}{(\log X)^k}, \quad \tilde{\lambda}(t \mid Z) = \lambda(t \mid Z) / \lambda_H^0.$$

For every fixed polymer A , uniformly for bounded diameter,

$$\frac{(\log X)^{d_H(A)} \rho^{(|A|)}(A \mid Z)}{\lambda(0 \mid Z) \prod_{t \in A \setminus \{0\}} \tilde{\lambda}(t \mid Z)} \longrightarrow \frac{\mathfrak{S}(U_A)}{\mathfrak{S}(H)}$$

in probability in the successive limits defining Z . Thus the conditional environment supplies only the one-point intensity, while every connected Palm activity is graded by $(\log X)^{-d_H(A)}$ and weighted by the complete-union singular series.

If admissible overlapping translates exist, put

$$\delta(H) = \min_{\substack{t \neq 0 \\ H \cup (H+t) \text{ admissible}}} (|H \cup (H+t)| - |H|), \quad \mathcal{T}_H = \{t \neq 0 : H \cup (H+t) \text{ admissible}, |H \cup (H+t)| - |H| = \delta(H)\}$$

Then the cluster-start fraction has the first nontrivial expansion

$$\theta_X(H) = 1 - \frac{c_H}{(\log X)^{\delta(H)}} + o((\log X)^{-\delta(H)}), \quad c_H = \sum_{\substack{t \in \mathcal{T}_H \\ t < 0}} \frac{\mathfrak{S}(H \cup (H+t))}{\mathfrak{S}(H)}.$$

The set $\mathcal{T}_H$ is symmetric under $t \mapsto -t$, and the cluster-start fraction counts, for each occurrence, only whether it has an overlapping predecessor, so the sum is restricted to $t < 0$ and equals half the symmetric sum over $\mathcal{T}_H$. If no admissible overlap exists, $\theta_X(H) = 1 + o((\log X)^{-M})$ for every fixed M . Higher corrections are the connected Ursell sums (the signed connected sums of cluster-expansion theory) of anchored polymers ordered by $d_H(A)$. These finite- X Gibbs expansions are projectively consistent and subcritical, and after division by the low-zero intensity the cluster-start process is Poisson to each fixed order of this triangular expansion.

Significance. The relevant rarity is the number of new constraints after one occurrence is given, not the enhancement relative to a product of all one-point intensities. For two translates sharing s of the k prime positions, the Palm scale is $(\log X)^{-(k-s)}$. A universal $1/\log X$ correction therefore holds only for special motifs, such as pair motifs that admit an overlapping translate. The law keeps polymers of every size, supplies a canonical extremal-index exponent $\delta(H)$, and links Hardy–Littlewood constants, low-zero environments, Palm theory, and extreme motif gaps. For $H = \{0, 6\}$ one has $\delta(H) = 1$ and the pair-motif coefficient is recovered. The first decisive theorem is the two-element polymer law for one pair motif, the Palm limit at $A = \{0, t\}$ with $t \in \mathcal{T}_H$ together with the expansion of $\theta_X(H)$ at order $(\log X)^{-\delta(H)}$. A failure would show that the low-mode environment Z supplies more than the one-point intensity, so connected Palm activities would not be graded by $(\log X)^{-d_H(A)}$ and weighted by $\mathfrak{S}(U_A)$ alone.

Conjecture 30 (Topological expansion of non-Gaussian prime cumulants). *For every connected exact-overlap diagram in Conjecture 26, expand its connected singular series over the linking primes and form the linkage graph of each term, whose vertices are the motif occurrences and whose edges join two occurrences sharing a linking prime in that term. Let g be the first Betti number of the linkage graph. Once all larger coincidence strata have been subtracted, the total contribution of the terms of Betti number g is smaller than that of the spanning-tree terms with the same external motifs by at least $(\log L)^{-g}$. The complete leading coefficient at each logarithmic order is the sum of the linked-term weights with the corresponding Betti number. In particular, the first nonzero odd cumulant of a balanced motif statistic, a combination Y_c whose coefficient vector annihilates every coincidence-stratum form of Conjecture 27, is carried by incidence trees. For the single-prime field the fully spread connected part $\kappa_{2m+1}^{\text{sp}}$, defined by subtracting every coincidence stratum as in Conjecture 27, obeys*

$$\kappa_{2m+1}^{\text{sp}} = (c_m + o(1))L^m(\log L)^{m+1},$$

where c_m is the total connected tree weight. The raw cumulant adds the explicit coincidence diagrams of Conjecture 26, which carry powers of $\log X$ and are computable term by term, so the tree law is a statement about the spread diagrams, exactly the regime of Kuperberg's singular-series sums.

Significance. This predicts a topological genus expansion for prime statistics: every independent cycle costs a logarithmic order. Kuperberg's odd-moment conjecture is the one-field calibration, and the incidence-homology filtration with its motif-dependent coefficients is the new mechanism [34]. The first theorem is the case $m = 1$ for the single-prime field, the spread third-cumulant tree law $\kappa_3^{\text{SP}} = (c_1 + o(1))L(\log L)^2$ with its one-cycle terms suppressed by $(\log L)^{-1}$. A failure would show that logarithmic suppression is controlled by an arithmetic invariant not visible in diagram topology.

7 Programme II: arithmetic first-arrival fields and class groups

For a prime modulus q and $a \in \mathbb{F}_q^\times$, let $p(q, a)$ be the least prime in the class a and put

$$T_q(a) = \frac{\text{Li}(p(q, a))}{q-1}, \quad \mathcal{N}_{q,a} = \sum_{p \equiv a \pmod{q}} \delta_{\text{Li}(p)/(q-1)}.$$

The probability space chooses q uniformly among primes in $[Q, 2Q]$ and then chooses a uniformly from $\mathbb{F}_q^\times$.

Conjecture 31 (Connected first-arrival functional). *Let $w_{r,Q}^c$ be the connected factorial-cumulant measure of the random point process $\mathcal{N}_{q,a}$. For every fixed $r \geq 2$ there is a signed measure K_r , locally finite on the off-diagonal region $\{u \in \mathbb{R}_+^r : u_i \text{ distinct}\}$, such that, on that region (the diagonal strata carry strictly smaller normalizations, as in the stratification of Conjecture 26, and are excluded here),*

$$(\log Q)^{r-1} w_{r,Q}^c \implies K_r$$

vaguely, and K_1 denotes the limiting intensity measure, which in this clock is Lebesgue measure on $\mathbb{R}_+$. The measure K_r is obtained by averaging, over the modulus aspect, the connected Hardy–Littlewood densities of r primes constrained to one residue class. For every nonnegative compactly supported f in a fixed uniform ball on which the Laplace functional is analytic,

$$\log \mathbb{E} \exp(-\langle f, \mathcal{N}_{q,a} \rangle) = \sum_{r \geq 1} \frac{(-1)^r}{r!} \int \prod_{j=1}^r (1 - e^{-f(u_j)}) dw_{r,Q}^c(\mathbf{u}),$$

and the asymptotic expansion obtained by inserting the limits K_r is uniform to every fixed connected order. In particular the entire survival function, not merely its first correction, is determined by the hierarchy $(K_r)_{r \geq 2}$.

Significance. This is an arithmetic point-process theory of least primes. The coupon-collector model of Li–Pratt–Shakan supplies the nearest extremal analogy, but not a connected Laplace functional [36]. Truth would determine waiting times, occupancy covariance, cover times, and terminal clustering from one hierarchy. The first theorem is the existence and explicit evaluation of K_2 on compact off-diagonal sets. A failure would show that least primes in residue classes cluster through a mechanism not captured by modulus-averaged connected Hardy–Littlewood densities, a connected correlation of $\mathcal{N}_{q,a}$ living outside the hierarchy $(K_r)_{r \geq 2}$.

Conjecture 32 (Tested Gauss-polyspectral reciprocity for the least-prime field). *Set $T_q(0) = \text{Li}(q)/(q-1)$ and centre the resulting field f_q on all of $\mathbb{F}_q$. Let $\mathbf{A}_q$ and $\mathbf{M}_q$ be its complete additive and multiplicative Fourier coefficient vectors, including the explicit rank-one coordinate needed to record the value at 0. There is then an explicit invertible Gauss matrix U_q with*

$$\mathbf{M}_q = U_q \mathbf{A}_q.$$

Choose q uniformly among primes in $[Q, 2Q]$. For every finite family of degrees $r_1, \dots, r_m$ and uniformly bounded fibrewise test tensors $\Phi_{j,q}$, define the scalar additive observables

$$X_{j,Q}(q) = \langle \Phi_{j,q}, \mathbf{A}_q^{\otimes r_j} \rangle$$

and the transported multiplicative observables

$$Y_{j,Q}(q) = \left\langle (U_q^{-1})^{*\otimes r_j} \Phi_{j,q}, \mathbf{M}_q^{\otimes r_j} \right\rangle.$$

For every modulus, $X_{j,Q}(q) = Y_{j,Q}(q)$, so all joint cumulants under the random-modulus law satisfy the exact finite identity

$$\text{Cum}_Q(X_{1,Q}, \dots, X_{m,Q}) = \text{Cum}_Q(Y_{1,Q}, \dots, Y_{m,Q}).$$

This is the probability-space-correct form of Gauss transport. It makes no reference to a cumulant of a deterministic coefficient vector.

The conjectural content is that, after normalization by the first nonzero connected scale, tests supported on additive zero-sum tensors converge to contractions of the Fourier transforms of the arrival kernels K_r from Conjecture 31. Transporting those tests fibrewise by the exact Gauss matrices and only then taking the modulus average gives the same limits as the corresponding multiple explicit-formula correlations of Dirichlet L -function zeros. The two descriptions determine the same continuous multilinear functional on the completion of the tested tensor class.

Significance. The elementary input is the finite Gauss matrix. The new claim is a complete higher-order tomography principle for the random-modulus least-prime field. Every scalar statistic is compared on a genuine probability space, while the varying finite fields are handled fibrewise before averaging. Failure would isolate a tensor class on which local same-class prime correlations and character-zero correlations disagree even though the underlying finite transforms are exact. The first theorem is the quadratic identity for tests localized to one additive lag and its transported character kernel.

Conjecture 33 (Rubinstein–Sarnak terminal chaos dichotomy). *Assume GRH and a q -aspect linear-independence/random-phase hypothesis for the low zeros of primitive Dirichlet L -functions. Let x_q satisfy $\text{Li}(x_q)/(q-1) = \log(q-1)$ and let $G_{q,A}(a)$ be the centred, variance-normalized low-zero field obtained from characters $\chi \neq \chi_0$ and zeros $|\gamma| \leq (\log q)^A$, with the standard compensator $-\frac{1}{2} \text{Var } G_{q,A}(a)$. Define the random empirical chaos measure*

$$\mathcal{M}_{q,A}(dg) = \frac{1}{q-1} \sum_{a \in \mathbb{F}_q^\times} \exp\left(G_{q,A}(a) - \frac{1}{2} \text{Var } G_{q,A}(a)\right) \delta_{G_{q,A}(a)}(dg).$$

Let b_Q be the smallest level with $\mathbb{E}_q \#\{a : T_q(a) > b_Q\} \leq 1$. In the successive limits $Q \rightarrow \infty$ and then $A \rightarrow \infty$, the pair consisting of $\mathcal{M}_{q,A}$ and

$$\sum_a \delta_{(G_{q,A}(a), T_q(a) - b_Q)}$$

converges jointly. Conditionally on the limiting chaos measure $\mathcal{M}$, the terminal cluster-start process is Poisson with directing measure

$$e^{-x} \mathcal{M}(dg) dx.$$

At finite modulus the cluster-start thinning enters through the terminal extremal index $\theta_{\text{occ}}(q)$, the analogue for the last classes of the cluster-start fraction $\theta_X(H)$ of Conjecture 29, together with the terminal connected decorations supplied by the hierarchy of Conjecture 31. Both corrections enter at the first nonzero connected scale and vanish in the successive limit, so $\theta_{\text{occ}}(q) \rightarrow 1$ and the limiting directing measure carries no cluster factor.

The nature of the directing measure is governed by the covariance-energy statistic

$$\mathfrak{E}_{q,A} = \frac{1}{(q-1)^2} \sum_{a,b \in \mathbb{F}_q^\times} \text{Cov}(G_{q,A}(a), G_{q,A}(b))^2.$$

If $\mathfrak{E}_{q,A} \rightarrow 0$ along the successive limit, $\mathcal{M}$ self-averages to the deterministic exponentially tilted Gaussian law and the terminal process is an ordinary decorated Poisson process. If $\liminf \mathfrak{E}_{q,A} > 0$ and the Gaussian-chaos second moments are uniformly integrable, $\mathcal{M}$ is nondegenerate and the limit is a genuine Cox process. No nondegeneracy is asserted without this criterion.

Significance. Rubinstein–Sarnak theory supplies the fixed-modulus prime-race environment [6]. The new object is its terminal first-arrival chaos measure in the modulus aspect. The statement now treats self-averaging as a serious competing model rather than assuming a Cox limit by terminology. It predicts the complete last class process, its prime-race marks, and an explicit spectral criterion separating a deterministic environment from persistent random mixing. The first theorem is the self-averaging branch at one fixed cutoff A , where $\mathfrak{E}_{q,A} \rightarrow 0$ forces the deterministic exponentially tilted Gaussian limit of $\mathcal{M}_{q,A}$ and an ordinary decorated Poisson law for the terminal cluster-start process.

Conjecture 34 (Nonlinear spectral response calculus for arithmetic first arrivals). *Let a sequence of modulus ensembles have a small explicit-formula perturbation of the arrival intensity*

$$V_Q(a, t) = \sum_{\chi \in \mathcal{C}_Q} v_{Q,\chi}(t) \chi(a), \quad \|V_Q\|_T \rightarrow 0,$$

where $\mathcal{C}_Q$ has uniformly bounded cardinality and, on every fixed interval $[0, T]$, the canonical norm is $\|V\|_T = \sup_{a, 0 \leq t \leq T} |V(a, t)|$. Assume that the connected hierarchy of Conjecture 31 satisfies an exponential cluster bound there, so its Laplace functional is analytic in a fixed $\|\cdot\|_T$ -ball. If $S_Q(a, t)$ and $S_Q^{(0)}(t)$ are the perturbed and ordinary survival functions, then for every fixed R

$$\log \frac{S_Q(a, t)}{S_Q^{(0)}(t)} = \sum_{m=1}^R \mathcal{R}_m[V_Q^{\otimes m}](a, t) + o(\|V_Q\|_T^R)$$

uniformly on compact t -ranges. The causal Volterra operator $\mathcal{R}_m$ is the universal linked-cluster contraction of the full hierarchy $(K_r)_{r \geq 1}$ with m marked perturbation insertions, and truncating the connected hierarchy at order J gives a convergent approximation as $J \rightarrow \infty$.

If the input character support is $\mathcal{C}_Q$, the m th response is supported exactly on products of at most m input characters (subject only to coincidences and the trivial character). In particular the linear response preserves character rank, while the first new harmonics occur at quadratic order with coefficients determined by connected three-point arrival correlations.

Significance. This is a nonlinear transfer calculus from zeros to first-hit statistics. A Siegel zero, a cluster of low zeros, and an ordinary prime-race bias become different inputs to the same response operators. The exceptional-zero rank-one statement is a special case, while the second-order harmonics provide a direct falsification test. The first theorem is the Fréchet derivative $\mathcal{R}_1$ from the two-point occupancy kernel.

Conjecture 35 (Local-information capacity of odd class groups). *Fix an odd prime ℓ . Sample negative fundamental discriminants $D \in [-2Y, -Y]$, conditioned on a fixed coarse two-primary invariant. For a set of odd primes S_Y , put $M_Y = \prod_{p \in S_Y} p$, let $\Sigma_Y(D) = ((D/p))_{p \in S_Y}$ be the attainable sign vector, and let H_Y be the logarithm of the number of attainable cells. Let $\delta_Y(\sigma)$ be the exact CRT density of the sign cell after imposing quadratic reciprocity and the fixed two-primary data (for a single odd unramified prime the density of each sign is $p/(2(p+1))$), and an unconditioned cell density is the corresponding product), and let $\mathcal{D}_Y$ be the ambient discriminant family. Write*

$$\Delta_Y(S_Y) = \sup_{\sigma} \left| \frac{\#\{D \in \mathcal{D}_Y : \Sigma_Y(D) = \sigma\}}{|\mathcal{D}_Y| \delta_Y(\sigma)} - 1 \right|$$

over attainable cells with $|\mathcal{D}_Y| \delta_Y(\sigma) \geq \log Y$.

There is a sequence $\varepsilon_Y \rightarrow 0$, independent of S_Y in the subcritical range, such that uniformly over those cells

$$d_{\text{TV}}(\mathcal{L}(\text{Cl}(D)[\ell^\infty] \mid \Sigma_Y = \sigma, 2\text{-data}), \mathcal{L}_{\text{CL}, \ell}) \leq \varepsilon_Y + O(\Delta_Y(S_Y)),$$

and the analogous bound holds for every fixed Cohen–Lenstra moment. In particular, for every $\varepsilon > 0$, the conclusion is uniform whenever

$$H_Y \leq (1 - \varepsilon) \log Y, \quad \log M_Y \leq (\tfrac{1}{2} - \varepsilon) \log Y, \quad \Delta_Y(S_Y) = o(1).$$

Since every odd prime contributes at most $\log 3 \leq \log p$ of cell entropy, $H_Y \leq \log M_Y$ throughout, so the conductor constraint is the binding one in the displayed range, while the entropy governs the impossibility clause below. At $H_Y \geq (1 + \varepsilon) \log Y$ no uniform statement is possible. Up to the conductor scale at which Δ_Y becomes macroscopic, every first-order failure of local independence factors through this discriminant-cell discrepancy, and there is no earlier odd-class-group obstruction.

Significance. Wood’s framework treats fixed local conditions [37]. The new statement proposes the full information capacity of an odd class group under a growing local sigma-algebra, with the entropy governing impossibility and the conductor governing equidistribution. It would transfer directly to Selmer groups, ray class groups, and unramified extension statistics. The first theorem is uniformity for $|S_Y| \rightarrow \infty$ with $M_Y = Y^{o(1)}$. The discrepancy half is not the obstruction: for cells of macroscopic mass the equidistribution of discriminants is elementary for $M_Y \leq Y^{1/2-\varepsilon}$, and squarefree-in-progressions technology carries it to $Y^{2/3-\varepsilon}$, so the conjectural content lies entirely in the class-group law over the growing cell family.

8 Programme III: finite logarithms, regulators, and algebraic tori

For $p \nmid a$, write

$$q_p(a) = \frac{a^{p-1} - 1}{p} \pmod{p}.$$

For a generating tuple $g = (g_1, \dots, g_s)$, let $R(g)$ be its multiplicative relation lattice and let $\mathbb{T}_g \subset (\mathbb{R}/\mathbb{Z})^s$ be the identity component of its annihilator. Its dimension is the saturated multiplicative rank. Write $e(x) = e^{2\pi i x}$.

Conjecture 36 (Kummer–Haar law on relation tori). *Let $\Gamma \leq \mathbb{Q}^\times$ be finitely generated and torsion-free, saturated modulo torsion (if $x^n \in \Gamma \cdot \{\pm 1\}$ for some $n \geq 1$ then $x \in \Gamma \cdot \{\pm 1\}$), and let g be any ordered generating tuple. As p varies in every fixed compatible cyclotomic–Kummer Chebotarev class, the vectors*

$$\frac{(q_p(g_1), \dots, q_p(g_s))}{p} \in (\mathbb{R}/\mathbb{Z})^s$$

equidistribute with Haar measure on $\mathbb{T}_g$. More strongly, for every finite collection of compatible Chebotarev restrictions, the conditional empirical measures converge to the Haar disintegration on the corresponding components, and the support is contained in no proper closed coset of $\mathbb{T}_g$. The law is invariant under changing generators and depends on Γ only through its saturation and the fixed Kummer conditioning data.

Significance. Katz formulated horizontal Wieferich equidistribution principles for algebraic groups, and Shparlinski proved estimates in different horizontal and averaged regimes [38, 39]. The new content here is the exact saturated relation torus, full generator invariance, and disintegration under fixed Kummer classes. It turns multiplicative rank into a geometric support theorem rather than a heuristic count of coordinates. The first decisive theorem is the Haar law on $\mathbb{T}_g$ for one rank-one saturated Γ presented by two dependent generators, restricted to a single fixed compatible cyclotomic–Kummer Chebotarev class. A failure would show that the vectors $q_p(g_i)/p$ remember more of Γ than its saturation, concentrating on a proper closed coset of $\mathbb{T}_g$ that no relation in $R(g)$ predicts.

Conjecture 37 (Rank-dimensional large sieve for finite logarithms). *Choose a Smith-normal-form basis of the character lattice $X^*(\mathbb{T}_g) = \mathbb{Z}^s/R(g)^{\text{sat}}$. There is a constant $\delta_\Gamma > 0$ such that, for every $\varepsilon > 0$, every $M \leq X^{\delta_\Gamma}$, and all complex weights a_p supported on $X < p \leq 2X$,*

$$\sum_{\substack{m \in X^*(\mathbb{T}_g) \\ \|m\|_\infty \leq M}} \left| \sum_{X < p \leq 2X} a_p e\left(\frac{\langle m, q_p(g) \rangle}{p}\right) \right|^2 \ll_{\Gamma, \varepsilon} (M^{\text{rank } \Gamma} + \pi(X)) X^\varepsilon \sum_p |a_p|^2.$$

The same inequality holds after every fixed compatible cyclotomic–Kummer restriction, and for matrix-valued weights with the Hilbert–Schmidt norm. The exponent δ_Γ is invariant under changing the generating tuple.

Significance. This is the quantitative mechanism behind Conjecture 36. It predicts that saturated rank is not only the support dimension but the exact dual dimension in a prime-varying large sieve. A proof for any positive power range would control genuine shrinking targets and would be far beyond currently available fixed-frequency tests. Failure would reveal an unrecognized spacing or energy obstruction among finite-logarithm vectors.

Conjecture 38 (Mesoscopic-to-lattice shrinking-target transition). *Let $\Gamma \leq \mathbb{Q}^\times$ be torsion-free of rank r and saturated modulo torsion, and choose a basis of Γ , identifying its relation torus with $(\mathbb{R}/\mathbb{Z})^r$. For integers w_p with $1 \leq w_p = o(p)$, let $E_p(w)$ be the event that every coordinate of the Fermat-quotient vector has a representative of absolute value at most w_p .*

Mesoscopic universality. If $w_p \rightarrow \infty$, then in every range in which the Haar mass

$$\lambda(X) = \sum_{p \leq X} \left(\frac{2w_p + 1}{p} \right)^r$$

tends to infinity, the count of $E_p(w)$ is asymptotic to $\lambda(X)$. On prime blocks with bounded Haar mass, the randomly translated event process converges to a Poisson process with that mass.

Lattice transition. For every fixed finite target set $B \subset \mathbb{Z}^r$, there is an arithmetic regulator intensity $\kappa_{\Gamma, B} \geq 0$ such that

$$\#\{p \leq X : q_p|_{\Gamma} \in B\} = (\kappa_{\Gamma, B} + o(1)) \sum_{p \leq X} p^{-r}$$

whenever the sum diverges, while only finitely many such primes occur when $r \geq 2$. As a separately falsifiable genericity clause, for Kummer-generic Γ (the Kummer fields $\mathbb{Q}(\zeta_m, \Gamma^{1/m})$ have the maximal degree $\varphi(m)m^r$ for every m) and $B = \{0\}$ one has $\kappa_{\Gamma, B} = 1$. The transition from Haar universality to the arithmetic intensity occurs only when w_p remains bounded.

Significance. This separates two universality classes that a single undifferentiated model conflates. Growing targets are controlled by Haar geometry, while exact Wieferich targets can carry a genuinely arithmetic regulator intensity. Gras's much rarer fixed-base model and the Haar model therefore disagree precisely at the lattice boundary [40, 41]. The first theorem is mesoscopic universality for $w_p = p^\eta$ with one rank-one base.

For a p -adic unit a , write $\langle a \rangle_p$ for the Teichmüller lift of its residue, and for a prime $\mathfrak{p} \mid p$ of a number field write $\tau_{\mathfrak{p}}$ for the Teichmüller lift in the completion at $\mathfrak{p}$. Define $q_{p,k}(a) = p^{-1} \log(a \langle a \rangle_p^{-1}) \pmod{p^k}$. The matrix entries below are the $\mathfrak{p}$ -adic analogues of $q_{p,k}$, and on $\mathbb{Q}$ with $k = 1$ one has $q_{p,1}(a) \equiv -q_p(a) \pmod{p}$, the Fermat quotient up to sign.

Conjecture 39 (Global-lattice horizontal p -adic regulator-matrix law). *Let $K/\mathbb{Q}$ be a fixed Galois number field with group G , and let $\Gamma \leq \mathcal{O}_K^\times$ be torsion-free and saturated modulo $\mu(K)$, with ordered basis $\epsilon_1, \dots, \epsilon_s$. Define the fixed global relation lattice*

$$\mathcal{R}_{\Gamma} = \left\{ (c_{\sigma j}) \in \mathbb{Z}^{G \times s} : \prod_{\sigma \in G} \prod_{j=1}^s \sigma(\epsilon_j)^{c_{\sigma j}} \in \mu(K) \right\}.$$

Let $\mathbb{T}_{\Gamma} \subset (\mathbb{R}/\mathbb{Z})^{G \times s}$ be its annihilator. This lattice contains the norm and Galois relations and is independent of the chosen basis up to the natural integral change of coordinates.

For a rational prime p splitting completely in K , label the primes above p by G , use the Teichmüller lift in each completion, and form the depth- k matrix

$$L_{p,k}(\Gamma) = \left(p^{-1} \log_{\mathfrak{p}}(\epsilon_j \tau_{\mathfrak{p}}(\bar{\epsilon}_j)^{-1}) \pmod{p^k} \right)_{\mathfrak{p} \mid p, 1 \leq j \leq s}.$$

It lies in the annihilator of $\mathcal{R}_{\Gamma} \otimes \mathbb{Z}/p^k \mathbb{Z}$. As p varies through any fixed compatible Frobenius class, $L_{p,1}/p$ equidistributes with Haar measure on $\mathbb{T}_{\Gamma}$, and the depth- k matrices are Haar at every fixed depth k , with the depth- $(k+1)$ law pushing forward to the depth- k law under reduction (the moduli vary with p , so this is a per-depth statement with compatible marginals, not Haar on one fixed group).

Let $\mathcal{D}$ be a basis-invariant determinantal or Smith stratum in this projective module, and let $\mu_{p,k}(\mathcal{D})$ be its exact local Haar mass. The primes for which $L_{p,k} \in \mathcal{D}$ have counting law governed by $\sum_{p \leq X} \mu_{p,k}(\mathcal{D})$. For a determinantal or Smith stratum of codimension c , whose point counts are polynomial in p , $\mu_{p,1}(\mathcal{D}) = p^{-c} + O(p^{-c-1})$, and codimension one has a Poisson point process in $\log \log p$ after uniform random translation, while codimension at least two has a summable large-prime tail together with the exact small-prime atoms.

Significance. The horizontal state space is one fixed global torus rather than a relation module that changes with p . The conjecture turns mod- p Leopoldt defects into a Galois-equivariant random-matrix problem, with exact Smith strata retaining higher depth. Existing mod- p Leopoldt and random-matrix theories provide neighbouring languages [41, 42, 43]. Recent number-field Wieferich results strengthen the arithmetic motivation without supplying this horizontal matrix law [57, 58]. The first theorem is the rank-one split-prime law for a real quadratic field. A failure would show that the matrices $L_{p,k}(\Gamma)$ favour some Smith stratum beyond its exact local Haar mass, so mod- p Leopoldt defects would be governed by an arithmetic invariant outside the fixed global torus $\mathbb{T}_\Gamma$.

Conjecture 40 (Functorial horizontal finite logarithms on algebraic tori). *Let $T/\mathbb{Q}$ be an algebraic torus, let $\mathcal{T}$ be its smooth integral model away from a finite set, and let $\Gamma \leq T(\mathbb{Q})$ be finitely generated. Let S be the identity component of the Zariski closure of Γ . At a good unramified prime p , let $\tau_p(\bar{P})$ be the canonical prime-to- p torsion lift of the reduction of $P \in \Gamma$ and define*

$$\ell_p(P) = p^{-1} \log_T(P \tau_p(\bar{P})^{-1}) \pmod{p} \in \text{Lie}(S)(\mathbb{F}_p).$$

Here $\text{Lie}(S)(\mathbb{F}_p)$ is identified with $(\mathbb{Z}/p)^{\dim S}$ through a fixed integral basis of the Lie algebra of $\mathcal{T}$, with representatives in $[0, p)$; a different basis changes the matrix by a fixed element of $\text{GL}_{\dim S}(\mathbb{Z})$, which preserves the Haar law and the shrinking-target intensities. For any generators $P_1, \dots, P_s$ of Γ , the normalized matrix $(\ell_p(P_i)/p)_i$ equidistributes, in every compatible Frobenius class, with Haar measure on the real relation subtorus cut out jointly by the algebraic subgroup S and the integral relations among the P_i . For every morphism of tori $\phi : T \rightarrow T'$,

$$\ell_p(\phi(P)) = d\phi(\ell_p(P)),$$

and the horizontal Haar measures push forward under the induced map. Isogenies preserve the shrinking-target intensities after the explicit finite kernel correction.

Significance. Katz already proposed Wieferich equidistribution beyond $\mathbb{G}_m$ [38]. The additional content is a matrix-valued law for a finitely generated global subgroup, relation-subtorus support, fixed-Frobenius disintegration, and exact functoriality under morphisms and isogenies. It turns horizontal finite logarithms into a functor on the category of tori with arithmetic subgroups. The first decisive theorem is the Haar law for the group generated by a single point of a one-dimensional nonsplit torus T , in a single fixed compatible Frobenius class. A failure would show that the matrices $(\ell_p(P_i)/p)_i$ concentrate on a proper closed subset of the relation subtorus, an arithmetic constraint beyond the algebraic subgroup S and the integral relations among the P_i .

9 Programme IV: arboreal arithmetic dynamics

For a degree- d map f and basepoint a , write $K_n = \mathbb{Q}(f^{-n}(a))$, $G_n = \text{Gal}(K_n/\mathbb{Q})$, and let $\mu_n(C) = |C|/|G_n|$ on conjugacy classes. For primes $p \leq X$ unramified in K_n , let $\hat{\mu}_{n,X}$ be the

empirical Frobenius measure.

Conjecture 41 (Entropy–conductor profile of arboreal resolution). *For a degree- d map f and level n , let $G_{n,f}$ be the arboreal quotient and $\mu_{n,f}$ its Haar measure on conjugacy classes. For N independent samples from $\mu_{n,f}$ define the exact sampling obstruction*

$$\mathcal{S}_{n,f}(N) = \mathbb{E} d_{\text{TV}} \left(N^{-1} \sum_{j=1}^N \delta_{Z_j}, \mu_{n,f} \right).$$

For each nontrivial irreducible Artin representation ρ of $G_{n,f}$, let $q(\rho)$ be its Artin conductor and define the GRH conductor energy

$$\mathcal{A}_{n,f}(X) = \frac{\log X}{\sqrt{X}} \left(\sum_{\rho \neq 1} (\dim \rho)^2 (\log q(\rho) + \dim(\rho) \log X)^2 \right)^{1/2}.$$

This is a canonical character-theoretic replacement for a single discriminant proxy. For a parameter family $\mathcal{F}(B)$ put

$$n_{\mathcal{F}}^*(B, X; \eta) = \max \left\{ n : \mathbb{E}_{f \in \mathcal{F}(B)} [\mathcal{S}_{n,f}(\pi(X)) + \mathcal{A}_{n,f}(X)] \leq \eta \right\}.$$

Assume the generic arboreal image is open, exceptional parameters have density zero, Artin holomorphy and GRH hold at the levels considered, and the family is in the large-sieve range of Conjecture 42 below. Then the actual empirical resolution depth, defined by the same total-variation threshold with prime Frobenius samples, differs from $n_{\mathcal{F}}^(B, X; \eta)$ by $O_{\mathcal{F}, \eta}(1)$.*

More precisely, suppose, uniformly for a density-one set of parameters, that the class measures have a limiting complete Rényi profile

$$\psi_{\mathcal{F}}(\alpha) = \lim_{n \rightarrow \infty} d^{-n} \log \sum_C \mu_{n,f}(C)^\alpha$$

uniformly for α in a fixed interval containing $[1/2, 2]$, and the corresponding Artin-conductor energies have finite positive exponential profiles on the same d^n scale. Then

$$n_{\mathcal{F}}^*(B, X; \eta) = \log_d \log X + O_{\mathcal{F}, \eta}(1).$$

Two families of the same degree with boundedly different complete Rényi and conductor profiles have resolution depths differing by $O(1)$, and in particular a fixed-index open subgroup changes only the bounded shift.

Significance. The observable depth is defined through the exact nonuniform sampling law and a representation-by-representation conductor energy. The $\log_d \log X$ scale follows from explicit Rényi and conductor profiles on the natural d^n complexity scale. Large arboreal images and effective Chebotarev are inputs [44, 45, 46]. Recent work on fixed-point profiles shows why the full class-measure tail, rather than group order alone, matters [59]. The first theorem is bounded-shift stability for one explicit index-two subgroup of the binary wreath tower. A failure would show that resolution depth responds to a family invariant beyond the complete Rényi profile $\psi_{\mathcal{F}}(\alpha)$ and the conductor energy $\mathcal{A}_{n,f}(X)$, so two families with matching profiles could separate by an unbounded shift.

Conjecture 42 (Arboreal family large sieve and cumulant independence). *Let $\mathcal{F} \rightarrow \mathbb{A}^1$ be a generically finite-index arboreal family. For each level n and irreducible representation ρ of the generic quotient G_n , let $\mathcal{V}_{n,\rho}$ be the associated middle-extension sheaf on the good parameter locus (the canonical extension of the corresponding local system across the bad parameters). Define the geometric conductor budgets*

$$\mathfrak{C}_{n,r} = \sum_{\rho_1, \dots, \rho_r} \left(\prod_{j=1}^r \dim \rho_j \right)^2 \operatorname{cond} \left(\boxtimes_{j=1}^r \mathcal{V}_{n,\rho_j} \right)^2, \quad \mathfrak{C}_n = \mathfrak{C}_{n,1},$$

where the external tensor product is taken on the CRT parameter space for distinct primes, and the conductor includes rank, the number of singular points, and the tame-drop and Swan terms (the tame and wild ramification invariants at those points).

Square-root tier. For every $\varepsilon > 0$ and all complex coefficients $\alpha_{p,\rho}$,

$$\sum_{|c| \leq B} \left| \sum_{p \leq X} \sum_{\rho \neq 1} \alpha_{p,\rho} \operatorname{tr} \rho(\operatorname{Frob}_{p,c}) \right|^2 \ll_{\varepsilon, \mathcal{F}} (B + X^2 \mathfrak{C}_n^{1+\varepsilon}) \sum_{p,\rho} |\alpha_{p,\rho}|^2,$$

uniformly when $X^2 \mathfrak{C}_n^{1+\varepsilon} \leq B^{1-\varepsilon}$, after deleting the singular fibres.

Connected tier. For every fixed $r \geq 3$, centred trace functions at distinct primes have family-averaged connected cumulants with square-root cancellation in the parameter-family count, uniformly in the range $X^r \mathfrak{C}_{n,r}^{1+\varepsilon} \leq B^{1-\varepsilon}$, and equivalently, after normalization, every fixed connected Frobenius cumulant tends to zero. An exponential connected-moment bound makes this convergence uniform in r over the range needed for occupancy functionals.

Consequently, whenever the analytic Chebotarev error is $o(1)$ but the iid sampling functional of μ_n is nonvanishing, the complete conjugacy-cylinder count vector, averaged over parameters, has the nonuniform multinomial occupancy law of independent samples from μ_n , including its missing-mass, collision, and total-variation profile. The multinomial conclusion is derived from the connected tier, not from the L^2 bound alone.

Significance. This separates the two analytic inputs that a single-tier statement compresses into one claim. The square-root tier is a growing-monodromy large sieve in the spirit of Kowalski [47]. The connected tier is the genuinely stronger cross-prime independence principle required for an occupancy limit. The conductor budget is attached to the actual sheaves and their tensor products rather than to an undefined scalar conductor of a cover. The first theorem is the connected tier at $r = 3$ for one fixed level n , square-root cancellation of the family-averaged third connected cumulant within the range governed by $\mathfrak{C}_{n,3}$. A failure identifies either geometric complexity missed by the sheaf conductors or arithmetic entanglement surviving all fixed-order trace tests.

Conjecture 43 (Coloured dynatomic Galois-factor process). *For $f_c(x) = x^d + c$, let the critical exact-period polynomial factor as $\Psi_n(c) = \prod_{j=1}^{s_n} F_{n,j}(c)$ over $\mathbb{Z}$. Sample c from a dyadic height interval away from postcritically finite, discriminant, and fixed-resultant loci.*

For every fixed n , condition on any fixed finite local valuation state and normalize the prime factors of each $F_{n,j}(c)$ by that component's residual logarithmic mass. The component-coloured factor processes converge jointly to independent Frobenius-marked Poisson–Dirichlet PD(1) processes, with the mark in colour j given by the fixed-point-size-biased conjugacy law in the splitting field of $F_{n,j}$. The total logarithmic mass of ramified factors tends to zero.

Uniformly for $n = n(H)$ with $\deg \Psi_n = o(\log H)$, every fixed correlation functional whose support lies inside the available level of distribution converges to the corresponding restricted-support correlation of these coloured marked processes. No full-process claim is made at growing degree without level-one distribution.

Significance. Exact-period components are genuine geometric colours, and their Galois marks distinguish dynamical families with similar unmarked factor sizes. The statement deliberately separates the full fixed-degree process from the restricted-support growing-degree law, in accordance with the distribution thresholds in Bharadwaj–Rodgers [51]. The first theorem is a two-colour correlation law at one fixed period. A failure would show that distinct exact-period components are entangled, with the splitting fields of two factors $F_{n,j}$ sharing Frobenius information that survives conditioning on the fixed finite local valuation state.

Conjecture 44 (Complexity-uniform primitive valuation process for dynatomic values). *For the critical exact-period polynomial $\Psi_n(c)$, remove its canonical greatest common divisor in $\mathbb{Q}[c]$ with the earlier critical-orbit product $\prod_{m < n} f_c^m(0)$, in the notation of Conjecture 43, and denote the result by Ψ_n^{new} . Define the logarithmic arithmetic complexity*

$$\mathfrak{H}_n = \deg \Psi_n^{\text{new}} + \log^+ H_{\text{coeff}}(\Psi_n^{\text{new}}) + \log^+ |\text{Disc}(\Psi_n^{\text{new}})| + \sum_{m < n} \log^+ |\text{Res}(\Psi_n^{\text{new}}, f_c^m(0))|,$$

where zero resultants have already been removed by the canonical gcd. For each prime p , let $\nu_{n,p}$ be the exact Haar law on $\mathbb{Z}_p$ of the primitive valuation

$$V_{n,p}(c) = v_p(\Psi_n^{\text{new}}(c)) \mathbf{1}_{p \nmid f_c^m(0) \ \forall m < n}.$$

All discriminant, resultant, and collision primes remain inside these local laws.

For every control function $\omega(H) = o(\log H)$, uniformly over levels with $\mathfrak{H}_n \leq \omega(H)$, the finite-prime valuation vectors for c in a dyadic height interval converge to the corresponding marginals of $\bigotimes_p \nu_{n,p}$. In addition, for every $\eta > 0$,

$$\lim_{z \rightarrow \infty} \limsup_{H \rightarrow \infty} \sup_{n: \mathfrak{H}_n \leq \omega(H)} \mathbb{P} \left(\sum_{p > z} (\log p)(V_{n,p} - 1)_+ > \eta \log |\Psi_n^{\text{new}}(c)| \right) = 0,$$

and the analogous unnormalized tail $\sum_{p > z} \mathbb{P}(V_{n,p} \geq 2)$ tends to zero whenever the local squarefree product is positive. These two clauses give convergence in the topology generated by bounded cylinder functions and the logarithmic squarefull-mass functional.

Consequently

$$\mathbb{P}(\Psi_n^{\text{new}}(c) \text{ is squarefree}) = \prod_p \tilde{\nu}_{n,p}(\{0, 1\}) + o(1)$$

uniformly in the stated complexity range, where $\tilde{\nu}_{n,p} = \nu_{n,p}$ except at the finitely many collision primes dividing one of the resultants $\text{Res}(\Psi_n^{\text{new}}, f_c^m(0))$, at which $\tilde{\nu}_{n,p}$ is the law of the raw valuation $v_p(\Psi_n^{\text{new}}(c))$, and the logarithmic squarefull mass converges to the law determined by the complete local product.

Significance. Degree alone does not control a growing polynomial family. The controlling range includes coefficient height, discriminant, and resultants, while the infinite-product conclusion is supported by an explicit uniform squarefull-tail assertion. The local process still retains every

bad-prime and collision state and remains level dependent, as primitive hits at a fixed prime occur by bounded time modulo p^k . For the unicritical families $x^d + c$ the resultants at the first levels are units, so the stated complexity range is nonempty there. The first decisive theorem is the primitive valuation law with its uniform squarefull tail at one fixed level of the unicritical family $x^d + c$, giving the squarefree asymptotic for that single Ψ_n^{new} . A failure would show that $\mathfrak{H}_n$ is not the right complexity gauge, with some growing-level family carrying squarefull mass or valuation correlations that degree, coefficient height, discriminant, and resultants together do not see. The nearest literature treats primitive divisors and fixed-polynomial squarefree values, not this complexity-uniform adelic process [49, 48].

Conjecture 45 (Divisor-sensitive classification of dynamical gcd height). *Let $F = (f, g)$ be a split map on $(\mathbb{P}^1)^2$, with f, g disintegrated (in the sense of Medvedev and Scanlon [56], so neither is linearly conjugate to a power map, a Chebyshev map, or a close relative of these) of the same degree, and let $P = (a, b)$ have positive canonical heights. Let D_1, D_2 be the two coordinate-zero divisors and let E be the exceptional divisor of the blow-up at $D_1 \cap D_2$. Then*

$$\limsup_{n \rightarrow \infty} \frac{h_E(F^n(P))}{\max\{h(f^n(a)), h(g^n(b))\}} > 0$$

if and only if there exist $m \geq 0$ and an irreducible F -periodic curve Z containing $F^m(P)$, with Z not contained in $D_1 \cup D_2$ (automatic when both coordinates have positive canonical height), such that, on the normalization of Z , the pullbacks of D_1 and D_2 have a common nonzero effective component. In the absence of such eventual entry into a divisor-compatible periodic curve, the normalized gcd height tends to zero.

The converse is unconditional in every setting where the required Vojta inequality for the blow-up is known, and otherwise is explicitly conditional on that inequality. The same criterion extends to split maps on $(\mathbb{P}^1)^r$ by replacing a common component with positive codimension-one intersection multiplicity among the pulled-back divisors.

Significance. The obstruction is not merely a common dynamical quotient. It is a periodic geometric relation carrying the actual divisors measured by the gcd height. Eventual entry is the right notion, since a preperiodic tail can carry the divisor relation before periodicity begins. Recent work resolves several set-theoretic dynamical-GCD problems, while this height-theoretic divisor classification is the additional boundary [50]. The first theorem is the positive-limsup direction, that eventual entry of the orbit into a divisor-compatible periodic curve Z forces $\limsup h_E(F^n(P)) / \max\{h(f^n(a)), h(g^n(b))\} > 0$. Failure would point to a new source of macroscopic Diophantine proximity outside invariant geometry.

10 Programme V: adelic factorization and sieve flow

Throughout this programme $P^+(m)$ and $P^-(m)$ denote the largest and smallest prime factors of m , the constant γ is Euler's, $\rho_f(p)$ counts the roots of f modulo p , and every sample $n \in [N, 2N]$ is drawn uniformly.

Conjecture 46 (Frobenius-marked Poisson–Dirichlet process). *Let $f \in \mathbb{Z}[x]$ be irreducible with splitting field L and transitive Galois group G . Sample $n \in [N, 2N]$. Mark each unramified prime factor p of $f(n)$ by its conjugacy class $C \subset G$ and give it mass $u = \log p / \log |f(n)|$. The*

macroscopic marked process converges to a marked PD(1) process with conditional mark law

$$\mathbb{P}(C \mid u) = \frac{|C|}{|G|} \text{fix}(C), \quad \text{fix}(C) \text{ the number of fixed roots of an element of } C,$$

independent of u . Given the masses and every fixed finite local valuation state, the marks of finitely many macroscopic factors are asymptotically independent. Under a normal quotient equipped with the induced permutation representation, the marked process pushes forward functorially.

Significance. Chebotarev contributes $|C|/|G|$, divisibility by a polynomial value size-biases by the number of fixed roots, and Burnside’s lemma normalizes the result. Bharadwaj–Rodgers supply the unmarked factor-process framework [51]. Simultaneous Galois marking, conditional independence, and functorial pushforward are the new content. It connects Galois theory to probabilistic factorization directly. The first theorem is the mark law for the largest prime factor of one irreducible quadratic f , the two-class case where the size bias $\text{fix}(C)$ already separates the marked law from the plain Chebotarev weight $|C|/|G|$. A failure would show that divisibility size-biases Frobenius by more than the fixed-root count, a coupling between factor masses u and Galois marks that survives conditioning on the finite local valuation state.

Conjecture 47 (Three-scale factorization of polynomial values). *Let $f \in \mathbb{Z}[x]$ be irreducible, sample $n \in [N, 2N]$, choose $y = y(N) \rightarrow \infty$ with $\log y = o(\log N)$, and write $R_y = f(n)/S_y(f(n))$, where $S_y(m) = \prod_{p \leq y} p^{v_p(m)}$ is the y -smooth part. Jointly with the complete small-prime valuation field $(v_p(f(n)))_{p \leq y}$, the residual logarithmic factor process*

$$\sum_{p \mid R_y} v_p(R_y) \delta_{(\log p / \log |R_y|, \text{Frob}_p)}$$

converges to the following conservation-corrected product object: the small valuations have their polynomial Kubilius law (independent exact local limit laws for the small-prime valuations), and conditional on their consumed logarithmic mass the residual process is a scale-invariant Poisson process of intensity du/u , marked by Conjecture 46, conditioned to have total mass one. The small field and the unconditioned residual Poisson process are asymptotically independent, and all leading dependence after conditioning is the single residual-mass constraint.

For disjoint normalized size bands bounded away from zero, the unconditioned residual process has independent increments. Its mesoscopic restriction and its ranked macroscopic atoms yield respectively the scale-invariant factor process and the Frobenius-marked PD(1) partition. Thus the local Kubilius field is the boundary variable fixing the available mass, while the mesoscopic and macroscopic laws are two projections of one residual process, and no additional cross-scale coupling survives.

Significance. Raw small–large independence is false because small factors consume mass. This conjecture identifies the complete conservation-corrected product structure and adds the mesoscopic scale that is absent from a two-scale statement. It would justify, in one limit theorem, the local sieve model, the scale-invariant factor process, and the macroscopic partition. The first decisive theorem is the conservation-corrected joint law at one fixed prime p and one normalized size band, where the only surviving dependence is the residual-mass constraint. A failure would show a cross-scale coupling between the small-prime valuation field and the residual process beyond the single residual-mass constraint, so smooth-part consumption would not exhaust the dependence between the sieve scale and the factor process.

Conjecture 48 (Adelic Gibbs gluing for reducible polynomial values). *Let $f = \prod_{i=1}^s g_i$ be squarefree with irreducible components, sample $n \in [N, 2N]$, and put $t = n/N$. For each prime p , let ν_p be the exact Haar law on $\mathbb{Z}_p$ of the valuation vector $(v_p(g_1(x)), \dots, v_p(g_s(x)))$. For $y \rightarrow \infty$ with $\log y = o(\log N)$, consider stable convergence jointly with the archimedean coordinate t and condition on $\mathcal{L}_y = (v_p(g_i(n)))_{p \leq y, i \leq s}$. After normalizing each component by its own residual logarithmic mass, the coloured Frobenius-marked factor processes converge to conditionally independent copies of the three-scale law of Conjecture 47, with the marks of Conjecture 46.*

Unconditionally, the joint process is the projective Gibbs mixture obtained by integrating those conditional product laws against Lebesgue measure in t and the adelic local law $\bigotimes_p \nu_p$, in the iterated order $N \rightarrow \infty$ and then $y \rightarrow \infty$. Every bounded finite-dimensional Laplace functional is determined by

$$Z_p(\mathbf{z}) = \int_{\mathbb{Z}_p} \prod_i z_i^{v_p(g_i(x))} dx$$

together with the archimedean size profile $(\log |g_i(Nt)|)_i$. No additional colour coupling survives after this adelic–archimedean mixture.

Significance. A scalar Euler factor cannot specify a point process. This statement gives the full local-to-global gluing rule and includes the archimedean variable that can couple component sizes. It is functorial under multiplying, splitting, or regrouping components. The first theorem is the case of two linear components, where $Z_p(\mathbf{z})$ is computable in closed form and the mixture over t and $\bigotimes_p \nu_p$ already yields the full two-colour gluing law. A persistent medium-prime or archimedean coupling after the stated mixture would identify a new nonadelic invariant.

Conjecture 49 (Full multivariate adelic saddle law for joint smoothness). *Let $f_1, \dots, f_s \in \mathbb{Z}[x]$ be pairwise coprime and primitive, sample $n \in [N, 2N]$, and put $t = n/N$. For complex variables $\mathbf{s}$ and smoothness bounds $\mathbf{y}$, define*

$$Z_{p,\mathbf{y}}(\mathbf{s}) = \int_{\mathbb{Z}_p} p^{-\sum_{i:p \leq y_i} (s_i-1)v_p(f_i(x))} dx, \quad \hat{Z}_{p,\mathbf{y}}(\mathbf{s}) = \frac{Z_{p,\mathbf{y}}(\mathbf{s})}{Z_{p,\mathbf{y}}(+\infty)}.$$

The denominator is the exact Haar probability that p divides none of the relevant $f_i(x)$. Put

$$\Phi_t(\mathbf{s}) = \sum_p \log \hat{Z}_{p,\mathbf{y}}(\mathbf{s}) + \sum_i (s_i - 1) \log |f_i(Nt)|.$$

For $f(x) = x$, this gives the usual truncated zeta factor $(1 - p^{-s})^{-1}$. Define the joint local probability by the Perron-calibrated multivariate inverse Mellin integral

$$\mathcal{P}_t^{\text{joint}} = \frac{1}{(2\pi i)^s} \int_{\sigma_t + i\mathbb{R}^s} e^{\Phi_t(\mathbf{s})} \prod_{i=1}^s \frac{ds_i}{s_i},$$

with the standard truncated-contour limit, where σ_t is the unique real saddle in the admissible region. The factors s_i^{-1} are part of the canonical cumulative smoothness event, and for one variable they recover the Hildebrand–Tenenbaum Perron normalization [55].

Uniformly when every $u_i(t) = \log |f_i(Nt)| / \log y_i$ lies in a fixed compact saddle-point region,

$$\mathbb{P}(P^+(f_i(n)) \leq y_i \ \forall i) \sim \int_1^2 \mathcal{P}_t^{\text{joint}} dt.$$

The integral has the full multivariate saddle expansion. Writing $H_t = \nabla^2 \Phi_t(\sigma_t)$, its leading term includes the joint action, the amplitude $\prod_i \sigma_{i,t}^{-1}$, and $(\det H_t)^{-1/2}$, with the usual lattice correction when the valuation span is proper.

If $\mathcal{P}_{i,t}^{\text{marg}}$ are the identically calibrated one-polynomial probabilities, then $\mathcal{P}_t^{\text{joint}} / \prod_i \mathcal{P}_{i,t}^{\text{marg}}$ is the complete ratio of joint and marginal actions, Perron amplitudes, Hessian determinants, and lattice corrections. Its Euler-product component is the ratio of the exact p -adic partition functions evaluated at the joint and marginal saddles. None of these factors may be replaced by a scalar Euler correction at the marginal saddle.

Significance. The statement is calibrated to the cumulative smoothness probability. It combines exact higher p -adic valuations, moving archimedean sizes, the joint saddle displacement, Perron amplitudes, mixed curvature, and possible lattice effects. Even one-polynomial smooth-value asymptotics are difficult [52, 53]. A complete multivariate adelic saddle law would be a major new local–global bridge. A failure would show a joint smoothness dependence not produced by the exact p -adic partition functions and the moving archimedean sizes, an arithmetic coupling among the f_i outside the saddle displacement, the Hessian H_t , and the lattice correction.

Conjecture 50 (Buchstab–Bateman–Horn component flow). *Let $f \in \mathbb{Z}[x]$ be primitive, irreducible, and admissible ($\rho_f(p) < p$ for every prime p), and sample $n \in [N, 2N]$. For y with $u_n = \log |f(n)| / \log y$ in a fixed compact subset of $(1, \infty)$, let $\Omega_y(f(n))$ be the number of prime factors of $f(n)$ exceeding y , counted with multiplicity, on the event $P^-(f(n)) > y$. Put*

$$V_f(y) = \prod_{p \leq y} \left(1 - \frac{\rho_f(p)}{p}\right).$$

Define the Buchstab component functions by

$$\omega_1(u) = \frac{1}{u}, \quad \omega_j(u) = \frac{1}{u(j-1)!} \int_{\substack{t_1, \dots, t_{j-1} \geq 1 \\ t_1 + \dots + t_{j-1} \leq u-1}} \frac{dt_1 \cdots dt_{j-1}}{t_1 \cdots t_{j-1}} \quad (j \geq 2),$$

so that $\sum_{j \geq 1} \omega_j(u) = \omega(u)$. Then for every fixed j ,

$$\mathbb{P}(P^-(f(n)) > y, \Omega_y(f(n)) = j) = V_f(y) e^\gamma \mathbb{E}_{n \in [N, 2N]} \omega_j(u_n) + o(V_f(y)),$$

uniformly on compact u -ranges. Conditional on the ordered logarithmic sizes of the j remaining factors, their unramified Frobenius marks are independent with the fixed-point-size-biased law of Conjecture 46. Summing over j gives the rough-value Buchstab law, $j = 1$ gives the Bateman–Horn prime density, and $j = 2$ gives the semiprime transition. Whenever $\max_{N \leq n \leq 2N} |f(n)| < y^2$, every rough value has exactly one prime factor above y , so the $j = 1$ component exhausts the rough set, while the density formula itself remains conjectural.

Significance. The conjecture resolves a rough polynomial value into its complete finite- u factor count and Galois mark content. Primes and semiprimes are not inserted as separate heuristics. They are components of one universal flow. Generalized Buchstab equations are the random-integer antecedent [60], and the fixed-polynomial, Galois-marked component law with its local factors is the new statement. It is strongly falsifiable even when total roughness follows Buchstab.

Part III

Six conjectures across fields

The six conjectures below leave the prime model. They range over multiplicative number theory, discrepancy theory, graph limits, additive information theory, and ergodic theory, and each is held to the same standard as the rest of the collection: a canonical object, a stated mechanism, the nearest literature boundary, a first decisive theorem where one exists, and an honest failure mode.

11 Relative polynomial pretentiousness

Let $P \in \mathbb{Z}[x]$ be primitive (the greatest common divisor of its coefficients is 1) and irreducible, with positive leading coefficient. For a prime p write

$$\rho_P(p) = \#\{a \in \mathbb{Z}/p\mathbb{Z} : P(a) \equiv 0 \pmod{p}\}.$$

The weight $\rho_P(p)/p$ is the first-order frequency with which p divides a polynomial value $P(n)$. A function f on the positive integers is completely multiplicative if $f(mn) = f(m)f(n)$ for all $m, n \geq 1$. For completely multiplicative f, g with values in the closed complex unit disc, define the truncated P -relative pretentious distance

$$\mathbb{D}_P(f, g; X)^2 = \sum_{p \leq X} \frac{\rho_P(p)}{p} (1 - \operatorname{Re}(f(p)\overline{g(p)})), \quad \mathbb{D}_P(f, g)^2 = \lim_{X \rightarrow \infty} \mathbb{D}_P(f, g; X)^2 \in [0, \infty].$$

Every summand is nonnegative, so the truncated distance is nondecreasing in X and the limit exists. Throughout, a primitive Dirichlet character χ of conductor c is regarded as the c -periodic completely multiplicative function on $\mathbb{Z}$ that vanishes on integers not coprime to c , so that expressions such as $\chi(P(a))$ are defined for every residue a modulo any multiple of c . For $t \in \mathbb{R}$, the comparison function $n \mapsto \chi(n)n^{it}$ is completely multiplicative with values in the closed unit disc.

Let $P_1, \dots, P_m \in \mathbb{Z}[x]$ be pairwise distinct primitive irreducible polynomials with positive leading coefficients and no fixed prime divisor, meaning $\rho_{P_j}(p) < p$ for every j and every prime p . Assume multiplicative independence modulo constants:

$$\prod_{j=1}^m P_j(x)^{e_j} \in \mathbb{Q}^\times, \quad e_j \in \mathbb{Z} \implies e_1 = \dots = e_m = 0.$$

Write $\mathbb{S}^1$ for the unit circle in the complex plane.

Conjecture 51 (Relative polynomial pretentiousness inverse principle). *Let $f_1, \dots, f_m : \{1, 2, 3, \dots\} \rightarrow \mathbb{S}^1$ be completely multiplicative. If*

$$\limsup_{X \rightarrow \infty} \left| \frac{1}{\log X} \sum_{\substack{n \leq X \\ P_j(n) > 0 \ \forall j}} \frac{\prod_{j=1}^m f_j(P_j(n))}{n} \right| > 0,$$

then there exist primitive Dirichlet characters $\chi_1, \dots, \chi_m$ and real numbers $t_1, \dots, t_m$ such that the following three conditions hold simultaneously:

- (i) $\mathbb{D}_{P_j}(f_j, \chi_j(n)n^{it_j}) < \infty$ for every $1 \leq j \leq m$,
- (ii) $\sum_{j=1}^m t_j \deg P_j = 0$,
- (iii) for some common multiple q of the conductors of $\chi_1, \dots, \chi_m$,

$$\frac{1}{q} \sum_{a \bmod q} \prod_{j=1}^m \chi_j(P_j(a)) \neq 0,$$

the characters being extended by 0 on nonunits as above.

Significance. The mechanism is that a persistent logarithmic correlation should have exactly two sources: an archimedean resonance carried by the factors n^{it_j} and a finite local resonance carried by the characters χ_j . The balance condition (ii) is forced by the expansion $P_j(n)^{it_j} = c_j^{it_j} n^{it_j \deg P_j} (1 + o(1))$, with c_j the leading coefficient, because logarithmic averaging annihilates any nonzero total power of n , and (iii) asserts that the finite-modulus resonances are jointly compatible, so that the model correlation does not vanish identically for local reasons. Packaging the conclusion as one existential block matters: the characters and archimedean parameters cannot be chosen independently for each j , and it is the joint system (i)–(iii) that reproduces the correlation. The decisive-theorem ledger for the linear case is as follows. For linear polynomials the two-point case is Tao’s logarithmically averaged two-point Elliott theorem [75]. The odd-order cases are the theorems of Tao–Teräväinen, and the shape of the conclusion here, one existential block coupling characters, archimedean parameters, and a nonvanishing local average, is the template of their structure theorem for logarithmically averaged correlations [76]. The general even-order linear case is precisely the logarithmically averaged Elliott conjecture. All of the novelty of the present statement therefore sits at $\max_j \deg P_j \geq 2$, where even the logarithmic mean value of $f(n^2 + 1)$ for a single nonpretentious completely multiplicative f is open, and a logarithmic mean value theorem for $f(n^2 + 1)$ with a conclusion of the stated relative-pretentious type would be the first decisive theorem. The failure mode is a system of functions nonpretentious in every relative metric whose logarithmic correlation has positive limsup, which would reveal a correlation mechanism beyond the archimedean and finite local resonances.

Remark 2. *The restriction to completely multiplicative f_j is not cosmetic. For 1-bounded multiplicative functions that are not completely multiplicative, the fixed-parameter form of the conclusion is false: Matomäki–Radziwiłł–Tao [71] constructed a 1-bounded multiplicative function that pretends to be n^{it_k} for different parameters t_k on different scales, refuting the original fixed-parameter Elliott conjecture, and the corrected conjecture must allow the comparison parameter to vary with the scale X . For the unimodular completely multiplicative f_j of the present statement the fixed-parameter form survives, and by a rigidity argument rather than by monotonicity alone. Fix a member f_j of the system, a primitive Dirichlet character χ , and scales $X_k \leq X_{k+1}$, the scale index k being unrelated to the function index j . If χn^{is_k} and $\chi n^{is_{k+1}}$ are within bounded truncated distance of f_j at the scales X_k and X_{k+1} respectively, monotonicity of the truncated distance in the scale places both within bounded distance of f_j at the common scale X_k , the triangle inequality places them within bounded distance of each other, and the growth in X of $\sum_{p \leq X} (1 - \cos(\delta \log p))/p$ for fixed $\delta \neq 0$ then forces $|s_{k+1} - s_k| = O(1/\log X_k)$. After passing to a geometrically growing subsequence of the scales, $\log X_{k+1} \geq 2 \log X_k$ say, which monotonicity permits without weakening the conclusion, the increments are summable, the parameters converge, and monotonicity converts boundedness along the scales into finiteness of the full distance at the*

limit parameter. Unimodularity at the primes is consumed at the triangle-inequality step: the passage through the middle function f_j rests on the identity $\chi n^{is_k} \overline{\chi n^{is_{k+1}}} = (\chi n^{is_k} \bar{f}_j)(f_j \overline{\chi n^{is_{k+1}}})$, valid at the primes exactly when $|f_j(p)| = 1$, and for values of modulus less than one the two comparisons decouple on the primes where the modulus drops, which is exactly the room the scale-switching construction occupies.

Remark 3. For each j , the finitely many primes dividing $\text{disc}(P_j) \text{lc}(P_j)$, the discriminant times the leading coefficient, may be weighted arbitrarily: replacing the weights $\rho_{P_j}(p)/p$ at those primes by any bounded values changes $\mathbb{D}_{P_j}(f, g)^2$ by a bounded amount and leaves the finiteness condition (i) unchanged. The relative metric is therefore canonical up to the bad primes, and the conjecture is insensitive to that choice.

12 Monotone threshold purification of vector paths

Let $d, m \geq 1$. Let $v_1, \dots, v_m \in \mathbb{R}^d$ satisfy $\|v_i\|_2 \leq 1$, where $\|\cdot\|_2$ is the Euclidean norm. Let $a_i : [0, 1] \rightarrow [0, 1]$ be continuous nondecreasing functions with $a_i(0) = 0$ and $a_i(1) = 1$, called a monotone activation schedule, and define the fractional path

$$F(t) = \sum_{i=1}^m a_i(t) v_i.$$

A threshold rounding of F is a choice of $\tau_1, \dots, \tau_m \in (0, 1]$ together with the pure path

$$G(t) = \sum_{i: \tau_i \leq t} v_i.$$

Conjecture 52 (Monotone threshold purification of vector paths). *There exists a universal constant $C < \infty$ such that for every $d, m \geq 1$, every $(v_i)_{i \leq m}$, and every monotone activation schedule $(a_i)_{i \leq m}$ as above, some threshold rounding satisfies*

$$\sup_{0 \leq t \leq 1} \|G(t) - F(t)\|_2 \leq C\sqrt{d}.$$

Significance. The headline attribution comes first. The special case $a_i(t) = t$ for all i with $\sum_i v_i = 0$ is exactly the open prefix-sum form of the Euclidean Steinitz problem at the scale $O(\sqrt{d})$: there $F \equiv 0$, and ordering the vectors by their thresholds turns the display into the assertion that some permutation π has all prefix sums bounded, $\max_{k \leq m} \|\sum_{i \leq k} v_{\pi(i)}\|_2 \leq C\sqrt{d}$. Grinberg–Sevast’yanov [65] proved the bound d in every norm, the best known Euclidean bound is Banaszczyk’s [61, 62] $O(\sqrt{d + \log m})$, and so already in this special case the conjecture asserts the removal of the logarithmic term uniformly in m . For the general threshold form, even the existence of any bound $C(d)$ depending on d alone and uniform in m appears to be open. The mechanism is that F is a monotone trajectory in the zonotope generated by the v_i , the set of sums $\sum_i \lambda_i v_i$ with $\lambda_i \in [0, 1]$, and the conjecture asserts that every monotone zonotope path admits an integral shadow at the optimal Euclidean discrepancy scale. The order $\sqrt{d}$ is verified sharp: take $m = d$, $v_i = e_i$ the standard basis, and $a_i(t) = t$, so that at $t = 1/2$ every coordinate of $G(1/2) - F(1/2)$ equals $\pm 1/2$ whatever the thresholds, giving $\|G(1/2) - F(1/2)\|_2 = \sqrt{d}/2$. No decisive theorem exists: the first would be any m -uniform bound $C(d)$ for the threshold form, even with a d -dependence far worse than $\sqrt{d}$. A falsification honesty note: since even

the existence of a finite bound at each fixed d is open, the conjecture can fail in two ways, at fixed dimension, through paths with d fixed and $m \rightarrow \infty$ whose best threshold roundings have unbounded discrepancy, or asymptotically, through an infinite family of monotone paths with $d \rightarrow \infty$ every threshold rounding of which has discrepancy $\omega(\sqrt{d})$. Small- d computer searches at bounded m cannot decide either failure mode.

13 Entropy dimension of random-free graphons

A graphon is a symmetric measurable function $W : [0, 1]^2 \rightarrow [0, 1]$, and W is random-free if it takes values in $\{0, 1\}$ almost everywhere. Define the row pseudometric

$$d_W(x, y) = \int_0^1 |W(x, z) - W(y, z)| dz.$$

Identifying points at zero distance and completing yields a metric-measure space (Ω_W, d_W, ρ_W) , the row space, where ρ_W is the pushforward of Lebesgue measure. The row space is Ahlfors s -regular if it is compact and there exist $c, C, r_0 > 0$ such that

$$cr^s \leq \rho_W(B(x, r)) \leq Cr^s$$

for every $x \in \text{supp } \rho_W$ and every $0 < r < r_0$. Let $G(n, W)$ be the random labelled graph on $[n] = \{1, \dots, n\}$ obtained from independent uniform $U_1, \dots, U_n$ on $[0, 1]$ by declaring ij an edge exactly when $W(U_i, U_j) = 1$, and let $H(G(n, W))$ be its Shannon entropy with natural logarithms.

Conjecture 53 (Entropy dimension of random-free graphons).

(i) [nonvacuity] *For every $s \in (0, 2]$ there exists a random-free graphon whose row space is Ahlfors s -regular.*

(ii) [refined entropy-dimension law] *If the row space of a random-free graphon W is Ahlfors s -regular for some $s > 0$, then*

$$H(G(n, W)) = sn \log n + O_W(n).$$

Proposition 1 (Entropy-dimension limit law). *If the row space of a random-free graphon W is Ahlfors s -regular for some $s > 0$, then*

$$H(G(n, W)) = sn \log n + O_W(n \log \log n),$$

and in particular $\lim_{n \rightarrow \infty} H(G(n, W)) / (n \log n) = s$.

Proof. Write $d = d_W$ and $\rho = \rho_W$, and use natural logarithms. For the lower bound, set $k = \lfloor n / \log n \rfloor$, fix the labelling, and designate the last k vertices as tests, with latent points $Z = (Z_1, \dots, Z_k)$ and remaining latent points $X_1, \dots, X_{n-k}$. The cross matrix $M = (W(X_i, Z_j))_{i,j}$ is a function of the labelled graph, and conditioning cannot increase entropy, so $H(G(n, W)) \geq H(M) \geq H(M \mid Z)$. Conditionally on Z the signature rows $S_Z(X_i) = (W(X_i, Z_1), \dots, W(X_i, Z_k))$ are independent with a common law, whence $H(M \mid Z) = (n - k) \mathbb{E}_Z H(S_Z(X))$ with X an independent uniform point. Let C_Z be the collision probability of the signature law. Shannon entropy dominates the Rényi entropy of order two, so $H(S_Z(X)) \geq -\log C_Z$, and convexity of $-\log$ gives $\mathbb{E}_Z H(S_Z(X)) \geq -\log \mathbb{E}_Z C_Z$. For independent uniform X and Y the tests are

independent uniform points, two rows agree at a test with probability $1 - d(X, Y)$, and Fubini gives the exact identity $\mathbb{E}_Z C_Z = \mathbb{E}_{X,Y} (1 - d(X, Y))^k$. Since $(1 - u)^k \leq e^{-ku}$ and the upper mass bound gives $\rho(B(x, r)) \leq Cr^s$, a dyadic shell decomposition yields, uniformly in x ,

$$\int e^{-k d(x,y)} d\rho(y) \leq \rho(B(x, 1/k)) + \sum_{m \geq 0} e^{-2^m} \rho(B(x, 2^{m+1}/k)) = O_W(k^{-s}).$$

Hence $\mathbb{E}_Z C_Z = O_W(k^{-s})$, so $\mathbb{E}_Z H(S_Z(X)) \geq s \log k - O_W(1)$ and $H(G(n, W)) \geq (n - k)(s \log k - O_W(1)) = s n \log n - O_W(n \log \log n)$.

For the upper bound, let $\delta \in (0, \frac{1}{2})$. The lower mass bound makes the balls of radius $\delta/2$ around a maximal δ -separated set disjoint, each of mass at least $c(\delta/2)^s$, so the row space admits a Borel partition into $N_\delta = O_W(\delta^{-s})$ cells of diameter at most 2δ . Let Q_i be the cell of the i th latent row. Subadditivity and conditioning give $H(G(n, W)) \leq n \log N_\delta + \binom{n}{2} \mathbb{E}_{(a,b)} h(p_{ab})$, where (a, b) is the cell pair of two independent rows, p_{ab} is the conditional edge probability, and h is the binary entropy. For a $\{0, 1\}$ -valued W , draw independent latent points X, X' with rows in the cell a and independent latent points Y, Y' with rows in the cell b , so that $W(X, Y)$ and $W(X', Y')$ are independent indicators of mean p_{ab} and $p_{ab}(1 - p_{ab}) = \frac{1}{2} \mathbb{P}(W(X, Y) \neq W(X', Y'))$, while the union bound gives $\mathbb{P}(W(X, Y) \neq W(X', Y')) \leq \mathbb{P}(W(X, Y) \neq W(X', Y)) + \mathbb{P}(W(X', Y) \neq W(X', Y'))$. Averaging each term over the opposite cell with its mass removes that cell restriction, so $\mathbb{E}_{(a,b)} [p_{ab}(1 - p_{ab})] \leq \frac{1}{2} (\mathbb{E}_a \mathbb{E}_{X, X' \in a} d(X, X') + \mathbb{E}_b \mathbb{E}_{Y, Y' \in b} d(Y, Y')) \leq 2\delta$. The bound $h(p) \leq \varphi(p(1 - p))$ with the concave increasing $\varphi(u) = 2u(1 + \log(1/u))$ and Jensen give $\mathbb{E}_{(a,b)} h(p_{ab}) \leq \varphi(2\delta) = O(\delta \log(1/\delta))$. Therefore $H(G(n, W)) \leq s n \log(1/\delta) + O_W(n) + O(n^2 \delta \log(1/\delta))$, and the choice $\delta = (n \log^2 n)^{-1}$ gives $H(G(n, W)) \leq s n \log n + O_W(n \log \log n)$. $\square$

Remark 4. *The conjecture is the refinement of the error term of proposition 1 to $O_W(n)$, which lies beyond both bounds of the proof: locating the mispredicted edges of a resolution-1/n code costs order $n \log n$ if done naively, and the two-scale correction that codes coarse types first stops at $O(n \log \log n)$.*

Significance. The mechanism is metric distinguishability of rows: two row types at d_W -distance ε disagree on an ε -fraction of potential neighbours and become statistically distinguishable against n sampled vertices at a scale comparable to $1/n$, an Ahlfors s -regular row space has about n^s distinguishable types at that scale, and the latent type of each of the n labelled vertices then carries $s \log n + O(1)$ nats. Clause (i) certifies that the hypothesis class is nonvacuous at every fractional s : it is expected to be provable by Cantor-type threshold constructions, in which W is the indicator of a threshold event read along a self-similar Cantor set, and it is included so that clause (ii) quantifies over a genuinely rich family. The nearest literature boundary is Hatami–Norine [66]: a graphon is random-free exactly when $H(G(n, W)) = o(n^2)$, and every subquadratic growth order occurs for suitable random-free graphons, so no hypothesis short of a metric-measure condition on the row space can pin the coefficient of $n \log n$, and Ahlfors regularity is exactly the condition that does. The decisive theorem half-exists: the limit law is provable from regularity by the collision-entropy and quantization bounds of proposition 1, so the first theorem for the conjecture proper is the linear error term for one explicit self-similar threshold graphon of noninteger s , together with a proof of clause (i). The failure mode is a genuinely growing correction: an Ahlfors regular random-free graphon whose entropy exceeds $s n \log n + C n$ for every C would show that the resolution-1/n coding loss is intrinsic rather than an artefact of the two-scale method.

14 A Hessian principle for isolated microcanonical graph phases

Fix a partition

$$[0, 1] = I_1 \sqcup \cdots \sqcup I_q, \quad |I_a| = \alpha_a > 0.$$

A coloured graphon is a symmetric measurable $U : [0, 1]^2 \rightarrow [0, 1]$ considered together with this fixed colour partition. Let $p(U) \in [0, 1]^D$ with $D = q(q+1)/2$ denote the block-average vector

$$p_{ab}(U) = \frac{1}{|I_a||I_b|} \int_{I_a \times I_b} U(x, y) \, dx \, dy, \quad 1 \leq a \leq b \leq q.$$

A coloured finite graph is a finite simple graph F with a colour $c(v) \in \{1, \dots, q\}$ attached to each vertex, and its coloured homomorphism density in U is

$$t(F, U) = \int_{[0, 1]^{V(F)}} \prod_{v \in V(F)} \alpha_{c(v)}^{-1} \mathbf{1}_{I_{c(v)}}(x_v) \prod_{uv \in E(F)} U(x_u, x_v) \prod_{v \in V(F)} dx_v.$$

Fix coloured finite graphs $F_1, \dots, F_k$ and put $\mathbf{t}(U) = (t(F_1, U), \dots, t(F_k, U))$. Define the graphon entropy

$$\mathcal{S}(U) = \frac{1}{2} \int_{[0, 1]^2} [-U \log U - (1 - U) \log(1 - U)] \, dx \, dy,$$

with $0 \log 0 = 0$, and for a feasible constraint vector $\mathbf{a}$ the contracted entropy

$$\Psi_{\mathbf{a}}(p) = \sup\{\mathcal{S}(U) : \mathbf{t}(U) = \mathbf{a}, p(U) = p\}.$$

The finite graphs are coloured with pinned classes. For each n set $n_a = \lfloor \alpha_a n \rfloor$ and $N_n = \sum_a n_a$, and fix a partition of the vertex set $[N_n]$ into colour classes $V_1, \dots, V_q$ with $|V_a| = \lfloor \alpha_a n \rfloor$ exactly, so that $N_n = n + O(q)$. For a graph G on $[N_n]$ with these colour classes, partition each I_a into n_a consecutive intervals of length $|I_a|/n_a$ assigned to the vertices of V_a , let U_G be the $\{0, 1\}$ -valued coloured graphon equal to 1 on a product of two vertex intervals exactly when the corresponding pair is an edge, and put $\mathbf{t}(G) = \mathbf{t}(U_G)$ and $\hat{p}(G) = p(U_G)$.

Fix $0 < \eta < 1$ and assume the following regularity hypotheses.

(H1) The supremum of $\mathcal{S}(U)$ over $\mathbf{t}(U) = \mathbf{a}$ is attained at a unique coloured graphon U_* , up to measure-preserving maps preserving each colour class.

(H2) Writing $p_* = p(U_*)$, the feasible block-average set $\{p(U) : \mathbf{t}(U) = \mathbf{a}\}$ is, near p_* , a C^3 embedded manifold $M \subset \mathbb{R}^D$.

(H3) The restriction $\Psi_{\mathbf{a}}|_M$ is C^3 near p_* and has there a unique local maximum.

(H4) The quadratic form $A = -D_M^2 \Psi_{\mathbf{a}}(p_*)$ is positive definite on the tangent space $T = T_{p_*} M$.

(H5) [transversally Lipschitz constraint correspondence] There exist $\delta_0 > 0$ and $L < \infty$ such that every coloured graphon U with $\|p(U) - p_*\| \leq \delta_0$ and $\|\mathbf{t}(U) - \mathbf{a}\|_{\infty} = \delta \leq \delta_0$ satisfies $\text{dist}(p(U), M) \leq L\delta$. In particular the conditioning window below confines $\hat{p}_n$, on the event $\|\hat{p}_n - p_*\| \leq \delta_0$, to an $o(n^{-1})$ thickening of M .

(H6) [nonemptiness] For every sufficiently large n the conditioning set $\{G : \|\mathbf{t}(G) - \mathbf{a}\|_{\infty} \leq n^{-1-\eta}\}$ over graphs G on $[N_n]$ with the pinned colour classes is nonempty.

Let G_n be uniform on the conditioning set of (H6), let $\hat{p}_n = \hat{p}(G_n)$, and let Π_T be orthogonal projection of $\mathbb{R}^D$ onto T .

Conjecture 54 (Hessian principle for an isolated microcanonical phase). *Under hypotheses (H1)–(H6),*

$$n \Pi_T(\hat{p}_n - \mathbb{E} \hat{p}_n) \xrightarrow{d} N_T(0, A^{-1}),$$

where $N_T(0, A^{-1})$ denotes the centred Gaussian law on T whose precision form is A .

Significance. The mechanism is a Laplace expansion at the dense-graph entropy exponent n^2 : on the feasible manifold, $\Psi_{\mathbf{a}}(p_* + u) = \Psi_{\mathbf{a}}(p_*) - \frac{1}{2}\langle Au, u \rangle + O(\|u\|^3)$, so the fluctuation scale is $u \asymp n^{-1}$ and the predicted covariance on the tangent space is A^{-1} , while (H5) suppresses the transversal directions below the fluctuation scale, which is exactly why the conclusion is stated only after projection by Π_T and centring at $\mathbb{E}\hat{p}_n$. The nearest literature boundary is the Chatterjee–Varadhan [63] large deviation principle, which supplies the n^2 normalization and the variational description of conditioned dense graphs, together with the multipodal structure theory of Kenyon–Radin–Ren–Sadun [67] and Neeman–Radin–Sadun [72] (constrained entropy maximizers are step-function graphons with finitely many blocks), which makes hypotheses (H1)–(H4) verifiable in concrete constrained ensembles. A scale audit disposes of one suspect: a single edge toggle moves each coordinate of $\mathbf{t}(G)$ by $O(n^{-2})$, so the lattice of attainable constraint values has spacing of order n^{-2} , far below the fluctuation scale n^{-1} , and the genuine risk is therefore Laplace-ratio control, namely uniform comparison of microcanonical counts with their Laplace approximations across windows of width $o(n^{-1})$, not arithmetic oscillation. Nonemptiness (H6) is a genuine hypothesis and may fail along subsequences of n for arithmetic reasons, in which case the statement is asserted only along the subsequence where it holds. No decisive theorem exists at this generality: the natural first one is the case $q = 1$ with the edge-triangle constraint pair in a region where the entropy maximizer is bipodal, a two-block step function, and (H1)–(H5) have been verified. The failure mode is an isolated nondegenerate phase whose tangent fluctuations are non-Gaussian or live at a different scale, which would necessarily come from a failure of Laplace-ratio control. The conjectural content is exactly that (H1)–(H6) already force this local regularity: a subexponential prefactor in the microcanonical counts varying across the conditioning window would leave every hypothesis intact while tilting the limit away from $N_T(0, A^{-1})$, and the conjecture asserts that isolated nondegenerate phases exclude it.

15 Sharp stability of entropy idempotence on finite abelian groups

Let G be a finite abelian group and μ a probability measure on G . Write $H(\mu)$ for Shannon entropy with natural logarithms, $\mu * \mu$ for the convolution, the law of $X + Y$ with X, Y independent of law μ , and

$$\Delta(\mu) = H(\mu * \mu) - H(\mu)$$

for the convolution entropy defect. If $H \leq G$ is a subgroup and $a \in G$, let u_{a+H} be the uniform measure on the coset $a + H$. Total variation distance is

$$\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sum_{x \in G} |\mu(x) - \nu(x)|.$$

Conjecture 55 (Sharp stability of entropy idempotence). *There exists a universal constant $C < \infty$ such that for every finite abelian group G and every probability measure μ on G ,*

$$\inf_{\substack{H \leq G \\ a \in G}} \|\mu - u_{a+H}\|_{\text{TV}}^2 \leq C(H(\mu * \mu) - H(\mu)).$$

Remark 5. *The equality case is verified. With X, Y independent of law μ ,*

$$\Delta(\mu) = H(X + Y) - H(X) = H(X + Y) - H(X + Y | Y) = I(X + Y; Y) \geq 0,$$

where I denotes mutual information. If $\Delta(\mu) = 0$ then $X + Y$ is independent of Y , which forces the translates of μ by the elements of its support to coincide, and hence forces μ to be uniform on a coset of a subgroup. The right-hand side therefore vanishes exactly on the proposed extremizers.

Significance. The mechanism is that the defect is the mutual information created by one self-convolution, so the conjecture is a dimension-free quantitative classification of near-idempotent measures with cosets as the exact equality case. The sharpness discussion identifies the extremal geometry. Coset perturbations $\mu = (1 + \varepsilon f)u_H$ with $\sum_{x \in H} f(x) = 0$ give, as $\varepsilon \rightarrow 0$, the ratio

$$\frac{\|\mu - u_H\|_{\text{TV}}^2}{\Delta(\mu)} \rightarrow \frac{(\mathbb{E}|f|)^2}{2 \mathbb{E}f^2} \leq \frac{1}{2}$$

by Cauchy–Schwarz, so perturbative examples certify at most the constant $1/2$. The near-extremizers are instead discretized Gaussians at intermediate scale on $\mathbb{Z}/p\mathbb{Z}$: for $1 \ll \sigma(p) \ll p$, the discretization μ_p of a centred Gaussian of standard deviation $\sigma(p)$ has $\Delta(\mu_p) \rightarrow \frac{1}{2} \log 2$, the entropy-power constant recording the entropy gain of doubling a Gaussian variance, while its total variation distance to every u_{a+H} tends to 1, since the only subgroups of $\mathbb{Z}/p\mathbb{Z}$ are trivial and full and μ_p is asymptotically singular to every point mass and to the uniform measure. Hence $C \geq 2/\log 2$, and we conjecture that the sharp constant is exactly $2/\log 2$. The positioning against the literature is as follows: the known entropic inverse theorems, Tao’s sumset entropy theory [74] and the entropic polynomial Freiman–Ruzsa theorem of Gowers–Green–Manners–Tao [64], conclude in Ruzsa distance, an entropic closeness of X to a subgroup-uniform variable, not in total variation, and the total variation form over all finite abelian groups is neither proven nor refuted by them. On $\mathbb{Z}/p\mathbb{Z}$ there are no proper nontrivial subgroups, so the conjecture specializes to a discrete entropy-power-type lower bound for spread measures: any μ far in total variation from every point mass and from the uniform measure must create a definite amount of entropy under self-convolution. No decisive theorem exists: the natural first one is precisely that $\mathbb{Z}/p\mathbb{Z}$ specialization with some constant. The defect also has the exact form $H(\mu * \mu) - H(\mu) = \mathbb{E}_Y D_{\text{KL}}(\tau_Y \mu \| \mu * \mu)$, with τ_y translation by y , so a small defect makes typical support-translates of μ close to $\mu * \mu$ in relative entropy, and Pinsker’s inequality converts that closeness to total variation: the route to a universal constant is to upgrade approximate invariance under many translations to proximity to a coset uniform. The failure mode is a sequence μ_n with $\Delta(\mu_n) \rightarrow 0$ whose coset distance is much larger than $\sqrt{\Delta(\mu_n)}$, and the natural hunting ground is measures straddling several subgroup scales inside groups with rich subgroup lattices, a family for which computational search through $\mathbb{Z}/2^8\mathbb{Z}$ finds no ratio $\|\mu - u_{a+H}\|_{\text{TV}}^2/\Delta(\mu)$ above the discretized-Gaussian value $2/\log 2$. The assertion has three layers of decreasing confidence: the existence of some universal constant, the sharp value $2/\log 2$, and the classification of near-extremizers as the discretized Gaussians at intermediate scale.

16 Stationary convolution-entropy rigidity

Let K be a finite abelian group and $\Omega = K^{\mathbb{Z}}$ with coordinatewise addition and the left shift σ . Let μ be a σ -invariant ergodic Borel probability measure on Ω , and write $h_\sigma(\mu)$ for its Kolmogorov–Sinai entropy. Define the translational stabilizer

$$H_\mu = \{h \in K^{\mathbb{Z}} : (\tau_h)_* \mu = \mu\}, \quad \tau_h(x) = x + h,$$

a closed shift-invariant subgroup of $K^{\mathbb{Z}}$, and let $\pi_\mu : K^{\mathbb{Z}} \rightarrow K^{\mathbb{Z}}/H_\mu$ be the quotient homomorphism. Let $\mu * \mu$ be the law of $X + Y$ for independent $X, Y \sim \mu$.

Conjecture 56 (Stationary convolution-entropy rigidity). *For every shift-ergodic μ ,*

$$h_\sigma(\mu * \mu) = h_\sigma(\mu) \iff h_\sigma((\pi_\mu)_*\mu) = 0.$$

Remark 6. *The binary specialization deserves prominence. Take $K = \mathbb{Z}/2\mathbb{Z}$. By Kitchens' structure theory of group shifts [68], every closed shift-invariant subgroup of $(\mathbb{Z}/2\mathbb{Z})^{\mathbb{Z}}$ is a group shift of finite type, and every proper one is the solution set of a nontrivial linear recurrence and hence finite. If H_μ is finite the quotient map has finite fibres and $h_\sigma((\pi_\mu)_*\mu) = h_\sigma(\mu)$, while if H_μ is everything then μ is the Haar measure, the uniform Bernoulli process, with entropy $\log 2$. Since $h_\sigma(\mu * \mu) \geq h_\sigma(\mu)$ always, by conditioning on one summand, the conjecture therefore asserts: every ergodic binary process with $0 < h_\sigma(\mu) < \log 2$ gains entropy under independent XOR, the coordinatewise sum modulo 2 with an independent copy.*

Proposition 2 (Easy direction, conditional). Suppose $h_\sigma((\pi_\mu)_*\mu) = 0$. Assume the following two ingredients.

(a) The Abramov–Rokhlin entropy addition formula holds for the Haar-fibred compact group extension $\pi_\mu : (\Omega, \mu, \sigma) \rightarrow (\Omega/H_\mu, (\pi_\mu)_*\mu, \sigma)$ and for the corresponding extension of $\mu * \mu$, so that each total entropy splits as the quotient entropy plus the common Haar fibre entropy, the conditional measures of μ and of $\mu * \mu$ over the quotient being Haar measures on cosets of H_μ .

(b) The entropy of a factor of an independent joining is bounded by the sum of the entropies of the factors, so that $(\pi_\mu)_*(\mu * \mu) = ((\pi_\mu)_*\mu) * ((\pi_\mu)_*\mu)$ has entropy at most $2h_\sigma((\pi_\mu)_*\mu) = 0$.

Then $h_\sigma(\mu * \mu) = h_\sigma(\mu)$.

Significance. The mechanism is the stationary analogue of entropy idempotence on a finite group: equality of entropy under independent self-convolution should force all positive entropy to be Haar-type randomness along a translation-invariant subgroup process, modulo a quotient that may retain arbitrarily rich zero-entropy dynamics such as rotations, substitutions, or Toeplitz systems. The easy direction is recorded above as a conditional proposition with its two ingredients named, the Abramov–Rokhlin formula for Haar-fibred compact group extensions and the factor-of-joining entropy bound, and the difficult direction is the rigidity claim that no other equality mechanism exists. The nearest literature boundary consists of Kułaga-Przymus–Lemańczyk [69] on the entropy of products of independent stationary processes and Lindenstrauss–Meiri–Peres [70] on the entropy of convolutions, neither of which decides the binary question of remark 6. No decisive theorem appears to be on record even for Markov processes, and strict XOR entropy gain for mixing two-state Markov chains would be the first. The failure-mode honesty note is this: the natural counterexample hunting ground is the Ornstein–Weiss [73] bilaterally deterministic processes, positive entropy processes whose distant past and distant future jointly determine the whole trajectory, together with T, T^{-1} -type constructions (skew products driven by a random walk, in the style of Kalikow), and no search has yet probed either family, so the negation of the conjecture has an unexplored natural habitat. The gap the rigidity must close is quantitative: equality of entropy rates asserts only that the block mutual information $I(X_{1:n} + Y_{1:n}; Y_{1:n})$ is $o(n)$, not that it is bounded, so sublinear information spread through arbitrarily long-range structure is the resource a counterexample would exploit. The natural staircase of evidence runs from independent letters through Markov and mixing processes to completely positive entropy.

Part IV

The class-counting polynomial of graphical Lie algebras

The twenty conjectures below concern a single canonical object, the class-counting polynomial of a finite graph, approached through the two-step nilpotent Lie algebras that realize it as a rank generating function. They form three programmes. A rigidity and reconstruction programme (Conjectures 57 to 66) asserts that coarse specializations of the polynomial already determine it and that the polynomial in turn determines classical invariants. A log-concavity programme (Conjectures 67 to 71) locates the exact strength of positivity that the coefficient arrays support, with the boundary calibrated by explicit counterexamples one level higher. A tree extremal programme (Conjectures 72 to 76) places the path and the star at the two ends of every ordering the polynomial induces on trees. The proved layer is separated from the conjectural one: the rank formula (Proposition 3) and the two lower bounds on the number of rank values in a tree (Lemma 2) are theorems with full proofs, and every conjecture is stated against that floor. Each statement is held to the standard of the rest of the collection: a canonical object, a stated mechanism, the nearest literature boundary, a first decisive theorem where one exists, and an honest failure mode. One statement, Conjecture 63, has since been resolved false by an explicit seven-vertex pair recorded with the statement, and it is retained with its resolution rather than renumbered.

For a finite simple graph G on vertex set $V(G)$ with edge set $E(G)$ and a field F , the *graphical Lie algebra* $L_G(F)$ is the two-step nilpotent Lie algebra over F with basis $\{v_i : i \in V(G)\}$ together with the central basis $\{z_e : e \in E(G)\}$, the bracket being $[v_a, v_b] = z_{ab}$ for each edge $\{a, b\}$ with $a < b$, $[v_b, v_a] = -z_{ab}$, and all other brackets of basis vectors zero. For $S \subseteq V(G)$ write $N_G[S]$ for the closed neighbourhood of S , the set of vertices lying in S or adjacent to it, and $c(G[S])$ for the number of connected components of the induced subgraph $G[S]$, and set

$$\rho_G(S) = |N_G[S]| - c(G[S]), \quad \rho_G(\emptyset) = 0.$$

Proposition 3 (Rank formula). Let G be a graph on n vertices, let F be a field, and let $x \in L_G(F)$ have vertex part supported on S , all coordinates x_i with $i \in S$ being nonzero. Then

$$\text{rank}(\text{ad}_x) = \rho_G(S),$$

independently of the field and of the nonzero values.

Proof. The map ad_x kills the centre and sends v_j to $\sum_{i \in S, i \sim j} \pm x_i z_{ij}$, so $\text{rank}(\text{ad}_x)$ is the rank of the linear map $y \mapsto [x, y]$ on the vertex space, whose coefficient at z_{ab} is $x_a y_b - x_b y_a$. A vector y lies in the kernel if and only if $x_a y_b = x_b y_a$ for every edge $\{a, b\}$. If both endpoints lie in S this forces the ratio y_i/x_i to be constant along edges of $G[S]$, one free parameter per connected component of $G[S]$. If exactly one endpoint a lies in S it forces $y_b = 0$, for every $b \in N(S) \setminus S$. Edges with neither endpoint in S impose nothing, so y_j is free for every j outside $N[S]$. Hence $\dim \ker = (n - |N[S]|) + c(G[S])$, and the rank is $|N[S]| - c(G[S])$. $\square$

The formula was verified before it was proved: on eighteen thousand random pairs of a graph and an element, the rank of ad_x computed by direct linear algebra over $\mathbb{F}_2$, $\mathbb{F}_3$, and $\mathbb{F}_5$ agreed with $\rho_G(S)$ in every case, and the proof above was written only afterwards.

The *class-counting polynomial* of G is

$$C_G(X, Y) = \sum_{S \subseteq V(G)} (X - 1)^{|S|} Y^{\rho_G(S)}.$$

The polynomial is Rossmann's [77]: he introduced it, proved that its specializations enumerate the conjugacy classes of graphical groups over $\mathbb{F}_q$ by size, proved the multiplicativity $C_{G \sqcup H} = C_G C_H$ over disjoint unions, and showed that the cardinalities of the connected dominating sets of G can be read off from it. Set

$$H_G(U, Y) = C_G(1 + U, Y), \quad F_G(X, Y) = X^{|E(G)|} C_G(X, X^{-1}Y), \quad f_G(X) = F_G(X, 1),$$

so that the coefficient of $U^k Y^r$ in H_G counts the sets S with $|S| = k$ and $\rho_G(S) = r$. At $X = q$ the polynomial records the distribution of $\text{rank}(\text{ad}_x)$ over $L_G(\mathbb{F}_q)$, by Proposition 3. The *subgraph-component polynomial* of Tittmann–Averbouch–Makowsky [78] is

$$Q_G(U, W) = \sum_{S \subseteq V(G)} U^{|S|} W^{c(G[S])}.$$

For a tree T on n vertices let $s(T)$ be the number of distinct values taken by ρ_T , let $\ell(T)$ be the number of leaves, and let $\text{diam}(T)$ be the diameter.

Lemma 2 (Two lower bounds on the number of rank values). *For every tree T on $n \geq 2$ vertices,*

$$s(T) \geq n - \ell(T) + 2 \quad \text{and} \quad s(T) \geq \text{diam}(T) + 1.$$

Proof. For the first bound, when $n = 2$ the values 0 and 1 occur. For $n \geq 3$ the internal vertices, those that are not leaves, induce a subtree with $m = n - \ell(T) \geq 1$ vertices. Order them $v_1, \dots, v_m$ by breadth-first search from an internal root, so that every prefix $S_j = \{v_1, \dots, v_j\}$ is connected and $\rho(S_j) = |N[S_j]| - 1$. Each v_{j+1} has degree at least two in T and its parent lies in the prefix, so it has a neighbour c on the far side, whose only neighbour inside $N[S_j]$ could be v_{j+1} itself, which is not in S_j . Hence c lies in $N[S_{j+1}]$ but not in $N[S_j]$, the sizes strictly increase, and the chain realizes m distinct values, all at least $\deg(v_1) \geq 2$. Together with $\rho(\emptyset) = 0$ and the value 1 at a single leaf this gives at least $m + 2 = n - \ell(T) + 2$ values. For the second bound take a diametral path $v_0, v_1, \dots, v_d$ with $d = \text{diam}(T)$, whose endpoint v_0 is a leaf. For the prefixes $S_j = \{v_0, \dots, v_{j-1}\}$ with $1 \leq j \leq d$, the vertex v_{j+1} witnesses strict growth of $|N[S_j]|$ as long as it exists, since a tree has no edge from v_{j+1} back into $\{v_0, \dots, v_{j-1}\}$. The chain realizes d distinct values starting from $\rho(S_1) = 1$, and $\rho(\emptyset) = 0$ gives $d + 1$ in total. $\square$

17 Rigidity and reconstruction

Conjecture 57 (Irreducibility and connectivity). *For every graph G with at least one vertex, $H_G(U, Y)$ is irreducible in $\mathbb{Z}[U, Y]$ if and only if G is connected.*

Significance. Rossmann's multiplicativity [77] makes the disconnected direction immediate, each factor being nonconstant since $H_{K_1} = 1 + U$, so the content is the connected direction. The mechanism is that a factorization of H_G would express the joint array of $(|S|, \rho_G(S))$ over all vertex subsets as a product of two smaller arrays, which is exactly what a separation of G into two vertex-disjoint parts produces, and the conjecture asserts that no other source of such a product

structure exists: connectivity of the graph should be readable as algebraic indecomposability of the polynomial. The nearest literature boundary is the multiplicativity theorem itself, which supplies one direction and gives the statement its shape. The verification is exact rather than probabilistic: all 996 connected graphs on at most seven vertices have irreducible H_G , by exact factorization over $\mathbb{Z}$. A first decisive theorem would be irreducibility of H_{P_n} for the paths, a one-parameter family at the bottom of the connectivity hierarchy. The failure mode is a connected graph whose polynomial factors, which would reveal a product structure in the rank array not induced by any decomposition of the graph and would make the polynomial blind to connectivity in the strongest possible way.

Conjecture 58 (Recovery of the subgraph-component polynomial). *For all graphs G and H , $C_G = C_H$ implies $Q_G = Q_H$.*

Significance. The two polynomials record transverse marginals of the induced-subgraph structure: C_G records the joint distribution of $(|S|, |N_G[S]| - c(G[S]))$ over all vertex subsets and Q_G the joint distribution of $(|S|, c(G[S]))$, and no functional relation between the two arrays is apparent, so the mechanism must be rigidity of the pairs of arrays that actually arise from graphs. The nearest literature boundary pairs the two sources: the class-counting polynomial and its enumerative theorems are Rossmann’s [77], the subgraph-component polynomial is Tittmann–Averbouch–Makowsky’s [78], and no bridge between them is on record. The polynomial C_G does not determine the graph, and every known collision class was tested: the 119 collision classes among the 1252 graphs on at most seven vertices, a randomized eight-vertex corpus, the unique collision class among all 3159 trees on fourteen vertices (there is none through thirteen, and the class was refound independently by exhaustive search), and the first unicyclic collision at ten vertices. In every class Q_G agrees, as do the domination invariants of Conjectures 65 and 66. A first decisive theorem would be a derivation of Q_T from C_T for trees. The failure mode is a C -collision pair with distinct Q , which would show that the closed-neighbourhood array and the component array are genuinely independent invariants and would break the entire recovery programme at its first joint.

Conjecture 59 (Two-field rigidity). *If $C_G(2, Y) = C_H(2, Y)$ and $C_G(3, Y) = C_H(3, Y)$, then $C_G = C_H$. Equivalently, the adjoint-rank distributions of the graphical Lie algebras over $\mathbb{F}_2$ and $\mathbb{F}_3$ determine them over every finite field.*

Significance. The two hypotheses are the unweighted and the $2^{|S|}$ -weighted rank statistics, $C_G(2, Y) = \sum_S Y^{\rho_G(S)}$ and $C_G(3, Y) = \sum_S 2^{|S|} Y^{\rho_G(S)}$, and recovering a nonnegative integer array from these two evaluations is impossible for general arrays, so the mechanism is rigidity of the arrays that graphs realize rather than linear algebra. The equivalence with the finite-field form is Proposition 3, and the nearest literature boundary is Rossmann’s enumeration theorem [77], through which the conjecture reads: two conjugacy-class censuses, over $\mathbb{F}_2$ and $\mathbb{F}_3$, determine the census over every finite field. The verification is exhaustive on at most seven vertices and covers six thousand random eight-vertex graphs. A first decisive theorem would be two-field rigidity for trees. The failure mode is a pair of graphs agreeing over $\mathbb{F}_2$ and $\mathbb{F}_3$ yet separated over some larger field, which would reveal genuinely field-dependent information in the adjoint-rank distribution and would locate it at a specific weight $(q - 1)^{|S|}$.

Conjecture 60 (Hybrid rigidity). *If $C_G(2, Y) = C_H(2, Y)$ and $Q_G = Q_H$, then $C_G = C_H$.*

Significance. The mechanism is complementarity of marginals: the binary specialization carries the rank distribution and Q_G carries the joint size-and-component array, and the conjecture asserts that these two projections of the induced-subgraph data jointly pin the full class-counting polynomial. The pairing is calibrated against a cheaper substitute, since $C_G(2, Y)$ together with the degree sequence does not determine C_G , so the component information in Q_G is doing work that no degree data can do. The nearest literature boundary is again the pair Rossmann [77] and Tittmann–Averbouch–Makowsky [78], whose polynomials the hypothesis mixes. The verification is exhaustive on at most seven vertices. A first decisive theorem would be the tree case. The failure mode is a pair agreeing in both hypotheses but not in C , which would show that the joint array holds correlation information invisible to this pair of its natural projections.

Conjecture 61 (Binary edge-deck reconstruction). *For graphs G and H with at least four edges each, if the multisets $\{C_{G-e}(2, Y) : e \in E(G)\}$ and $\{C_{H-e}(2, Y) : e \in E(H)\}$ coincide, then G and H are isomorphic.*

Significance. This is the high-risk member of the part, in the orbit of the edge reconstruction conjecture, for which Bondy–Hemminger [79] is the survey. Each card here retains only the binary rank distribution of an edge-deleted subgraph rather than its isomorphism type, so the deck is a drastic coarsening of the classical edge deck, and the mechanism is that rank distributions vary so tightly under single-edge deletion that the multiset of shadows should already separate graphs. The nearest literature boundary is the reconstruction literature surveyed in [79], against which the present statement trades weaker cards for the same conclusion. The verification is exhaustive on at most seven vertices among graphs with at least four edges. A first decisive theorem would be reconstruction of trees from the binary deck. The failure mode is a nonisomorphic pair with equal binary decks, which would measure exactly how much the rank distribution forgets under edge deletion, and a counterexample here would be the least surprising in the part, which is why the statement is flagged as its stress point.

Conjecture 62 (Bipartite binary rigidity). *For bipartite graphs G and H , $C_G(2, Y) = C_H(2, Y)$ implies $C_G = C_H$.*

Significance. A single specialization cannot determine the polynomial in general, since the implication fails already for connected unicyclic graphs on seven vertices (the resolution of Conjecture 63), so the mechanism is that a hereditary class restriction removes the room in which the unweighted rank distribution and the size-weighted array can decouple. The nearest literature boundary is Rossmann’s specialization theorem [77], which gives the hypothesis its meaning as the $\mathbb{F}_2$ census. The verification is a six-thousand-graph random holdout of bipartite graphs on nine to twelve vertices, spanning 2747 distinct binary specializations, with no violation. A first decisive theorem would be the tree case, shared with the surviving tree layer of Conjecture 63. The failure mode is a bipartite pair with equal binary specialization and distinct C , which would locate the boundary of one-field rigidity strictly inside the bipartite world and would redirect the class-restriction programme towards sparser families.

Conjecture 63 (Connected-pseudoforest binary rigidity, resolved false). *For G and H each a tree or a connected unicyclic graph, $C_G(2, Y) = C_H(2, Y)$ implies $C_G = C_H$.*

Remark 7 (Resolution). *The conjecture is false as stated. The connected unicyclic graphs on vertex set $\{0, \dots, 6\}$ with edge sets $\{\{0, 3\}, \{0, 4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{3, 6\}, \{4, 5\}\}$ and $\{\{0, 1\}, \{0, 3\}, \{0, 4\}, \{1, 2\}, \{2, 3\}, \{2, 4\}, \{3, 6\}, \{4, 5\}\}$ have the same binary specialization but are not isomorphic.*

a four-cycle carrying three leaves against a triangle carrying a three-vertex path with two further leaves, share the binary specialization $C(2, Y) = 1 + 3Y + 7Y^2 + 29Y^3 + 32Y^4 + 40Y^5 + 16Y^6$ while their class-counting polynomials differ already in the singleton stratum, whose coefficients record the degree multisets $\{1, 1, 1, 2, 3, 3, 3\}$ and $\{1, 1, 2, 2, 2, 3, 3\}$. The pair was communicated to us and confirmed by the independent implementation, and an exhaustive scan shows it is the unique violating class among all trees and connected unicyclic graphs through nine vertices. One member is bipartite and not chordal and the other chordal and not bipartite, and the ternary specializations differ, so Conjectures 62, 64, and 59 are untouched. The tree case survives exhaustively through fourteen vertices and remains open. Connectedness was in any case not the failure boundary, since the implication also fails for disconnected pseudoforests.

Significance. The statement is retained with its resolution because the failure is itself informative. The refuting pair shows that on unicyclic graphs the binary rank distribution can forget the degree multiset, the coarsest datum beyond the vertex and edge counts, so one cycle already leaves enough freedom to decouple the unweighted rank census from the size-weighted array. The failure calibrates the neighbouring statements rather than undermining them: the pair straddles the classes of Conjectures 62 and 64, one member bipartite and one chordal, so the bipartite and chordal rigidity claims stand on graphs the counterexample cannot reach, and the two-field claim of Conjecture 59 absorbs it, since the ternary specializations differ. The surviving positive content is the tree case, exhaustively unrefuted through fourteen vertices, which is exactly the first decisive theorem the original statement named.

Conjecture 64 (Chordal binary rigidity). *For chordal graphs G and H , $C_G(2, Y) = C_H(2, Y)$ implies $C_G = C_H$.*

Significance. Chordal graphs, those in which every cycle of length at least four has a chord, form the natural dense counterpart to the sparse classes of Conjectures 62 and 63, and the mechanism is the same class rigidity tested against a family with unbounded clique structure rather than bounded cycle structure. The nearest literature boundary is once more the specialization theorem [77] that interprets the hypothesis over $\mathbb{F}_2$. The verification is a six-thousand-graph random holdout of chordal graphs on nine to twelve vertices, spanning 5598 distinct binary specializations, with no violation. A first decisive theorem would be the case of trees, which are chordal, followed by block graphs. The failure mode is a chordal pair separating the binary specialization from the polynomial. The resolution of Conjecture 63 sharpens rather than settles the question here, since the refuting pair there has exactly one chordal member, so chordal rigidity survives it and its status is genuinely open.

Conjecture 65 (Recovery of the domination number). *For all graphs G and H , $C_G = C_H$ implies $\gamma(G) = \gamma(H)$.*

Significance. A set is dominating exactly when its closed neighbourhood is the whole vertex set, and the array behind C_G records $|N_G[S]|$ only through the difference $\rho_G(S) = |N_G[S]| - c(G[S])$, so the dominating sets are entangled with component counts and the mechanism is that the entanglement can be undone at the extreme where γ lives. The conjecture sits strictly between a theorem and a false analogue, which is the nearest literature boundary: the cardinalities of connected dominating sets are readable from the coefficients, by Rossmann [77], while the total domination number is not determined by C_G . Every known collision class of C , the corpus

recorded at Conjecture 58, has agreeing domination number. A first decisive theorem would be the tree case, where the unique fourteen-vertex collision class is the only known test. The failure mode is a collision class with distinct domination numbers, which would place γ on the undetermined side of the boundary alongside total domination and would show that the connected-domination theorem is the exact limit of what the polynomial sees.

Conjecture 66 (Recovery of the independent domination number). *For all graphs G and H , $C_G = C_H$ implies $i(G) = i(H)$, where i is the independent domination number, the least cardinality of a set that is simultaneously independent and dominating.*

Significance. Independent dominating sets impose the two conditions the array treats most asymmetrically, domination through $|N_G[S]|$ and independence through $c(G[S]) = |S|$, and the mechanism is that their interaction leaves a trace in the joint distribution of $(|S|, \rho_G(S))$ even though neither condition is separately readable. The nearest literature boundary is the same theorem-versus-false-analogue frame as Conjecture 65: connected domination is readable [77], total domination is not determined, and i is conjectured to fall on the determined side. Every known collision class of C , the corpus recorded at Conjecture 58, has agreeing independent domination number. A first decisive theorem would be recovery of i for trees. The failure mode is a collision class with distinct i , which would split the domination invariants into determined and undetermined families along a line that no present structural explanation predicts.

18 Log-concavity and support

Conjecture 67 (Global double log-concavity). *For every graph G the coefficient sequence a of $f_G(1 + Z)$ is log-concave, and so is its transform $L(a)$ defined by $L(a)_i = a_i^2 - a_{i-1}a_{i+1}$, out-of-range terms being zero.*

Significance. The expansion point is canonical: at $X = 1 + Z$ the polynomial is expanded about the degenerate point $X = 1$, where the weights $(X - 1)^{|S|}$ vanish for every nonempty set, and f_G folds the rank variable into the size variable at the same point, so the sequence a is the most compressed image of the whole array. The statement is calibrated to the strongest form the data support: log-concavity is asserted, its once-iterated form is asserted, and the calibration is exact because the third iterate of L fails for a thirteen-vertex graph. The nearest literature boundary is Rossmann’s paper [77], whose theorems about this polynomial are enumerative, the specialization, multiplicativity, and dominating-set results, so the positivity layer opens a new face of the object. The verification covers the full atlas of graphs on at most seven vertices and twenty-five hundred random graphs on eight to ten vertices. A first decisive theorem would be log-concavity of $f_T(1 + Z)$ for trees. The failure mode is graded: a failure of the second level would compress the true boundary into the narrow band between one and three iterates, while a failure of bare log-concavity on some large graph would show that the seven-vertex atlas sits below the threshold where the array’s geometry turns.

Conjecture 68 (Strict log-concavity for connected graphs). *For connected G with at least one edge, $a_i^2 > a_{i-1}a_{i+1}$ at every interior position of the support of a , where a is the coefficient sequence of $f_G(1 + Z)$.*

Significance. The mechanism is that equality cases of log-concavity should be confined to the degenerate stratum, and in this part the degenerate stratum is disconnectedness, exactly as in Conjecture 57: multiplicativity [77] builds the polynomials of disconnected graphs as products, and products are where interior coincidences are manufactured. The nearest literature boundary is that same multiplicativity theorem, which marks the hypothesis as necessary in spirit. The verification is the corpus of Conjecture 67, the full seven-vertex atlas and twenty-five hundred random graphs on eight to ten vertices, with strictness observed throughout the connected stratum. A first decisive theorem would be strictness for paths and stars, the two ends of the tree order of Conjecture 72. The failure mode is a connected graph with an interior equality, an exact coincidence $a_i^2 = a_{i-1}a_{i+1}$, which would demand an algebraic explanation and would show that connectivity does not exhaust the sources of degeneracy.

Conjecture 69 (Strict second-order log-concavity). *For connected G with at least one edge, the transform $L(a)$ of Conjecture 67 is strictly log-concave at every interior position of its support.*

Significance. The statement pushes the strictness mechanism of Conjecture 68 one level up the L -hierarchy, and its position is the sharpest the evidence permits, since the third iterate of L fails already for a thirteen-vertex graph and the second-order slice strengthening fails for an eight-vertex graph (Conjecture 70), so the conjecture threads a boundary with counterexamples visible on two sides. The nearest literature boundary is as in Conjecture 67: the polynomial's literature [77] is enumerative and the positivity hierarchy is uncharted. The verification is the same corpus, the full seven-vertex atlas and twenty-five hundred random graphs on eight to ten vertices. A first decisive theorem would be second-order strictness for paths. The failure mode is a connected graph whose $L(a)$ has an interior equality or reversal, which by the two flanking counterexamples would pin the exact rung of the hierarchy at which graphs stop being log-concave, the most informative single outcome the programme could produce.

Conjecture 70 (Rank-slice strict log-concavity). *For connected G and every attainable rank r , the slice $A_{G,r}(Z) = [Y^r] F_G(1 + Z, Y)$, when it is not a monomial, has strictly log-concave coefficients at every interior position of its support.*

Significance. The mechanism refines Conjecture 68 from the folded sequence to each rank slice separately: the sets of a fixed rank r are counted by size, and strict log-concavity asserts that each of these level sets is unimodal in the strong multiplicative sense, so that positivity holds fibrewise and not only in aggregate. The calibration is exact here too, since the second-order strengthening of this slice statement fails for an eight-vertex graph, a counterexample reconfirmed by the independent implementation, so the slice hierarchy stops one level below the global hierarchy of Conjecture 69. The nearest literature boundary is Rossmann's [77] reading of the slices as class data over finite fields. The verification is the full seven-vertex atlas and twenty-five hundred random graphs on eight to ten vertices. A first decisive theorem would be strict log-concavity of every slice for stars. The failure mode is a connected graph with a non-log-concave slice, which would show that aggregate log-concavity, if it survives in Conjecture 67, is an artefact of summation rather than a fibrewise phenomenon.

Conjecture 71 (Interval support at fixed rank). *For every graph G and every attainable rank r , the set $K_r(G) = \{|S| : \rho_G(S) = r\}$ is an interval of integers.*

Significance. This is the support statement beneath the positivity statements: strict log-concavity of a slice presupposes gapless support, and the conjecture asserts gaplessness for every graph, not only connected ones. The mechanism would be an exchange argument, passing between witnesses of the same rank whose sizes differ by one, and the direction of the statement is essential, since the transposed assertion is false: the attainable ranks at a fixed support size need not form an interval. The interval structure, if real, is therefore a property of the size direction alone. The nearest literature boundary is the support analysis implicit in Rossmann’s connected-dominating-set theorem [77], which reads off one extreme stratum of the array. The verification is the full seven-vertex atlas and twenty-five hundred random graphs on eight to ten vertices. A first decisive theorem would be the interval property for trees. The failure mode is a graph with a gapped rank level set, which would break the fibrewise programme of Conjecture 70 at the level of supports and, given the falsity of the transposed statement, would show that neither direction of the array has interval structure.

19 Tree extremality and rigidity

Conjecture 72 (Coefficientwise tree extremality). *For every tree T on n vertices,*

$$f_{P_n}(1 + Z) \leq f_T(1 + Z) \leq f_{K_{1,n-1}}(1 + Z)$$

coefficientwise, with equality only for the path and the star respectively.

Significance. The mechanism is visible in Lemma 2: the rank landscape of a tree is driven by its internal vertices and its diameter, and the path and the star are the two trees that extremize both statistics simultaneously, the path with two leaves and maximal diameter, the star with maximal leaf count and diameter two. The conjecture asserts that this extremality is not merely a matter of the range of ρ_T but holds coefficient by coefficient in the shifted polynomial, the strongest ordering the object admits. The nearest literature boundary is Rossmann’s enumeration [77], which interprets each coefficientwise inequality as a family of inequalities between class censuses over every finite field at once. The verification is exhaustive over all trees through fourteen vertices. A first decisive theorem would be either inequality for one nontrivial family, say the star bound for caterpillars. The failure mode is a tree escaping the sandwich at a single coefficient, which would show that the coefficientwise order on trees is not anchored at the degree extremes and would force the extremal programme to work slice by slice instead.

Conjecture 73 (Star rank-majorization). *For every $n \geq 5$ and every prime power q , the decreasingly ordered adjoint-rank distribution of $L_{K_{1,n-1}}(\mathbb{F}_q)$ majorizes that of $L_T(\mathbb{F}_q)$ for every n -vertex tree T .*

Remark 8. *The hypothesis $n \geq 5$ cannot be dropped: at $n = 4$ and $q = 2$ the star fails to majorize the path. The dual assertion, that the path is majorization-minimal among n -vertex trees, is false. The failure at $(n, q) = (4, 2)$ was found in the verification reported below.*

Significance. Majorization compares two probability vectors through all partial sums of their decreasingly ordered entries, so the conjecture asserts that the star’s adjoint-rank distribution is the most concentrated among trees uniformly across the whole partial-sum hierarchy and across every finite field at once. The mechanism is concentration: by Proposition 3 the rank of a

random element is a statistic of its vertex support, and the star compresses the possible values harder than any other tree. The remark shows the inequality system is genuinely tight, which is the honest register for a majorization claim: at $(n, q) = (4, 2)$ one partial sum reverses, and the dual minimality of the path fails outright, so the surviving statement is exactly the one the data leave standing. The nearest literature boundary is Rossmann’s [77] reading of these distributions as class data of graphical groups. The verification is exhaustive for $q \in \{2, 3, 4, 5\}$ over all trees through twelve vertices, for $q \in \{2, 3\}$ at thirteen, and for $q = 2$ at fourteen. A first decisive theorem would be the fixed comparison of star against path for all $n \geq 5$ and all q . The failure mode is a tree and a prime power breaking one partial sum at some $n \geq 5$, which would show that the small- n anomaly is the visible end of a persistent phenomenon rather than a boundary effect.

Conjecture 74 (Tree rank-variance extremality). *For every $n \geq 4$ and every prime power q , among n -vertex trees the variance of $\text{rank}(\text{ad}_x)$ over uniformly random $x \in L_T(\mathbb{F}_q)$ is uniquely minimized by the path and uniquely maximized by the star.*

Significance. The two tree orderings of this programme are not redundant: majorization compares the ordered probabilities while the variance measures the spread of the rank values themselves, and the star sits at the top of both scales, most concentrated in the sense of Conjecture 73 and most spread in the sense of variance, with the path now appearing as the genuine minimum. The mechanism is again Proposition 3, which makes the variance a purely combinatorial functional of the array $(|S|, \rho_T(S))$ weighted by $(q - 1)^{|S|}$. The nearest literature boundary is the enumerative frame of Rossmann [77], in which the variance is a second moment of the class census. The verification matches Conjecture 73: exhaustive for $q \in \{2, 3, 4, 5\}$ through twelve vertices, for $q \in \{2, 3\}$ at thirteen, and for $q = 2$ at fourteen. A first decisive theorem would be the uniqueness of the star maximum at $q = 2$. The failure mode is a tree overtaking an extreme at some (n, q) , and since the statement begins at $n = 4$ where the majorization statement already fails, a violation would separate the variance order from the majorization order and show that the two concentration scales genuinely diverge on trees.

Conjecture 75 (Leaf-count rigidity). *For a tree T on $n \geq 2$ vertices, equality holds in the bound $s(T) \geq n - \ell(T) + 2$ of Lemma 2 exactly when T is a path or a star.*

Significance. The lemma’s proof realizes $n - \ell(T) + 2$ rank values along a breadth-first chain of internal vertices, and equality demands that every subset of vertices lands on one of the chain’s values, so the mechanism is rigidity of the equality case: only the two extreme trees should leave no room for excess values off the chain. The nearest literature boundary is internal, the lemma itself, whose bound the conjecture upgrades to a characterization, in the same relation as an isoperimetric equality case stands to its inequality. The verification is exhaustive over all trees through fourteen vertices. A first decisive theorem would be the forward half, that the path and the star do achieve equality, a concrete computation of s for two explicit families. The failure mode is a third tree achieving equality, which would show that the leaf count fails to see some genuinely different extremal shape and would refute the rigidity reading of Lemma 2 while leaving the lemma intact.

Conjecture 76 (Diameter rigidity). *For a tree T on $n \geq 2$ vertices, equality holds in the bound $s(T) \geq \text{diam}(T) + 1$ of Lemma 2 exactly when T is a path or a star.*

Significance. The diametral chain of the lemma realizes $\text{diam}(T) + 1$ values, and equality requires the whole subset lattice to produce nothing beyond the chain, so the mechanism parallels Conjecture 75 with the diameter replacing the internal vertex count, and the two rigidity statements pin the same extremal pair from two independent directions, each serving as a cross-check on the other. The nearest literature boundary is again Lemma 2 itself. The verification is exhaustive over all trees through fourteen vertices. A first decisive theorem would be the forward half for the path, where the diameter is $n - 1$ and the claimed equality reads $s(P_n) = n$. The failure mode is a tree of small diameter and few rank values outside the pair, which would show the diameter bound to be loose in a structured way and would decouple the two rigidity statements that the exhaustive search through fourteen vertices keeps locked together.

20 Stress tests and independent recomputation

Each statement was tested three ways: against the literature, against computation well beyond its original range, and against an independent reimplement.

Layer 1: literature. Five independent searches combed OEIS, arXiv, and the standard references for prior art, producing the novelty labels and the attributions used throughout. Each search was run at neighbourhood depth: defining objects, derived sequences, OEIS comments, and the abstracts and bodies of near-neighbour papers read in full rather than skimmed from search summaries—the depth at which, for instance, OEIS A088054’s intersection comment (Conjecture 21(ii)) and the second half of Lillie’s abstract (the primordial companion) are found, neither being visible in the defining sequences or in a search snippet. This layer also fixes two attributions used above: the $k \geq 1$ Stern list of Conjecture 12 is OEIS A060003 verbatim, with 1493 the largest known Stern prime, and the primordial-twin statement is Lillie’s [5].

Layer 2: scripted refutation. Every falsifiable-by-instance statement was pushed at least a decade past its original bound: exception hunts across $(10^8, 10^9]$ for Conjectures 5, 12, 17, 22; the uniformity of Conjecture 8 stressed on $d \leq 6000$; the statistical laws re-tested at 4×10^9 (Conjectures 1, 4, 9, 11, and 13, and the quintuplet calibration count reported with Conjecture 8); the recovery clause of Conjecture 3 tested on fresh moduli $q \in (3000, 6000]$. No statement moved. The same layer settles one statement negatively: a completeness list for Fibonacci–Lucas twins is destroyed by OEIS A080327’s catalogued index 148091 (conditionally on the probable-prime classifications of F_{148091} and L_{148091} alike, neither being a proven prime), which is why Conjecture 20 conjectures finiteness without a completeness clause and carries that index as its calibration clause.

Each statement was also verified at large scale: the Conjecture 10 profile across 150 shifts and the moment law for its constants $C(d)$ (derived mean 2.7456 and sd 1.6840 against measured 2.7434 and 1.6726); the cubic family moments and uniformity of Conjecture 15 (294 constants, 57 count profiles); the triplet and sexy-pair races of Conjectures 18 and 19 at 10^9 and the Stern lanes of Conjecture 12 on 2,500 samples at 10^8 ; the chain of Conjecture 24 over five decades to 10^7 ; the null race of Conjecture 14 at 10^7 ; the Fibonacci–Lucas joint scan to $p \leq 10^4$; factorial twins to $n = 700$; the CRT-exact factorization of Conjecture 6 through a joint period of 1.2×10^8 , with the scan extended to $n = 6000$; the pinned average $G(H)$ of Conjecture 2 computed exactly to $H = 3000$; the boundary trichotomy of Conjecture 22 with two new lanes counted at 10^6 ; the least-summand law of Conjecture 5 on 2,000 samples; the covariance kernels of Conjectures 8(ii) and 16(ii) evaluated over 870 and 1,225 pairs, with the moving-window

randomization of Conjecture 8(ii) over 2000 windows (empirical correlation matrix matching the predicted kernel entrywise at 0.86); the orientation-resolved twin-member profile of Conjecture 17 on 150 samples; the race autocorrelation and running-maximum measurements of Conjecture 9 at 10^9 against simulated nulls; the singular-series waiting-time clause of Conjecture 11(iii) (regression coefficient -0.466 against the predicted $-\frac{1}{2}$); the stratified experiment of Conjecture 3 on 80 moduli; the multibase quotient tests of Conjecture 7 on 664,577 primes; and the weighted drift and internal null lane of Conjecture 4 on 400 fresh samples, together with the cousin-race predictions of the contamination calculus, Conjecture 1(v), at 10^9 (leadership log-densities 0.99 and 0.92 on the predicted sides). In each case the prediction was derived before the corresponding data were taken.

Layer 3: independent replication. Constants were recomputed from their definitions, and sequences recounted, by implementations sharing no code with the primary computation and working from the bare statements alone. All recomputed constants landed inside the stated truncation error, and all recounts agreed. The same layer reproduced the Fibonacci deficit recorded at Conjecture 20 and the ordering anomaly of Remark 1, confirming that both are properties of the primes rather than of a single implementation.

One methodological conclusion is worth recording. The mathematical failure modes in this subject are not statistical but *algebraic*: factorizations and congruence collapses living on density-zero or positive-density-but-structured subsequences—the cubes obstructing $n = p + k^3$ (Theorem 1), the composite- k factorization of D_k (Conjecture 23(i)), the $k = 4$ parity branch of Conjecture 25, the inadmissible naive chain of Conjecture 24. Probabilistic sanity checks, however extensive, integrate over density and cannot see such families; only structural tests—testing a claimed exception census on special subsequences, for instance—find them. A verification that only re-runs an existing count at a larger bound is blind to every one of them.

Part IV. The Part IV verification programme follows the same standard. The rank formula of Proposition 3 was checked against direct linear algebra over $\mathbb{F}_2$, $\mathbb{F}_3$, and $\mathbb{F}_5$ on eighteen thousand random pairs of a graph and an element before the proof was written. The irreducibility claim of Conjecture 57 was verified by exact factorization over $\mathbb{Z}$ for all 996 connected graphs on at most seven vertices. The recovery statements of Conjectures 58, 65, and 66 were tested on every known collision class of C_G : the 119 classes among the 1252 graphs on at most seven vertices, a randomized eight-vertex corpus, the unique class among all 3159 trees on fourteen vertices, refound independently by exhaustive search, and the first unicyclic collision at ten vertices, with Q_G , γ , and i agreeing in every class. The rigidity statements were scanned exhaustively on at most seven vertices, with six thousand random eight-vertex graphs added for Conjecture 59 and the scan for Conjecture 61 restricted to graphs with at least four edges, and the class-restricted statements were tested on six-thousand-graph random holdouts for each of Conjectures 62, 63, and 64, spanning 2747, 3192, and 5598 distinct binary specializations respectively, together with all 656 binary classes of ten-vertex unicyclic graphs for Conjecture 63. The log-concavity and support statements of Conjectures 67 to 71 were verified on the full atlas of graphs on at most seven vertices and twenty-five hundred random graphs on eight to ten vertices, with the eight-vertex counterexample to the second-order slice strengthening reconfirmed by the independent implementation. The tree programme of Conjectures 72 to 76 was verified exhaustively over all trees through fourteen vertices, for Conjectures 73 and 74 with $q \in \{2, 3, 4, 5\}$ through twelve vertices, $q \in \{2, 3\}$ at thirteen, and $q = 2$ at fourteen, and the sharpness failure at $(n, q) = (4, 2)$ recorded at Conjecture 73 was found in this verification rather than assumed. A seven-vertex pair refuting Conjecture 63 was communicated to us after these holdouts were run, outside their

ten-to-thirteen-vertex range. The pair was confirmed by the independent implementation, an exhaustive rescan located it as the unique violating class among trees and connected unicyclic graphs through nine vertices, and the tree case was reverified exhaustively through fourteen vertices.

21 An open question

Question 1. Determine $\Theta(q)$ of Conjecture 3: derive, from the Hardy–Littlewood correlations or otherwise, the first-order deficit of $\mathbb{E}_a \text{Li}(p(a, q))/\varphi(q)$ below its random-model value 1—equivalently, explain quantitatively why the primes in their natural order occupy residue classes faster than any exchangeable resampling of themselves.

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